Geometric singularities of K-trivial regular surfaces

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Abstract

Let X be a regular projective surface over a field k of characteristic p such that $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. We show that X is geometrically normal if $p \geq 13$ and smooth if $p \geq 23$. For $p \in \{13, 17, 19\}$, the geometric base change $X_{\bar{k}}$ is either smooth or has exactly one singularity, of type A_{p-1}. Both bounds are optimal: we construct an example in characteristic 11 that is not geometrically normal and, for every prime $13 \leq p \leq 19$, an example that is not smooth. The proofs were mainly done by AI, and the author is still checking and revising the manuscript.

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1 Introduction

Over an imperfect field, a regular variety need not be smooth, and its base change to an algebraic closure need not be normal or even reduced. Let k be a field of characteristic $p > 0$, and let X be a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. Wang and Zhang proved that such a surface is geometrically integral if $p \geq 7$ [WZ26, Theorem 1.2]. If moreover $p \geq 7$ and X is not geometrically normal, then every numerically trivial line bundle on X is trivial, in particular $K_X \sim 0$ [CWZ26, Corollary 6.2]. Wang and Zhang also exhibit, for $p = 5$ and $p = 7$, examples with $K_X \equiv 0$ which are geometrically integral but not geometrically normal, and they ask whether geometric normality holds in large characteristic [WZ26, Question 1.1 and Remark 1.5(2)]. We prove that it does for $p \geq 13$.

Theorem 1.1 (Theorem 7.6)**.** *Let k be a field of characteristic $p \geq 13$, and let X be a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. Then X is geometrically normal over k.*

The bound $p \geq 13$ is optimal.

Theorem 1.2 (Proposition 10.1)**.** *Let k be the field $\mathbb{F}_{11}(c)$, with c transcendental over $\mathbb{F}_{11}$. Then there is a regular projective surface X over k with $H^0(X, \mathcal{O}_X) = k$ and $\omega_X \cong \mathcal{O}_X$ which is not geometrically normal over k.*

The surface X is fibred over $\mathbb{P}^1_k$, with coordinate u, in curves of genus one. Its generic fibre is a twist, by an α_{11}-torsor, of the elliptic curve $y^2 = x^3 + u^5$. The fibre over $u = 0$ is an integral curve, but $X_{k(c^{1/11})}$ is not normal along it. Over this point the Jacobian has reduction of type II*, which is precisely the configuration that the argument of Section 6 leaves open for $p = 11$ (Remark 7.7).

For $p \geq 23$, we obtain the stronger conclusion of smoothness.

Theorem 1.3 (Theorem 8.9)**.** *Let k be a field of characteristic $p \geq 23$, and let X be a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. Then X is smooth over k.*

The bound $p \geq 23$ is optimal.

Theorem 1.4 (Proposition 9.1 and Example 11.1)**.** *Let k be a field of characteristic $p \in \{13, 17, 19\}$, and let X be a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. Then $X_{\bar{k}}$ is either smooth or has exactly one singular point, of type A_{p-1}. For each such p, the singular case occurs over $k = \mathbb{F}_p(c)$, with c transcendental over $\mathbb{F}_p$.*

Remark 1.5. For $p = 19$, the example in Theorem 1.4 is the special fibre of a regular flat projective model over $\mathbb{Z}[c]_{(19)}$ with K3 generic fibre and trivial relative canonical sheaf (Proposition 11.2). This model shows that the bound $p \geq 23$ in Corollary 11.3 is optimal in mixed characteristic, even when both the total space and the special fibre are regular.

For completeness, we show that regular surfaces with nef anticanonical class and $K_X \not\equiv 0$ are geometrically normal for $p \geq 5$ and smooth for $p \geq 11$.

Theorem 1.6 (Proposition 12.3)**.** *Let k be a field of characteristic $p \geq 5$, and let X be a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ such that $-K_X$ is nef and $K_X \not\equiv 0$. Then X is geometrically normal over k, and $X_{\bar{k}}$ has at worst rational double points. If moreover $p \geq 11$, then X is smooth over k.*

Both bounds in Theorem 1.6 are optimal, already for del Pezzo surfaces. In characteristics 2 and 3 there are regular del Pezzo surfaces which are not geometrically normal [Ber24, Example 2.3]. In characteristic 7 there is a regular del Pezzo surface of degree 2 whose geometric base change has a singular point of type A_6, so it is not smooth [MS24, Example 2.3].

The genus-one argument uses a local result for minimal regular models over excellent discrete valuation rings of equal characteristic $p > 0$. If the Jacobian has good reduction and the closed fibre is reduced, then the discrepancy b between the lattices of relative differentials is either zero, in which case the model is smooth, or at least $p-1$ (Corollary 5.5). For the surfaces considered here, a tame base change gives $b \leq 11$, so this local result supplies the contradiction for $p \geq 13$ in the potentially good reduction case.

1.1 Proof strategy of Theorem 1.1

Suppose that $p \geq 13$ and that X is not geometrically normal. We may assume that k is F-finite and separably closed. The main step in the genus-one case compares the relative dualizing sheaves of the fibration and its Jacobian after tame base change. In the anisotropic case, we contract a negative curve while preserving failure of geometric normality, then exclude the resulting surface of Picard number one. Figure 1 shows the structure of the argument.

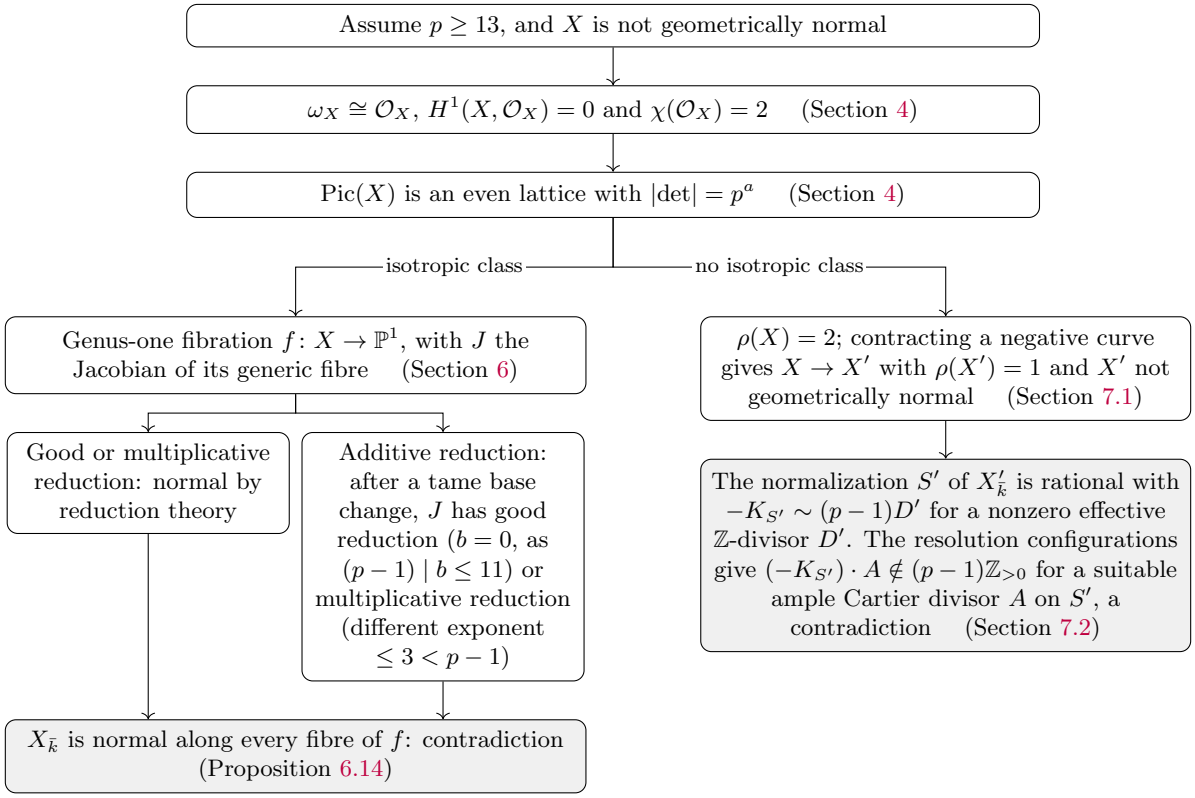


Figure 1: Structure of the proof of Theorem 1.1.

1.2 Proof strategy of Theorem 1.3

Assume now that $p \geq 23$. By Theorem 1.1, the geometric surface $Y := X_{\bar{k}}$ is normal. We may assume that k is F-finite and separably closed. The elliptic case is excluded by combining a local bound $d \leq 3$ on the degree of the singularity with a computer-assisted enumeration forcing $d = 4$. The computation is needed

only when $p \equiv 1 \pmod 4$ and $p \leq 6912$. Figure 2 shows the structure of the argument.

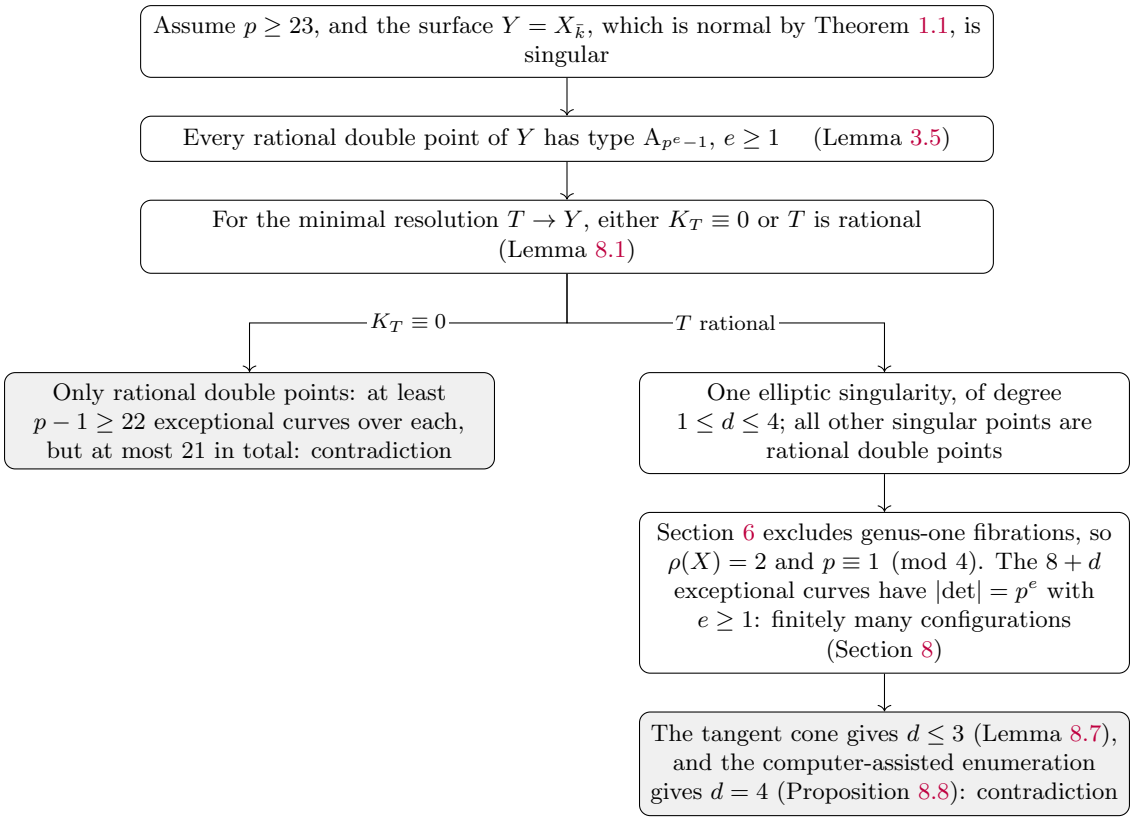


Figure 2: Structure of the proof of Theorem 1.3.

Acknowledgements

The author thanks Lei Zhang for sharing an initial draft of this work, which was generated by Jie Liu using ChatGPT. That draft showed that the surfaces studied here are smooth for $p \geq p_0$, with an ineffective constant p_0. The author improved this result with the help of Claude Code, and contributed mainly by reorganizing the exposition.

2 Conventions and preliminaries

2.1 Conventions

- Throughout, let k be the base field, of characteristic $p > 0$.
- Surfaces and curves are integral and of finite type over k, unless stated otherwise.
- For a line bundle $\mathcal{L}$ on a proper curve over a field F, we write $\deg_F(\mathcal{L})$ for its degree over F. For a Cartier divisor D on a surface X and an integral curve $C \subset X$ proper over F, we put $(D \cdot C)_F := \deg_F(\mathcal{O}_X(D)|_C)$ and write $(D^2)_F$ for self-intersection. Unless indicated otherwise, intersection numbers on a surface over F are computed over F, and intersections with vertical curves on a model over a discrete valuation ring are computed over its residue field. We omit the subscript when the field is clear.
- If a curve C is proper over a finite extension ℓ/k, then $\deg_k(\mathcal{L}) = [\ell : k] \deg_\ell(\mathcal{L})$ for every line bundle $\mathcal{L}$ on C. For an integral proper curve $C \subset X$ over k, with normalization $\nu\colon C^\nu \to C$ and $\ell = H^0(C^\nu, \mathcal{O}_{C^\nu})$, we also write $(D \cdot C)_\ell := \deg_\ell\big(\nu^*(\mathcal{O}_X(D)|_C)\big)$, so
$$(D \cdot C)_k = [\ell : k](D \cdot C)_\ell.$$

- On a normal projective surface we use Mumford's rational intersection pairing of Weil divisors, computed over the base field of the surface.
- We write $\sim$ for linear equivalence and $\equiv$ for numerical equivalence.
- For a $\mathbb{Q}$-factorial normal projective surface Z, we write $N^1(Z)_{\mathbb{Q}}$ for the $\mathbb{Q}$-vector space of Weil divisors with rational coefficients modulo numerical equivalence, and $\rho(Z) = \dim N^1(Z)_{\mathbb{Q}}$.
- The intersection form on $N^1(Z)_{\mathbb{Q}}$ is *isotropic* if $D^2 = 0$ for some nonzero $D \in N^1(Z)_{\mathbb{Q}}$, and *anisotropic* otherwise.
- A *negative curve* on a surface is a curve with negative self-intersection.
- A *fibration* is a proper morphism $f\colon W \to B$ of schemes with $f_*\mathcal{O}_W = \mathcal{O}_B$.
- For a scheme W over k and a field extension k'/k we write $W_{k'} = W \times_k \operatorname{Spec} k'$.
- If W is geometrically integral over k, its *geometric normalization* is the normalization of $W_{\bar{k}}$.
- A locally Noetherian scheme W over a field k is *geometrically regular* at a point w if, for every finite extension k'/k, the scheme $W_{k'}$ is regular at every point over w [GD67, Définition 6.7.6]. If $W_{\bar{k}}$ is locally Noetherian, for instance if W is locally of finite type over k, then W is geometrically regular if and only if $W_{\bar{k}}$ is regular, by [GD67, Proposition 5.13.7] and faithfully flat descent of regularity [Sta26, Tag 07NG]; cf. [GD67, Proposition 6.7.7].
- We write F_W^e for the e-th power of the absolute Frobenius morphism of an $\mathbb{F}_p$-scheme W.
- For a finite surjective morphism $g\colon W_1 \to W_2$ of normal Noetherian integral schemes and a Weil divisor D on W_2, the pullback g^*D is defined as the closure of the pullback of D restricted to the regular locus of W_2; the complement of the regular locus has codimension at least two, and so does its preimage. This pullback is compatible with composition, because this may be checked at points of codimension one, where all Weil divisors are Cartier.
- For a local ring $(R, \mathfrak{m}, \kappa)$, we write $\widehat{R}$ for its $\mathfrak{m}$-adic completion.
- For a point z of a scheme Z, we write $\mathcal{O}_{Z,z}^{\mathrm{sh}}$ for the *strict henselization* of $\mathcal{O}_{Z,z}$ with respect to a separable closure of $\kappa(z)$. The ring $\mathcal{O}_{Z,z}^{\mathrm{sh}}$ is a filtered colimit of étale $\mathcal{O}_{Z,z}$-algebras, and the map $\mathcal{O}_{Z,z} \to \mathcal{O}_{Z,z}^{\mathrm{sh}}$ is flat and local [Sta26, Tag 03QL].

2.2 Purely inseparable base change

We record the behavior of canonical divisors under base change; cf. [Tan18a] and [PW22, Theorems 1.1 and 1.2]. For our application, we need the case where the scheme X/k is not necessarily of finite type, which is due to [NT24]. For simplicity, we only consider the case where X is geometrically integral.

Lemma 2.1. *Let A be an excellent Gorenstein Noetherian k-algebra of finite Krull dimension. Let $X \to \operatorname{Spec} A$ be a flat Gorenstein morphism of finite type, with X normal and geometrically integral over k. For a finite field extension k'/k, put $A' = A \otimes_k k'$ and $X_{A'} = X \times_A \operatorname{Spec} A'$. Let $\nu\colon X' \to X_{A'}$ be the normalization of the integral scheme $X_{A'} = X_{k'}$. Then the divisorial conductor of ν is $(p-1)C_{k'}$ for an effective integral Weil divisor $C_{k'}$, and*

$$\omega_{X'/A'} \cong \mathcal{O}_{X'}\big(-(p-1)C_{k'}\big) \otimes \nu^*\omega_{X_{A'}/A'}, \tag{2.1}$$

where $\omega_{X'/A'}$ denotes the relative canonical sheaf, a rank-one reflexive sheaf, and $C_{k'} = 0$ if and only if $X_{A'}$ is normal.

Proof. The normalization ν is finite since A is excellent. The ring A' is Gorenstein by base change, and $X_{A'}$ is Gorenstein because $X_{A'} \to \operatorname{Spec} A'$ is Gorenstein [Sta26, Tags 0C03 and 0BJL]. Since X is of finite type over the excellent ring A, it is Noetherian and Japanese, so [NT24, Corollary 5.10] shows that the divisorial conductor of ν is $(p-1)C_{k'}$ for an effective integral Weil divisor $C_{k'}$. Since $\omega_{X_{A'}/A'}$ is invertible, finite duality gives (2.1) in codimension one, and both sides are reflexive on X', so the isomorphism extends globally. Finally, $C_{k'} = 0$ precisely when $X_{A'}$ is normal in codimension one; since $X_{A'}$ is Gorenstein and hence S_2, this is equivalent to normality by Serre's criterion. □

For extensions of height one, the vanishing of this conductor divisor admits the following local criterion, which can be seen as an algebraic version of Ji–Waldron's description of normalized base changes by foliations [JW21, Sections 3 and 4].

Lemma 2.2. *Let A be a normal k-domain, and let k'/k be finite with $k'^p \subseteq k$. Suppose that $B := A \otimes_k k'$ is a domain, with normalization $\widetilde{B}$. For $D \in \mathrm{Der}_k(k')$, write $D_B = \mathrm{id}_A \otimes D$ and use the same notation for its extension to $\mathrm{Frac}(B)$. Then B is normal if and only if D_B preserves $\widetilde{B}$ for every $D \in \mathrm{Der}_k(k')$.*

Proof. The forward implication is immediate. For the converse, put $L = \mathrm{Frac}(A)$ and $K = L \otimes_k k' = \mathrm{Frac}(B)$. Choose a p-basis $\gamma_1, \dots, \gamma_r$ of k'/k and let δ_i be the L-derivation of K with $\delta_i(\gamma_j) = \delta_{ij}$. The commuting derivations δ_i satisfy $\delta_i^p = 0$ and preserve $\widetilde{B}$ by hypothesis. They therefore define an action of $G = \alpha_p^r$ on $\operatorname{Spec} \widetilde{B}$ [ST23, Section 4]. Writing $t_1, \dots, t_r$ for the coordinates on G, its coaction on k' sends γ_i to $\gamma_i + t_i$, so the inclusion $k' \hookrightarrow \widetilde{B}$ is equivariant. Since $\operatorname{Spec} k' \to \operatorname{Spec} k$ is a G-torsor, faithfully flat descent for algebras gives

$$\widetilde{B} \cong (\widetilde{B})^G \otimes_k k'.$$

The common kernel of the δ_i in K is L, so $(\widetilde{B})^G = \widetilde{B} \cap L$. Every element of this intersection is integral over A, since B is integral over A. Normality of A therefore gives $(\widetilde{B})^G = A$, and the displayed isomorphism is the natural map $B \to \widetilde{B}$. Hence B is normal. □

2.3 Killing a cohomology class by a generic α_p-torsor

Lemma 2.3. *Let X/k be a regular projective geometrically integral variety with $H^1(X, \mathcal{O}_X) \neq 0$. Suppose that $H^1((X_{k^{1/p^e}})^\nu, \mathcal{O}) = 0$ for some $e \geq 1$. Then over $K = k(a)$, with a transcendental over k, there is an α_p-torsor $\pi\colon V \to X_K$, where V is a regular projective geometrically non-reduced variety with $H^0(V, \mathcal{O}_V) = K$ and $\omega_V \cong \pi^* \omega_{X_K}$.*

Proof. Put $\ell = k^{1/p^e}$ and $T = (X_\ell)^\nu$. Then $H^0(T, \mathcal{O}_T) = \ell$. The absolute Frobenius F_X^e factors through $T \to X$. Since $H^1(T, \mathcal{O}_T) = 0$, we have $(F_X^*)^e = 0$ on $H^1(X, \mathcal{O}_X)$. Choose $\eta \neq 0$ in $H^1(X, \mathcal{O}_X)$ and let $r \geq 1$ be minimal with $(F_X^*)^r \eta = 0$. Then $\xi := (F_X^*)^{r-1}\eta$ is nonzero and satisfies $F_X^*(\xi) = 0$. Choose a finite affine open cover $\{U_i = \operatorname{Spec} A_i\}$ and a Čech cocycle (g_{ij}) representing ξ. Since $F_X^*(\xi) = 0$, the cocycle (g_{ij}^p) is a coboundary, so there are $f_i \in A_i$ with $g_{ij}^p = f_i - f_j$. Write $\xi_K = \xi \otimes 1 \in H^1(X_K, \mathcal{O}_{X_K})$.

Glue

$$V_i = \operatorname{Spec}\big((A_i \otimes_k K)[t_i]/(t_i^p - f_i - a)\big)$$

by $t_i = t_j + g_{ij}$. These transitions preserve the equations because $g_{ij}^p = f_i - f_j$, and they satisfy the cocycle condition. Translations $t_i \mapsto t_i + \varepsilon$ define an α_p-torsor $\pi\colon V \to X_K$. Since $g_{ij} = t_i - t_j$ on V, we have $\pi^*\xi_K = 0$ in $H^1(V, \mathcal{O}_V)$. Each V_i is regular, since its coordinate ring is a localization of the regular ring

$$A_i[a, t_i]/(t_i^p - f_i - a) \cong A_i[t_i].$$

Hence V is regular. Since π is a universal homeomorphism and X_K is integral, V is integral. The morphism π is finite, so V is projective over K.

To show that $H^0(V, \mathcal{O}_V) = K$, let $s \in H^0(V, \mathcal{O}_V)$ be nonzero, and represent $s|_{V_i}$ by a polynomial $s_i(t_i) \in (A_i \otimes_k K)[t_i]$ of degree $< p$. The transitions preserve degree, so let m be the common degree of the s_i. Their leading coefficients glue to an element $b \in H^0(X_K, \mathcal{O}_{X_K}) = K$. If $m > 0$, write their two leading terms as $bt_i^m + c_i t_i^{m-1}$, where $b \in K^\times$ and $c_i \in A_i \otimes_k K$. Comparing coefficients of t_j^{m-1} in $s_i(t_j + g_{ij}) = s_j(t_j)$ gives $c_j - c_i = mbg_{ij}$, so $mb\xi_K = 0$ in $H^1(X_K, \mathcal{O}_{X_K})$. This contradicts $\xi_K \neq 0$. Thus the s_i have degree zero and glue to an element of K, so $H^0(V, \mathcal{O}_V) = K$.

Let W_i be the inverse image of U_i under $T \to X$. Since $H^1(T, \mathcal{O}_T) = 0$, the pulled-back cocycle has the form $g_{ij} = h_i - h_j$ with $h_i \in H^0(W_i, \mathcal{O}_T)$. The functions $f_i - h_i^p$ glue to a constant $\lambda \in H^0(T, \mathcal{O}_T) = \ell$. Put $L = \ell(a)$ and $T_L = T \times_\ell L$. Consider the Cartesian square

$$\begin{array}{ccc} V \times_{X_K} T_L & \longrightarrow & V \\ \downarrow & & \downarrow \pi \\ T_L & \longrightarrow & X_K. \end{array}$$

The coordinates $z_i = t_i - h_i$ give an isomorphism

$$V \times_{X_K} T_L \cong T_L \times_L \operatorname{Spec} L[z]/(z^p - a - \lambda).$$

Put $L' = L((a+\lambda)^{1/p})$ and $T_{L'} = T \times_\ell L'$. Then the projection $V \times_{X_K} T_{L'} \to T_{L'}$ is a trivial α_p-torsor. Since $T_{L'} \to X_{L'}$ is birational, $V_{L'}$ is generically nonreduced. Hence V is not geometrically reduced over K.

Put $B_i = H^0(V_i, \mathcal{O}_{V_i})$. Let $\varphi_i \colon B_i \to A_i \otimes_k K$ be the coefficient functional $\sum_{r<p} b_r t_i^r \mapsto b_{p-1}$. For $0 \le r < p$, the multiple $t_i^{p-1-r}\varphi_i$ extracts the coefficient of t_i^r. These multiples form the dual basis to $1, t_i, \dots, t_i^{p-1}$, so φ_i generates $\operatorname{Hom}_{A_i \otimes_k K}(B_i, A_i \otimes_k K)$ as a B_i-module. The translations $t_i = t_j + g_{ij}$ preserve the coefficient of degree $p-1$, so these generators glue to trivialize ω_{V/X_K}. Finite duality then gives $\omega_V \cong \pi^* \omega_{X_K}$. □

2.4 Tangent cones

Let X be a variety over k, let $x \in X$ be a closed point at which X is regular, with $\kappa(x)$ purely inseparable over k, and let $\bar{x} \in X_{\bar{k}}$ be the geometric point over x. Let $\mathfrak{m}_{\bar{x}}$ be the maximal ideal of $\mathcal{O}_{X_{\bar{k}},\bar{x}}$, and let $T = (\mathfrak{m}_{\bar{x}}/\mathfrak{m}_{\bar{x}}^2)^\vee$ be the Zariski tangent space of $X_{\bar{k}}$ at $\bar{x}$. Write $C = \operatorname{Spec}\bigl(\bigoplus_{j\ge 0} \mathfrak{m}_{\bar{x}}^j/\mathfrak{m}_{\bar{x}}^{j+1}\bigr) \subseteq T$ for the tangent cone and I_C for its homogeneous ideal. The tangent cone C has the same dimension as X, and $C = T$ if $\bar{x}$ is regular. If $\bar{x}$ is a singular point, let e be the lowest degree of a nonzero form in I_C. We define the space of *vertical directions* to be

$$L = \{v \in T : v(f) = 0 \text{ for all } f \in \mathfrak{m}_x\},$$

where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$, mapped to $\mathfrak{m}_{\bar{x}}/\mathfrak{m}_{\bar{x}}^2$ by the projection $\pi \colon X_{\bar{k}} \to X$. Equivalently, L is the Zariski tangent space at $\bar{x}$ of the one-point fibre $\pi^{-1}(x) = \operatorname{Spec}(\kappa(x) \otimes_k \bar{k})$.

Lemma 2.4. *In this situation, the following hold.*

(1) $\dim L = \dim_{\kappa(x)} \Omega_{\kappa(x)/k}$ *and* $\dim T \le \dim X + \dim L$*; in particular,* $\dim T \le \dim X + 1$ *if* $\kappa(x)/k$ *is simple.*

(2) *If* $y_1, \dots, y_N$ *is a basis of* $\mathfrak{m}_{\bar{x}}/\mathfrak{m}_{\bar{x}}^2$ *such that* $y_1, \dots, y_u$ *span the image of* $\mathfrak{m}_x$*, so that* $L = \{v \in T : v(y_1) = \cdots = v(y_u) = 0\}$*, then* I_C *is stable under* $\partial/\partial y_j$ *for* $j > u$*.*

(3) *Suppose that* $e < p$*, and let* $y_1, \dots, y_N$ *be as in* (2)*. Then every form of degree* e *in* I_C *is a polynomial in* $y_1, \dots, y_u$ *alone, so it does not involve the coordinates along* L*. In particular, if* I_C *is generated by one form, then* C *is the preimage of a cone in* T/L *under the projection* $T \to T/L$*.*

Proof. Put $\kappa = \kappa(x)$. For (1), the description of L as the tangent space of the fibre gives $L^\vee \cong \Omega_{\kappa/k} \otimes_\kappa \bar{k}$. Since $\mathfrak{m}_x$ is generated by $\dim X$ elements and L is the annihilator of its image in the cotangent space, $\dim T - \dim L \le \dim X$.

For (2), put $A = \mathcal{O}_{X,x}$ and $B = A \otimes_k \kappa$. The ring B is finite over A, and $B/\mathfrak{m}_x B = \kappa \otimes_k \kappa$ is local with residue field κ because κ/k is purely inseparable. Hence B is local with residue field κ, and it is the local ring of X_κ at the unique point over x, which is κ-rational. Let $\mathfrak{m}$ be the maximal ideal of B, let $\Sigma \subseteq \mathfrak{m}/\mathfrak{m}^2$ be the κ-span of the image of $\mathfrak{m}_x$, and write the associated graded ring $\operatorname{gr}_\mathfrak{m} B$ as S/I^* with $S = \operatorname{Sym}_\kappa(\mathfrak{m}/\mathfrak{m}^2)$, where I^* is the ideal of initial forms. The maximal ideal of $\mathcal{O}_{X_{\bar{k}},\bar{x}}$ is generated by $\mathfrak{m}$, and $\operatorname{gr}_{\mathfrak{m}_{\bar{x}}} \mathcal{O}_{X_{\bar{k}},\bar{x}} = \operatorname{gr}_\mathfrak{m} B \otimes_\kappa \bar{k}$. Hence $T = (\mathfrak{m}/\mathfrak{m}^2)^\vee \otimes_\kappa \bar{k}$, $I_C = I^* \otimes_\kappa \bar{k}$, and L is the annihilator of Σ.

For $b \in \kappa$ with a lift $\tilde{b} \in A$, put $w_b = \tilde{b} \otimes 1 - 1 \otimes b$, which lies in $\mathfrak{m}$. Let $\delta \in \operatorname{Der}_k(\kappa)$. Then $D = \mathrm{id}_A \otimes \delta$ is a derivation of B which vanishes on $\mathfrak{m}_x$ and satisfies $D(w_b) = -1 \otimes \delta(b)$, so $D(\mathfrak{m}^j) \subseteq \mathfrak{m}^{j-1}$ for $j \ge 1$. The induced maps $\mathfrak{m}^j/\mathfrak{m}^{j+1} \to \mathfrak{m}^{j-1}/\mathfrak{m}^j$ form a derivation of degree -1 of S/I^*, which is κ-linear because $f\,D(1 \otimes c) \in \mathfrak{m}^j$ for $f \in \mathfrak{m}^j$ and $c \in \kappa$. Hence it is induced by the κ-linear derivation ∂_λ of S with $\partial_\lambda(y) = \lambda(y)$ for $y \in \mathfrak{m}/\mathfrak{m}^2$, where λ is the functional $f \mapsto D(f) \bmod \mathfrak{m}$ on $\mathfrak{m}/\mathfrak{m}^2$, and therefore $\partial_\lambda(I^*) \subseteq I^*$. The functional λ vanishes on Σ, and $\lambda(w_b) = -\delta(b)$.

The cotangent-space identification for the fibre $\operatorname{Spec}(\kappa \otimes_k \kappa)$ gives an isomorphism $\Omega_{\kappa/k} \cong (\mathfrak{m}/\mathfrak{m}^2)/\Sigma$, sending db to the class of w_b. Under this identification, λ corresponds to $-\delta$, so these functionals exhaust the dual of $(\mathfrak{m}/\mathfrak{m}^2)/\Sigma$ as δ varies. Thus I^* is stable under ∂_λ for every functional λ on $\mathfrak{m}/\mathfrak{m}^2$ vanishing on Σ.

Choose a basis $y_1, \dots, y_N$ of $\mathfrak{m}/\mathfrak{m}^2$ such that $y_1, \dots, y_u$ span Σ. Then $L = \{v \in T : v(y_1) = \cdots = v(y_u) = 0\}$, and I^* is stable under $\partial/\partial y_j$ for $j > u$. The derivations $\partial/\partial y_j$ with $j > u$ are the derivatives along a basis of L, so this condition does not depend on the choice of coordinates.

For (3), let $Q \in I^*$ be homogeneous of degree $e < p$. For $j > u$ we have $\partial Q/\partial y_j \in I^*_{e-1} = 0$, and every exponent occurring in Q is at most $e < p$, so Q is a polynomial in $y_1, \dots, y_u$. The forms of degree e in I_C are the $\overline{k}$-linear combinations of such Q, and $\overline{k}[y_1, \dots, y_u]$ is the coordinate ring of T/L for every choice of coordinates as in (2), which proves the first assertion of (3). If I_C is generated by one form, then this form has degree e, so C is the preimage of the cone in T/L defined by the same form. □

2.5 Birational contractions

Definition 2.5. Let X be a regular integral scheme of dimension two, proper and flat over S, where S is either the spectrum of a field or a Dedekind scheme of dimension one. A *vertical curve* on X is an integral closed subscheme $C \subset X$ of dimension one contained in a fibre X_s over a closed point $s \in S$. We call such a curve C a (-1)-*curve* if

$$(K_{X/S} \cdot C)_{k_C} = (C^2)_{k_C} = -1,$$

where $K_{X/S}$ is a relative canonical divisor and $k_C = H^0(C, \mathcal{O}_C)$. For such a curve C, adjunction gives $H^1(C, \mathcal{O}_C) = 0$. By Castelnuovo's criterion [Liu02, Theorem 9.3.8 and Proposition 9.3.10], these are exactly the vertical curves that can be contracted to a regular point.

We record standard facts about contracting negative curves, including the residue-field details needed below.

Lemma 2.6. *Let k be a separably closed field of characteristic $p > 3$, and let X be a regular projective surface over k with $K_X \equiv 0$. Let $C \subset X$ be a negative curve, and put $\ell = H^0(C, \mathcal{O}_C)$.*

(1) *$C \cong \mathbb{P}^1_\ell$, $(C^2)_\ell = -2$, and $\mathcal{O}_C(C) \cong \mathcal{O}_{\mathbb{P}^1_\ell}(-2)$.*

(2) *There is a birational morphism $f\colon X \to X'$ onto a normal projective surface which contracts exactly C, to a point P with $\kappa(P) = \ell$. Moreover $R^1 f_* \mathcal{O}_X = 0$ and $\rho(X') = \rho(X) - 1$.*

(3) *The surface X' is Gorenstein, and $f^*\omega_{X'} \cong \omega_X$.*

(4) *$2D$ is Cartier for every Weil divisor D on X'.*

(5) *There is an isomorphism of rings $\widehat{\mathcal{O}}_{X',P} \cong \ell[[u, v, w]]/(uv - w^2)$; in particular X' is a local complete intersection over k. (This isomorphism need not be an isomorphism of k-algebras.)*

Proof. (1) The curve C is a Cartier divisor on the regular surface X, hence Gorenstein. Put $g = \dim_\ell H^1(C, \mathcal{O}_C)$. Adjunction gives

$$(C^2)_\ell = ((K_X + C) \cdot C)_\ell = 2(g - 1) < 0,$$

so $g = 0$ and $(C^2)_\ell = -2$. By [Kol13, Lemma 10.6(3)], C is a conic or $\mathbb{P}^1$ over ℓ. Since ℓ is separably closed and $p \neq 2$, an integral conic over ℓ is smooth and split, so $C \cong \mathbb{P}^1_\ell$. The normal bundle has degree $(C^2)_\ell = -2$, hence is $\mathcal{O}_{\mathbb{P}^1_\ell}(-2)$, proving (1).

(2) The negative curve C spans a $(K_X + C)$-negative extremal ray. Applying [Tan18b, Theorem 4.4] with boundary C gives a projective birational morphism $f\colon X \to X'$ contracting exactly C, with X' normal and projective and $\rho(X') = \rho(X) - 1$. Since $K_X \cdot C = 0$ and $-C$ is f-nef, [Kol13, Theorem 10.4] gives $R^1 f_* \mathcal{O}_X(-jC) = 0$ for every $j \geq 0$. Thus P is a rational singularity, and [Kol13, Proposition 10.9(4)] gives $\kappa(P) = \ell$.

(3) The numerical discrepancy of C is zero because $K_X \cdot C = 0$. Hence X' is Gorenstein and f is crepant by [Kol13, Theorem 2.29(2)].

(4) The intersection matrix over ℓ is (-2), so [Kol13, Proposition 10.9(3)] shows that $2D$ is Cartier at P for every Weil divisor D on X'. Since X' is regular away from P, the divisor $2D$ is Cartier everywhere.

(5) Put $R = \mathcal{O}_{X',P}$ and $I_j = (f_* \mathcal{O}_X(-jC))_P$. Pushing forward $0 \to \mathcal{O}_X(-C) \to \mathcal{O}_X \to \mathcal{O}_C \to 0$ and taking the stalk at P gives $I_1 = \ker(R \to H^0(C, \mathcal{O}_C) = \kappa(P)) = \mathfrak{m}_P$. Similarly, since $R^1 f_* \mathcal{O}_X(-jC) = 0$ for all $j \geq 0$, we have $I_j/I_{j+1} \cong H^0(\mathbb{P}^1_\ell, \mathcal{O}(2j))$, compatibly with multiplication. The multiplication map $H^0(\mathbb{P}^1_\ell, \mathcal{O}(2))^{\otimes \ell j} \to H^0(\mathbb{P}^1_\ell, \mathcal{O}(2j))$ is surjective, since every monomial of degree $2j$ is a product of j quadratic monomials. Hence every class in I_j/I_{j+1} is a sum of products of j classes from I_1/I_2, so $I_j = I_1^j + I_{j+1} =$

$\mathfrak{m}_P^j + I_{j+1}$. Iterating this equality gives $I_j = \mathfrak{m}_P^j + I_n$ for all $n \geq j$. By formal functions, $\widehat{R} \cong \varprojlim_n R/I_n$; taking limits gives $I_j\widehat{R} = \mathfrak{m}_P^j\widehat{R}$ and hence

$$\operatorname{gr}_{\mathfrak{m}_P} \widehat{R} \cong \bigoplus_{j\geq 0} H^0(\mathbb{P}^1_\ell, \mathcal{O}(2j)) \cong \ell[u,v,w]/(uv-w^2).$$

The displayed graded ring gives $\operatorname{length}_R(R/\mathfrak{m}_P^n) = \sum_{j=0}^{n-1}(2j+1) = n^2$ for $n \geq 1$. After choosing a coefficient field identified with ℓ, [Lip69, Lemma 23.4] therefore gives a quadratic hypersurface presentation $\widehat{R} \cong \ell[[u,v,w]]/(g)$. Comparing associated graded rings, we may choose the variables u, v, w and rescale g so that $g - (uv - w^2) \in (u,v,w)^3$. Since $p \neq 2$, the quadratic form $uv - w^2$ is nondegenerate. The formal Morse lemma [Kol23, Lemma 10.43], applied with coefficient ring ℓ and maximal ideal (0), therefore gives $\widehat{R} \cong \ell[[u,v,w]]/(uv-w^2)$ as rings. The hypersurface presentation and regularity of X' away from P imply that X' is a local complete intersection over k [Sta26, Tag 09Q6]. □

2.6 Even lattices

An *integral lattice* is a finitely generated free abelian group N equipped with a nondegenerate symmetric bilinear pairing $(\ ,\ ) \colon N \times N \to \mathbb{Z}$. It is *even* if $(v,v) \in 2\mathbb{Z}$ for every $v \in N$. For a $\mathbb{Z}$-basis $e_1, \ldots, e_r$ of N, the matrix $G = ((e_i, e_j))_{1\leq i,j\leq r}$ is called a *Gram matrix* of N. Its determinant $\det N := \det G$, called the *Gram determinant*, is independent of the chosen $\mathbb{Z}$-basis. The lattice N is *unimodular* if $|\det N| = 1$, equivalently if the pairing induces an isomorphism $N \cong \operatorname{Hom}_{\mathbb{Z}}(N, \mathbb{Z})$. For a lattice over $\mathbb{Z}_\ell$, unimodularity means that its Gram determinant is a unit in $\mathbb{Z}_\ell$. The pairing induces a quadratic form $q(v) = (v,v)$ on $N \otimes \mathbb{Q}$. An nonzero element $v \in N \otimes \mathbb{Q}$ is called *anisotropic* if $q(v) \neq 0$, and *isotropic* if $q(v) = 0$. The quadratic form q (or the corresponding pairing) is called *anisotropic* if it has no isotropic elements, and *isotropic* otherwise.

Lemma 2.7. *Let p be an odd prime and let N be an even integral lattice of signature $(1, r-1)$ with $|\det N| = p^a$ for an integer $a \geq 0$. Then r is even, and $N \otimes \mathbb{Q}$ is anisotropic if and only if $r = 2$ and a is odd. In this case $p \equiv 1 \pmod 4$.*

Proof. Since N is even of odd determinant, the induced bilinear form on $N/2N$ with values in $\mathbb{F}_2$ is nondegenerate and alternating, so r is even. In particularly, $r \geq 2$. Then the quadratic form on $N \otimes \mathbb{Q}$ is indefinite by the signature. If $r \geq 5$, then this implies that the form is isotropic by Meyer's theorem [Ser73, Chapter IV, Section 3.2, Corollary 2 to Theorem 8].

Suppose that $r = 4$. By the Hasse–Minkowski theorem [Ser73, Chapter IV, Section 3.2, Theorem 8], to show that $N \otimes \mathbb{Q}$ is isotropic, it suffices to show that $N \otimes \mathbb{R}$ is isotropic and that $N \otimes \mathbb{Q}_\ell$ is isotropic for every prime ℓ.

At $\ell = 2$, $\det N$ is a unit in $\mathbb{Z}_2$ because $|\det N| = p^a$ and p is odd, so $N \otimes \mathbb{Z}_2$ is *unimodular*. Since it is also even, it is an orthogonal sum of two binary lattices with Gram matrices [Cas62, Hilfssatz 5]

$$B_1 = \begin{pmatrix} 2a_1 & 1 \\ 1 & 2b_1 \end{pmatrix}, \qquad B_2 = \begin{pmatrix} 2a_2 & 1 \\ 1 & 2b_2 \end{pmatrix},$$

where $a_1, b_1, a_2, b_2 \in \mathbb{Z}_2$. Since $\det B_1 \equiv \det B_2 \equiv -1 \pmod 4$ and squares of 2-adic units are 1 modulo 4, we have $\det N \equiv 1 \pmod 4$. Since $\det N = -p^a$, we get $p \equiv -1 \pmod 4$ and that a is odd. In these coordinates, the quadratic form satisfies $q = 2Q$, where

$$Q(x_1, x_2, x_3, x_4) = a_1x_1^2 + x_1x_2 + b_1x_2^2 + a_2x_3^2 + x_3x_4 + b_2x_4^2.$$

The polynomial Q has coefficients in $\mathbb{Z}_2$ and has the same zeros as q over $\mathbb{Q}_2$. The values of Q at $(1,0,0,0)$, $(0,0,1,0)$ and $(1,0,1,0)$ are a_1, a_2 and $a_1 + a_2$, respectively. So modulo 2, the form Q has a zero at one of these points. The gradient of Q modulo 2 is (x_2, x_1, x_4, x_3), so this zero is nonsingular and lifts by Hensel's lemma to a nonzero isotropic vector over $\mathbb{Q}_2$.

At the real place, $N \otimes \mathbb{R}$ has signature $(1,3)$ and is therefore isotropic. At an odd prime $\ell \neq p$, the determinant is a unit in $\mathbb{Z}_\ell$, so the reduction of the lattice is a nondegenerate quaternary form over $\mathbb{F}_\ell$. Since ℓ is odd, [Ser73, Chapter IV, Section 1.4, Theorem 1] gives coordinates in which this reduction is $\bar{q} = \sum_{i=1}^4 c_i x_i^2$ with $c_i \in \mathbb{F}_\ell^\times$. The sets $\{c_1x^2 : x \in \mathbb{F}_\ell\}$ and $\{-c_3 - c_2y^2 : y \in \mathbb{F}_\ell\}$ each have $(\ell+1)/2$

elements, so they intersect. Thus $\bar{q}(x,y,1,0)=0$ for some $x,y\in\mathbb{F}_\ell$, and the partial derivative with respect to x_3 at this zero is $2c_3\neq 0$. Hensel's lemma therefore lifts this zero to a nonzero isotropic vector over $\mathbb{Q}_\ell$. At $\ell=p$, an anisotropic quaternary quadratic form has square discriminant [Ser73, Chapter IV, Section 2.2, Theorem 6(iii)], whereas $\det N$ has p-adic valuation a, which is odd, a contradiction.

Suppose $r=2$. Choose an orthogonal basis of $N\otimes\mathbb{Q}$ and write $q(x,y)=ux^2+vy^2$ with $u,v\in\mathbb{Q}^\times$. A nonzero zero must have $y\neq 0$, and $q(x,y)=0$ is then equivalent to $(x/y)^2=-v/u$. Thus $N\otimes\mathbb{Q}$ is isotropic if and only if $-v/u$ is a rational square, or equivalently if $-uv$ is a rational square. A change of basis multiplies the Gram determinant by a rational square, so uv and $\det N$ have the same square class, and the criterion is equivalent to $-\det N$ being a rational square. The signature $(1,1)$ gives $\det N<0$, so $-\det N=p^a$ is a rational square if and only if a is even. Therefore, under the assumption that $N\otimes\mathbb{Q}$ is anisotropic, a is odd. Since N is even, its Gram matrix in an integral basis has the form $\begin{pmatrix}2b & c\\ c & 2d\end{pmatrix}$ with $b,c,d\in\mathbb{Z}$. Its determinant $4bd-c^2$ is odd, so c is odd and $\det N\equiv -1 \pmod 4$. Since $\det N=-p^a$, this gives $p^a\equiv 1 \pmod 4$. As a is odd, $p^a\equiv p \pmod 4$, hence $p\equiv 1 \pmod 4$. □

2.7 Genus-one fibrations

Lemma 2.8. *Let X be a regular projective surface over k with $H^0(X,\mathcal{O}_X)=k$, $K_X\equiv 0$ and $\chi(\mathcal{O}_X)=2$. If the intersection form on $N^1(X)_\mathbb{Q}$ is isotropic, then there is a fibration $f\colon X\to B$ onto a regular projective curve.*

Proof. Choose an ample divisor A and a divisor D with $D^2=0$ and $A\cdot D>0$. Riemann–Roch gives $h^0(D)+h^0(K_X-D)\geq\chi(\mathcal{O}_X(D))=\chi(\mathcal{O}_X)=2$. Hence $h^0(D)\geq 2$.

Suppose that D is not nef, and let C be a curve with $D\cdot C<0$. Then C is a component of every member of $|D|$, so $C^2<0$. As in the proof of Lemma 2.6(1), with $h=[H^0(C,\mathcal{O}_C):k]$ we have $(C^2)_k=-2h$, and $(D\cdot C)_k\in h\mathbb{Z}$. Hence $r_C(D):=D+\frac{(D\cdot C)_k}{h}C$ is a $\mathbb{Z}$-divisor with $(r_C(D)^2)_k=0$. The reflection r_C is an isometry of $N^1(X)_\mathbb{Q}$ preserving the component of the positive cone containing A; hence $A\cdot r_C(D)>0$, while $A\cdot r_C(D)<A\cdot D$. By Riemann–Roch again, $h^0(r_C(D))\geq 2$. Since the positive integer $A\cdot D$ decreases, after finitely many reflections we obtain a nef divisor D with $D^2=0$ and $h^0(D)\geq 2$.

It follows that $\kappa(X,D)=1$, so the nef divisor D is semiample (cf. [Tan20, Lemma 2.7]). The Stein factorization of the morphism defined by a sufficiently divisible multiple of D gives a fibration $f\colon X\to B$ onto a regular projective curve. □

3 Local fundamental groups and exceptional lattices

In this section all punctured local schemes are formed after strict henselization.

For a closed point z of a normal surface Z, write $U_z^\circ=\operatorname{Spec}\mathcal{O}^{\mathrm{sh}}_{Z,z}\setminus\{z\}$ and let $\pi_1^{\mathrm{et}}(U_z^\circ)$ denote its étale fundamental group. We recall the following standard comparison lemma.

Lemma 3.1. *Let k be a separably closed field, let X be a normal surface over k, and let $\nu\colon S\to(X_{\bar{k}})_{\mathrm{red}}$ be the normalization, with induced morphism $\phi\colon S\to X$. Let $s\in S$ be a closed point with image $x\in X$. Then ϕ is a universal homeomorphism and induces a universal homeomorphism $U_s^\circ\to U_x^\circ$, hence an isomorphism $\pi_1^{\mathrm{et}}(U_s^\circ)\cong\pi_1^{\mathrm{et}}(U_x^\circ)$. If X is regular at x, then $\pi_1^{\mathrm{et}}(U_s^\circ)=1$.*

Proof. Since $\bar{k}/k$ is purely inseparable, ϕ is a universal homeomorphism by [Tan18a, Lemma 2.2(2)]. The scheme $S\times_X\operatorname{Spec}\mathcal{O}^{\mathrm{sh}}_{X,x}$ is strictly henselian and ind-étale over S, hence isomorphic to $\operatorname{Spec}\mathcal{O}^{\mathrm{sh}}_{S,s}$. Removing the closed points gives a universal homeomorphism $U_s^\circ\to U_x^\circ$, so topological invariance gives the asserted isomorphism [Sta26, Tag 04DY]. The regular-point case follows from [Sch08, Corollary 2.4]. □

We next construct a regular double cover of the singularity produced by the contraction in Lemma 2.6.

Lemma 3.2. *Let k be a separably closed field of characteristic $p>3$, let X be a regular projective surface over k with $K_X\equiv 0$, and let $f\colon X\to X'$ be the contraction of a negative curve C as in Lemma 2.6. Let S be the normalization of $(X'_{\bar{k}})_{\mathrm{red}}$ and let $s\in S$ lie over $P=f(C)$. Then $\pi_1^{\mathrm{et}}(U_P^\circ)\cong\pi_1^{\mathrm{et}}(U_s^\circ)\cong\mathbb{Z}/2$.*

Proof. Put $k_C = H^0(C, \mathcal{O}_C)$ and $A = \mathcal{O}^{\mathrm{sh}}_{X',P}$, a strictly henselian local ring with residue field k_C. Define T by the Cartesian square

$$\begin{array}{ccc} T & \longrightarrow & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ \operatorname{Spec} A & \longrightarrow & X'. \end{array}$$

The projection $T \to X$ is ind-étale, so T is regular [Sta26, Tags 07EP and 0H7S], with exceptional curve $C \cong \mathbb{P}^1_{k_C}$.

We first construct a double cover of T. Put $T_0 = T \times_A k_C$. By proper base change [AGV73, Exp. XII, Corollaire 5.5(iii)], $H^2_{\text{ét}}(T, \mu_2) \cong H^2_{\text{ét}}(T_0, \mu_2)$. Since $C = (T_0)_{\mathrm{red}}$, restriction induces $H^2_{\text{ét}}(T_0, \mu_2) \cong H^2_{\text{ét}}(C, \mu_2)$. After the purely inseparable extension $\overline{k_C}/k_C$, the standard cohomology calculation for $C_{\overline{k_C}} \cong \mathbb{P}^1_{\overline{k_C}}$ gives $H^2_{\text{ét}}(C, \mu_2) \cong \mathbb{Z}/2$ [Sta26, Tag 03RQ]. The Kummer sequence induces an exact sequence [Sta26, Tag 03PL]

$$\operatorname{Pic}(T) \xrightarrow{(-)^{\otimes 2}} \operatorname{Pic}(T) \xrightarrow{\delta_T} H^2_{\text{ét}}(T, \mu_2). \tag{3.1}$$

For a line bundle $\mathcal{L}$ on T, functoriality of the Kummer sequence shows that the image of $\delta_T(\mathcal{L})$ under $H^2_{\text{ét}}(T, \mu_2) \xrightarrow{\sim} H^2_{\text{ét}}(C, \mu_2) \cong \mathbb{Z}/2$ is $\deg_{k_C}(\mathcal{L}|_C) \bmod 2$. Since $\deg_{k_C}(\mathcal{O}_T(C)|_C) = -2$, exactness of (3.1) gives a line bundle M on T with $M^{\otimes 2} \cong \mathcal{O}_T(C)$. Let

$$\varpi \colon V = \mathbf{Spec}_T(\mathcal{O}_T \oplus M^{-1}) \to T$$

be the double cover obtained locally by adjoining a square root of a local equation $t = 0$ for C. At a closed point $x \in C$, regularity of T and C allows us to choose regular parameters t, u in $\mathcal{O}_{T,x}$. The local ring of V above x has dimension two and maximal ideal (z, u), since $t = z^2$, so it is regular. Over $T \setminus C$, the derivative $2z$ is invertible, so the cover is étale and V is regular there. The cover is integral because t has order one along C. The ramification curve $D = \varpi^{-1}(C)_{\mathrm{red}}$ satisfies $D \cong C$ and $2D = \varpi^* C$. Thus $\mathcal{O}_V(D)|_D \cong \varpi^* M|_D$ has degree -1 over k_C, giving $\mathcal{O}_D(-jD) \cong \mathcal{O}_{\mathbb{P}^1_{k_C}}(j)$ for $j \geq 0$.

We next show that $B = H^0(V, \mathcal{O}_V)$, the normalization of A in the function field of V, is regular. Since B is finite over the henselian ring A and is a domain, it is local and henselian. The closed fibre of $V \to \operatorname{Spec} B$ is supported on D. Since $-C$ is ample over $\operatorname{Spec} A$ and $2D = \varpi^* C$, the divisor $-D$ is ample over $\operatorname{Spec} B$. Relative Serre vanishing and descending induction in the exact sequences

$$0 \to \mathcal{O}_V(-(j+1)D) \to \mathcal{O}_V(-jD) \to \mathcal{O}_D(-jD) \to 0$$

give $H^1(V, \mathcal{O}_V(-jD)) = 0$ for all $j \geq 0$, since $H^1(D, \mathcal{O}_D(-jD)) = 0$. By taking cohomology of the above sequence with $j = 0$, the restriction $B \to H^0(D, \mathcal{O}_D) = k_C$ is surjective, so the residue field of B is k_C. By the same argument as in the proof of Lemma 2.6 (5), we have

$$\operatorname{gr}_{\mathfrak{m}_B} \widehat{B} \cong \bigoplus_{j \geq 0} H^0(\mathbb{P}^1_{k_C}, \mathcal{O}(j)) \cong k_C[u, v].$$

Hence B is a regular local ring of dimension two and is strictly henselian.

Finally, we compute the local fundamental groups. Over U°_P, the finite morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ is the étale double cover $V \setminus D \to T \setminus C$. Put $U^\circ_B = \operatorname{Spec} B \setminus \{\mathfrak{m}_B\}$. Then $U^\circ_B \to U^\circ_P$ is a connected étale double cover. By purity [Gro71, Exp. X, Corollaire 3.3], $\pi_1^{\mathrm{et}}(U^\circ_B) \cong \pi_1^{\mathrm{et}}(\operatorname{Spec} B) = 1$, since B is strictly henselian. It follows that $\pi_1^{\mathrm{et}}(U^\circ_P) \cong \mathbb{Z}/2$. Lemma 3.1 gives $\pi_1^{\mathrm{et}}(U^\circ_s) \cong \pi_1^{\mathrm{et}}(U^\circ_P)$. □

The next lemma relates the local fundamental group to the exceptional curves and their intersection matrix, without assuming that the singularity is rational. It is an étale analogue of Mumford's computation of the first homology of the link of a complex normal surface singularity [Mum61, Part I, pp. 10–11]. In particular, it restricts the possible intersection matrices in the regular-point case of Lemma 3.1 and in the setting of Lemma 3.2.

Lemma 3.3. *Let S be a normal surface over an algebraically closed field of characteristic p, let $s \in S$ be a closed point, let $A \supseteq \mathcal{O}^{\mathrm{sh}}_{S,s}$ be a normal local domain finite over $\mathcal{O}^{\mathrm{sh}}_{S,s}$, and let $g \colon T \to \operatorname{Spec} A$ be a resolution of singularities. Let $E = E_1 \cup \cdots \cup E_r$ be the reduced exceptional curve of g, let $M = (E_i \cdot E_j)_{\kappa(s)}$, and let $U^\circ_s = \operatorname{Spec} A \setminus \{\mathfrak{m}_A\}$. The following hold for every positive integer n prime to p.*

(1) *There is an exact sequence of étale cohomology groups*

$$0 \to H^1_{\mathrm{et}}(E,\mu_n) \to H^1_{\mathrm{et}}(U_s^\circ,\mu_n) \to \ker\bigl(M\otimes\mathbb{Z}/n\colon (\mathbb{Z}/n)^r \to (\mathbb{Z}/n)^r\bigr) \to 0. \tag{3.2}$$

An invertible sheaf $\mathcal{L}$ on T has an n-th root in $\mathrm{Pic}(T)$ if and only if $n \mid \deg(\mathcal{L}|_{E_i})$ for every i.

(2) *If $\pi_1^{\mathrm{et}}(U_s^\circ) = 1$, then $H^1_{\mathrm{et}}(E,\mu_n) = 0$, every E_i has geometric genus zero, and $|\det M|$ is a power of p.*

(3) *If $p \neq 2$, $\pi_1^{\mathrm{et}}(U_s^\circ) \cong \mathbb{Z}/2$ and $H^1_{\mathrm{et}}(E,\mu_4) = 0$, then $|\det M| = 2p^e$ for some $e \geq 0$. The vanishing assumption holds if E is a tree of smooth rational curves meeting transversally.*

Proof. For (1), the ring A is strictly henselian with residue field $\kappa(s)$. Hensel's lemma shows that every unit of A is an n-th power, so A^* is n-divisible. Normality gives $H^0(T,\mathcal{O}_T) = H^0(U_s^\circ,\mathcal{O}_{U_s^\circ}) = A$. As in the proof of Lemma 3.2, we have $H^i_{\mathrm{et}}(T,\mu_n) \cong H^i_{\mathrm{et}}(E,\mu_n)$ for $i = 1,2$. Let $\pi\colon \widetilde{E} \to E$ be the normalization, with components $\widetilde{E}_i$ over the E_i. Since π is finite and the cokernel of $\mu_n \to \pi_*\mu_n$ is supported at finitely many points, the normalization sequence gives a surjection $H^1_{\mathrm{et}}(E,\mu_n) \to H^1_{\mathrm{et}}(\widetilde{E},\mu_n)$ and an isomorphism $H^2_{\mathrm{et}}(E,\mu_n) \cong H^2_{\mathrm{et}}(\widetilde{E},\mu_n) \cong (\mathbb{Z}/n)^r$ [Sta26, Tag 03RQ]. For $Y = T$ or $Y = U_s^\circ$, we have $H^0(Y,\mathcal{O}_Y)^* = A^*$, so the Kummer sequence gives

$$0 \to A^*/(A^*)^n \to H^1_{\mathrm{et}}(Y,\mu_n) \to \mathrm{Pic}(Y)[n] \to 0.$$

Since A^* is n-divisible, the quotient $A^*/(A^*)^n$ vanishes, and we obtain

$$\begin{aligned} \mathrm{Pic}(T)[n] &\cong H^1_{\mathrm{et}}(T,\mu_n) \cong H^1_{\mathrm{et}}(E,\mu_n), \\ \mathrm{Pic}(U_s^\circ)[n] &\cong H^1_{\mathrm{et}}(U_s^\circ,\mu_n). \end{aligned} \tag{3.3}$$

The Kummer boundary map $\mathrm{Pic}(T) \to H^2_{\mathrm{et}}(T,\mu_n)$ has kernel $n\,\mathrm{Pic}(T)$, so it induces an injection $\mathrm{Pic}(T)/n \hookrightarrow H^2_{\mathrm{et}}(T,\mu_n) \cong (\mathbb{Z}/n)^r$. Since the Kummer boundary map commutes with pullback, the image of $[\mathcal{L}]$ has i-th coordinate $\deg(\mathcal{L}|_{E_i}) \bmod n$. Thus $\mathcal{L}$ has an n-th root if and only if all these coordinates vanish. This proves the assertion about n-th roots of line bundles.

Every line bundle on U_s° is represented by a divisor whose closure is Cartier on T, since T is regular. Thus the restriction map $\mathrm{res}\colon \mathrm{Pic}(T) \to \mathrm{Pic}(U_s^\circ)$ is surjective. For $[\mathcal{L}] \in \ker(\mathrm{res})$, a trivialization of $\mathcal{L}|_{U_s^\circ}$ defines a rational section of $\mathcal{L}$ whose divisor is supported on E. Thus $\ker(\mathrm{res})$ consists of the classes $\mathcal{O}_T(\sum_i a_iE_i)$ with $a_1,\dots,a_r \in \mathbb{Z}$. If $\mathcal{O}_T(\sum_i a_iE_i)$ is trivial, then the negative definiteness of M forces all $a_i = 0$. We therefore have an exact sequence

$$0 \to \mathbb{Z}^r \to \mathrm{Pic}(T) \xrightarrow{\mathrm{res}} \mathrm{Pic}(U_s^\circ) \to 0,$$

whose first map sends $(a_i)_i$ to $\mathcal{O}_T(\sum_i a_iE_i)$. The long exact sequence for $\mathrm{Tor}_i^{\mathbb{Z}}(-,\mathbb{Z}/n)$ gives $0 \to \mathrm{Pic}(T)[n] \to \mathrm{Pic}(U_s^\circ)[n] \to \ker\bigl((\mathbb{Z}/n)^r \to \mathrm{Pic}(T)/n\bigr) \to 0$. The map $(\mathbb{Z}/n)^r \to \mathrm{Pic}(T)/n$ sends the reduction of $(a_i)_i \in \mathbb{Z}^r$ to $[\mathcal{O}_T(\sum_i a_iE_i)]$. For each i, the degree of this line bundle on E_i is

$$\deg\bigl(\mathcal{O}_T(\textstyle\sum_j a_jE_j)|_{E_i}\bigr) = \sum_j (E_i\cdot E_j)a_j.$$

The composite $(\mathbb{Z}/n)^r \to \mathrm{Pic}(T)/n \hookrightarrow (\mathbb{Z}/n)^r$ is therefore $M\otimes\mathbb{Z}/n$. Composing with the injection constructed above preserves the kernel. Together with the Kummer identifications in (3.3), this proves (3.2).

To prove (2) and (3), first note that choosing a primitive n-th root of unity in the ground field identifies μ_n with the constant sheaf $\mathbb{Z}/n$ on U_s°. As U_s° is connected, we have $H^1_{\mathrm{et}}(U_s^\circ,\mu_n) \cong \mathrm{Hom}(\pi_1^{\mathrm{et}}(U_s^\circ),\mathbb{Z}/n)$.

For (2), the exact sequence (3.2) gives $H^1_{\mathrm{et}}(E,\mu_n) = 0$ and $\ker(M\otimes\mathbb{Z}/n) = 0$ for every such n. For a prime $\ell \neq p$, the normalization surjection then gives $H^1_{\mathrm{et}}(\widetilde{E}_i,\mu_\ell) = 0$, so each E_i has geometric genus zero. Moreover, $\ker(M\otimes\mathbb{Z}/\ell) = 0$ implies $\ell \nmid \det M$, proving that $|\det M|$ is a power of p.

For (3), if $\ell \neq p$ is an odd prime, then $\ker(M\otimes\mathbb{Z}/\ell)$ is a quotient of $\mathrm{Hom}(\mathbb{Z}/2,\mathbb{Z}/\ell) = 0$, so $\ell \nmid \det M$. As $H^1_{\mathrm{et}}(E,\mu_4) = 0$, the group $\ker(M\otimes\mathbb{Z}/4) \cong \mathrm{Hom}(\mathbb{Z}/2,\mathbb{Z}/4)$ has order two. Writing $d_1,\dots,d_r$ for the invariant factors of M, we have $\ker(M\otimes\mathbb{Z}/4) \cong \bigoplus_i \mathbb{Z}/\gcd(d_i,4)$. Thus exactly one d_i is even and it is not divisible by four, so $\det M$ has 2-adic valuation one, proving the determinant formula. If E is a tree of smooth rational curves meeting transversally, then $\mathrm{Pic}(E) \cong \mathbb{Z}^r$ and $H^0(E,\mathcal{O}_E)^* = \kappa(s)^*$. The latter group is 4-divisible, so the Kummer sequence gives $H^1_{\mathrm{et}}(E,\mu_4) \cong \mathrm{Pic}(E)[4] = 0$. □

Corollary 3.4. *Let k, W and $\phi\colon S \to W$ be as in Lemma 3.1, let $s \in S$ be a singular point with image $w \in W$, and let $g\colon T \to \operatorname{Spec} \mathcal{O}^{\mathrm{sh}}_{S,s}$ be a resolution with reduced exceptional curve $E = E_1 \cup \cdots \cup E_r$ and intersection matrix $M = (E_i \cdot E_j)$.*

(1) *If W is regular at w, then $H^1_{\mathrm{et}}(E, \mu_n) = 0$ for every integer n prime to p, every E_i has geometric genus zero, and $|\det M| = p^e$ for some $e \geq 0$.*

(2) *If s is a rational singularity and g is minimal, then E is a tree of smooth rational curves meeting transversally. If moreover $W = X'$ and $w = P$ are as in Lemma 3.2, then $|\det M| = 2p^e$ for some $e \geq 0$.*

Proof. Apply Lemma 3.3 with $A = \mathcal{O}^{\mathrm{sh}}_{S,s}$. Part (1) follows from Lemmas 3.1 and 3.3(2). In (2), the description of E follows from rationality [Lip69], and the determinant assertion follows from Lemmas 3.2 and 3.3(3). □

Let s be a rational double point on a normal surface S over an algebraically closed field k of characteristic p, and put $A = \mathcal{O}^{\mathrm{sh}}_{S,s}$. Let $T \to \operatorname{Spec} A$ be the minimal resolution, with exceptional curves $E_1, \ldots, E_r$ and intersection matrix M. Table 1 records the equations and the absolute determinants $|\det M|$. The determinants follow from the ADE intersection matrices and are valid in every characteristic. If $p > 5$, then $\widehat{A} \cong k[[x, y, z]]/(f)$ with f as in the table [Art77, Section 3].

Type	Equation f	$\lvert\det M\rvert$
A_n, $n \geq 1$	$x^{n+1} + yz$	$n+1$
D_n, $n \geq 4$	$z^2 + x^2 y + y^{n-1}$	4
E_6	$z^2 + x^3 + y^4$	3
E_7	$z^2 + x^3 + xy^3$	2
E_8	$z^2 + x^3 + y^5$	1

Table 1: Equations for $p > 5$ and exceptional-lattice determinants of rational double points.

Combining Lemma 3.1 and Corollary 3.4(1) with the known local fundamental groups of rational double points gives the following restriction. Hirokado [Hir04, Theorem 3.4] proved a special case; the general form below is due to Schröer.

Corollary 3.5 ([Sch08, Theorem 6.1])**.** *Let X be a regular surface over a field k of characteristic $p > 5$. Then every rational double point of $X_{\bar{k}}$ has type A_{p^e-1} for some integer $e \geq 1$.*

4 First properties

Assumption 4.1. In this section, k is an F-finite separably closed field of characteristic $p \geq 7$, and X is a regular projective geometrically integral surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$ which is not geometrically normal. We write $\nu\colon S \to X_{\bar{k}}$ for the normalization and $\phi\colon S \to X$ for its composite with the projection.

Proposition 4.2. (1) *every numerically trivial line bundle on X is trivial; in particular $\omega_X \cong \mathcal{O}_X$;*

(2) *$K_S \sim -(p-1)D$ for some nonzero effective $\mathbb{Z}$-divisor D on S supported over the non-normal locus of $X_{\bar{k}}$;*

(3) *S is a rational surface with rational singularities;*

(4) *$H^1(X, \mathcal{O}_X) = 0$;*

(5) *$\chi(\mathcal{O}_X) = 2$;*

(6) *the group $N := \operatorname{Pic}(X)$ injects into $N^1(X)_{\mathbb{Q}}$ and, with the intersection form, is an even lattice of signature $(1, \rho(X) - 1)$;*

(7) *$\det N = (-1)^{\rho(X)-1} p^a$ for some $a \geq 0$.*

Proof. (1) This is [CWZ26, Corollary 6.2].

(2) By Equation (2.1), we have $K_S \sim -(p-1)D + \phi^* K_X$, where $D > 0$ is support on the non-normal locus of $X_{\bar{k}}$. Then the conclusion follows from (1).

(3) The surface S is rational by [BF25, Proposition 5.21]; in particular $H^1(S, \mathcal{O}_S) = 0$. Since $D > 0$, we have $H^0(S, \omega_S) = 0$, which implies $H^2(S, \mathcal{O}_S) = 0$ by Serre duality. Let $\mu\colon T \to S$ be a resolution; since S is rational, T is a smooth projective rational surface, so $H^1(T, \mathcal{O}_T) = 0$. The Leray spectral sequence and $H^1(T, \mathcal{O}_T) = 0$ give an injection $H^0(S, R^1\mu_* \mathcal{O}_T) \hookrightarrow H^2(S, \mathcal{O}_S) = 0$. As $R^1\mu_* \mathcal{O}_T$ has finite support, it vanishes; that is, S has rational singularities.

(4) Suppose that $H^1(X, \mathcal{O}_X) \neq 0$. By [Tan21, Remark 5.2], choose $e \geq 1$ such that, for $\ell = k^{1/p^e}$, the normalization $S_0 = (X_\ell)^\nu$ is geometrically normal. Then $(S_0)_{\bar{k}} \cong S$, so flat base change and (3) give $H^1(S_0, \mathcal{O}_{S_0}) = 0$. Lemma 2.3 then gives an α_p-torsor $\pi\colon V \to X_K$, where $K = k(a)$, such that V is a regular projective surface over K with $H^0(V, \mathcal{O}_V) = K$ which is not geometrically reduced. The identity $\omega_V \cong \pi^* \omega_{X_K}$ gives $K_V \equiv 0$. This contradicts [WZ26, Theorem 1.2], which asserts that V is geometrically integral because $p \geq 7$.

(5) By Serre duality, $h^2(\mathcal{O}_X) = h^0(\omega_X) = 1$; with (4) we get $\chi(\mathcal{O}_X) = 2$.

(6) By (1), N injects into $N^1(X)_{\mathbb{Q}}$; in particular it is torsion free. For every divisor G on X, Riemann–Roch and (5) give $G^2 = 2\chi(\mathcal{O}_X(G)) - 4$, so N is even. Its signature is $(1, \rho(X) - 1)$ by the Hodge index theorem.

(7) Let $\mu\colon T \to S$ be the minimal resolution. By (3), T is a smooth projective rational surface, so $\Lambda = \mathrm{Pic}(T)$ is a unimodular lattice. Let $E \subset \Lambda$ be the sublattice spanned by the exceptional curves of μ. It is the orthogonal sum of the exceptional lattices of the singular points of S, and each of these has determinant $\pm$ a power of p by Corollary 3.4 (1), because X is regular. Let

$$E^{\mathrm{sat}} := (E \otimes_{\mathbb{Z}} \mathbb{Q}) \cap \Lambda = \{x \in \Lambda : mx \in E \text{ for some integer } m \geq 1\}$$

be the saturation of E in Λ. Since $|\det E| = [E^{\mathrm{sat}} : E]^2 \, |\det E^{\mathrm{sat}}|$, the determinant of E^{sat} is $\pm$ a power of p. Let $Q = (E^{\mathrm{sat}})^\perp$. Since Λ is unimodular and E^{sat} is primitive, E^{sat} and Q have isomorphic discriminant groups, so $|\det Q|$ is a power of p.

For divisors A and B on X, we have $\left(\mu^*\phi^* A \cdot \mu^*\phi^* B\right)_{\bar{k}} = (A \cdot B)_k$, where degrees are taken over $\bar{k}$ on T and over k on X. Thus the homomorphism $\mu^*\phi^*\colon N \to \Lambda$ preserves the intersection pairing. Its image lies in Q, since pullbacks from S have zero intersection with every μ-exceptional curve. Let $v \in Q$ be represented by a divisor D_T on T. Descending S and $\mu_* D_T$ to a finite purely inseparable extension of k and clearing the ramification indices, which are powers of p, gives $e \geq 0$ and a divisor G on X with $p^e \mu_* D_T = \phi^* G$. Put $\Delta = \mu^*\phi^* G - p^e D_T$. The divisor Δ is μ-exceptional, so its class lies in E. Its class also lies in $Q = (E^{\mathrm{sat}})^\perp$, so $\Delta^2 = 0$. Negative definiteness of E therefore gives $\Delta = 0$. Hence $Q/\mu^*\phi^* N$ is a finite p-group, and

$$|\det N| = [Q : \mu^*\phi^* N]^2 \, |\det Q|$$

is a power of p. The sign of $\det N$ is determined by the signature in (6). □

Let Y be the normalization of $X_{k^{1/p}}$, which is finite over X because k is F-finite, and let $C_1 = C_{k^{1/p}}$ be the effective $\mathbb{Z}$-divisor on Y in (2.1), with $k' = k^{1/p}$. Since $\omega_X \cong \mathcal{O}_X$ by Proposition 4.2(1), we have

$$K_Y \sim -(p-1)C_1,$$

and $C_1 \neq 0$, since [Tan21, Proposition 2.10(3)] shows that $X_{k^{1/p}}$ is not normal. The ramification indices of the height-one morphism $Y \to X$ divide p. Hence pC_1 descends to an effective nonzero Weil divisor D_1 on X, which is Cartier because X is regular. Intersection numbers on X, Y and S are computed over k, $k^{1/p}$ and $\bar{k}$, respectively. Pullback from X to Y preserves these intersection numbers of Cartier divisors, since $Y \to X_{k^{1/p}}$ is finite and birational. We say that C_1 is ample if D_1 is ample; equivalently, pC_1 is ample on Y.

Proposition 4.3. *The divisor C_1 is not ample.*

Proof. Suppose that D_1 is ample. Let $\mu\colon T \to S$ be the minimal resolution, and let H be the pullback of D_1 to T under $T \to S \to X$. Then H is nef, and

$$(H^2)_{\bar{k}} = (D_1^2)_k = ((pC_1)^2)_{k^{1/p}} = p\left((pC_1) \cdot C_1\right)_{k^{1/p}} \geq p,$$

because $\big((pC_1)\cdot C_1\big)_{k^{1/p}}$ is the degree over $k^{1/p}$ of the line bundle $\mathcal{O}_Y(pC_1)$ on the nonzero effective divisor C_1, hence a positive integer. Since $(D_1^2)_k$ is even by Proposition 4.2(6) and p is odd, we have $H^2 \geq 2p$. The curves contracted by $T \to X_{\bar{k}}$ are exactly the μ-exceptional curves, since $S \to X_{\bar{k}}$ is finite; so H is positive on every curve which is not μ-exceptional.

By Proposition 4.2(2), $-K_S \sim (p-1)D$, where $(p-1)D$ is the divisorial conductor of $S \to X_{\bar{k}}$. Let $\mu_1 \colon S \to Y$ be the natural morphism. The conductor of $S \to X_{\bar{k}}$ is contained in the extension to S of the conductor of $Y \to X_{k^{1/p}}$, so $D \geq \mu_1^* C_1$. Hence $-K_S = \frac{p-1}{p} H_S + (p-1)(D - \mu_1^* C_1)$, where H_S is the pullback of D_1 to S and the second summand is effective. By [Kol13, Theorem 10.5 and Claim 10.3.5], $K_T = \mu^* K_S - Z$ with $Z \geq 0$ exceptional. Therefore $-K_T = \frac{p-1}{p} H + Z'$ with Z' effective, and, H being nef,

$$K_T \cdot H \leq -\tfrac{p-1}{p} H^2.$$

The surface T is a smooth projective rational surface, so $K_T^2 \leq 9$. If $K_T + H$ were nef, then

$$0 \leq (K_T + H)^2 = K_T^2 + 2K_T \cdot H + H^2 \leq 9 - \tfrac{p-2}{p} H^2 \leq 9 - 2(p-2) = 13 - 2p < 0,$$

a contradiction. So there is a $(K_T + H)$-negative extremal ray R of $\overline{\mathrm{NE}}(T)$, and $K_T \cdot R < 0$ because H is nef. If R were spanned by a (-1)-curve E, then E would not be μ-exceptional, since μ is minimal, so $H \cdot E \geq 1$ and $(K_T + H) \cdot E \geq 0$, which is impossible. By the cone theorem for smooth projective surfaces, $T \cong \mathbb{P}^2$ or T is a geometrically ruled surface over a curve, which is $\mathbb{P}^1$ because T is rational; so $T \cong \mathbb{P}^2$ or $T \cong \mathbb{F}_n$ for some $n \geq 0$.

On $\mathbb{P}^2$ there is no curve of negative self-intersection, and on $\mathbb{F}_n$ the only one is the negative section, which has self-intersection $-n$. Hence S is $\mathbb{P}^2$, a surface $\mathbb{F}_n$, or the surface $\mathbb{P}(1,1,n)$ obtained by contracting the negative section of $\mathbb{F}_n$ with $n \geq 2$. In the last case, the vertex of $\mathbb{P}(1,1,n)$ lies over a regular point of X, and its exceptional curve has self-intersection $-n$; so n is a power of p by Corollary 3.4(1). Since $-K_S \sim (p-1)D$ and D is a $\mathbb{Z}$-divisor, the class of K_S in $\mathrm{Cl}(S)$ is divisible by $p-1$. For $S = \mathbb{P}^2$ this gives $(p-1) \mid 3$. For $S = \mathbb{F}_n$, with $\mathrm{Cl}(S)$ freely generated by the negative section σ and a fibre F and $K_S = -2\sigma - (n+2)F$, it gives $(p-1) \mid 2$. For $S = \mathbb{P}(1,1,n)$, with $\mathrm{Cl}(S) \cong \mathbb{Z}$ generated by the image $\bar{F}$ of a fibre and $K_S = -(n+2)\bar{F}$, it gives $(p-1) \mid (p^e + 2)$ with $n = p^e$, hence $(p-1) \mid 3$. All three conclusions contradict $p \geq 7$. □

Proposition 4.4. *One of the following holds:*

(1) *there is a fibration $f \colon X \to B$ onto a regular projective curve whose generic fibre is a smooth curve of genus one;*

(2) *$\rho(X) = 2$, the intersection form on $N^1(X)_{\mathbb{Q}}$ is anisotropic, $\det \mathrm{Pic}(X) = -p^{2a+1}$ for some $a \geq 0$, and $p \equiv 1 \pmod 4$. Moreover, X contains a negative curve.*

Proof. By Proposition 4.2, the lattice $N := \mathrm{Pic}(X)$ satisfies the hypotheses of Lemma 2.7, and X satisfies those of Lemma 2.8. Suppose that $N \otimes \mathbb{Q}$ is anisotropic. The numerical assertions in (2) follow from Lemma 2.7. Suppose that X contains no negative curve. The divisor D_1 is effective and nonzero, and each of its components has nonnegative self-intersection; hence D_1 is nef and $D_1^2 \geq 0$. Anisotropy gives $D_1^2 > 0$. If a curve Γ satisfied $D_1 \cdot \Gamma = 0$, the Hodge index theorem and anisotropy would give $\Gamma^2 < 0$. Hence D_1 is positive on every curve, and it is ample by the Nakai–Moishezon criterion. This contradicts Proposition 4.3, proving (2).

Now suppose that $N \otimes \mathbb{Q}$ is isotropic. Lemma 2.8 gives a fibration $f \colon X \to B$ onto a regular projective curve. The generic fibre X_η is a regular proper curve over $K = k(B)$ with $H^0(X_\eta, \mathcal{O}) = K$. Since f is flat and ω_{X_η} is the restriction of ω_X, we have $[k(P) : k] \deg_K \omega_{X_\eta} = (K_X \cdot X_P)_k = 0$ for every closed point $P \in B$, where X_P is the fibre. Hence $\chi(\mathcal{O}_{X_\eta}) = 0$ and $h^0(\omega_{X_\eta}) = h^1(\mathcal{O}_{X_\eta}) = 1$; a degree-zero line bundle with a nonzero section on an integral curve is trivial, so $\omega_{X_\eta} \cong \mathcal{O}_{X_\eta}$. By [Sch22, Theorem 2.3(i)] and $p > 3$, the curve X_η is geometrically regular, that is, smooth, so (1) holds. □

In case (1) of Proposition 4.4, the genus-one fibration allows us to study geometric normality fibre by fibre using the reduction theory of its Jacobian (Section 6). In case (2), treated in Section 7, let $f \colon X \to X'$ be the contraction of the negative curve. Then $\rho(X') = \rho(X) - 1 = 1$. If X' remains geometrically non-normal, the condition $\rho(X') = 1$ makes the conductor divisor on its geometric normalization ample, providing the positivity used in Section 7.2.

5 Local genus-one models

We recall the comparison of relative differentials from [LLR04] and prove a congruence used in the genus-one case of the main theorem.

5.1 Local models and relative differentials

Let R be a discrete valuation ring with uniformizer π, residue field $k = R/\pi R$, and fraction field K. Let Y be a smooth projective curve of genus one over K with $H^0(Y, \mathcal{O}_Y) = K$. Let J be its Jacobian. Let $\mathcal{Y}$ and $\mathcal{J}$ be the *minimal regular models* of Y and J over R, respectively; see for example [Liu02, Section 9.3]. We say that J has *good reduction* over R if it extends to an elliptic scheme over R, equivalently if $\mathcal{J}$ is smooth over R. There is a canonical cohomology isomorphism [LLR04, 1.4]

$$\tau_Y \colon H^1(Y, \mathcal{O}_Y) \xrightarrow{\sim} H^1(J, \mathcal{O}_J). \tag{5.1}$$

This isomorphism is compatible with field extension and, when $Y(K) \neq \varnothing$, is induced by a choice of K-point identifying J with Y [LLR04, 1.4 and Corollary 1.5]. The comparison theorem [LLR04, Theorem 3.8] shows that τ_Y extends to an R-linear homomorphism

$$\tau_{\mathcal{Y}} \colon H^1(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}}) \to H^1(\mathcal{J}, \mathcal{O}_{\mathcal{J}}).$$

The target is a free R-module of rank one [LLR04, Corollary 5.8]. Since $\tau_{\mathcal{Y}}$ becomes an isomorphism over K, its kernel is exactly the R-torsion submodule $H^1(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}})_{\mathrm{tors}}$ of $H^1(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}})$, and its cokernel is also torsion. Both have finite R-length because the cohomology groups are finitely generated over R.

Definition 5.1. We define

$$a(\mathcal{Y}) = \operatorname{length}_R \ker(\tau_{\mathcal{Y}}), \qquad b(\mathcal{Y}) = \operatorname{length}_R \operatorname{coker}(\tau_{\mathcal{Y}}).$$

By relative Serre duality, $H^0(\mathcal{Y}, \omega_{\mathcal{Y}/R})$ is the R-dual of $H^1(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}})$, and similarly for $\mathcal{J}$; these modules are free of rank one. Let $\eta_{\mathcal{Y}}$ and η_J be generators of $H^0(\mathcal{Y}, \omega_{\mathcal{Y}/R})$ and $H^0(\mathcal{J}, \omega_{\mathcal{J}/R})$, respectively, and identify $H^0(Y, \omega)$ with $H^0(J, \omega)$ by the dual of τ_Y. Write $\mathcal{Y}_s$ for the closed fibre of $\mathcal{Y}$.

Lemma 5.2. (1) *$a(\mathcal{Y}) = 0$ if and only if $H^0(\mathcal{Y}_s, \mathcal{O}_{\mathcal{Y}_s}) = k$.*

(2) *If $\mathcal{Y}$ is smooth over R, then $a(\mathcal{Y}) = b(\mathcal{Y}) = 0$.*

(3) *$\mathcal{Y} \otimes_R \widehat{R}$ and $\mathcal{J} \otimes_R \widehat{R}$ are the minimal regular models over the completion $\widehat{R}$, and $a(\mathcal{Y} \otimes_R \widehat{R}) = a(\mathcal{Y})$, $b(\mathcal{Y} \otimes_R \widehat{R}) = b(\mathcal{Y})$.*

(4) *If k is perfect, or J has split multiplicative reduction, then $a(\mathcal{Y}) = b(\mathcal{Y})$.*

(5) (Comparison of relative differentials) *The inclusion $H^0(\mathcal{J}, \omega_{\mathcal{J}/R}) \hookrightarrow H^0(\mathcal{Y}, \omega_{\mathcal{Y}/R})$ has cokernel of R-length $b(\mathcal{Y})$. In particular,*

$$\eta_J = \pi^{b(\mathcal{Y})} \eta_{\mathcal{Y}} \quad \text{up to a unit of } R. \tag{5.2}$$

This identity persists in $H^0(Y_{K'}, \omega)$ after any flat base change $R \to R'$ of discrete valuation rings, where $K' = \operatorname{Frac}(R')$.

Proof. (1) Since $\mathcal{Y}$ is integral and dominates $\operatorname{Spec} R$, multiplication by π is injective on $\mathcal{O}_{\mathcal{Y}}$. The ring $H^0(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}})$ is a finite R-subalgebra of $H^0(Y, \mathcal{O}_Y) = K$, so it equals R because R is normal. The exact sequence $0 \to \mathcal{O}_{\mathcal{Y}} \xrightarrow{\pi} \mathcal{O}_{\mathcal{Y}} \to \mathcal{O}_{\mathcal{Y}_s} \to 0$ therefore gives

$$0 \to k \to H^0(\mathcal{Y}_s, \mathcal{O}_{\mathcal{Y}_s}) \to H^1(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}})[\pi] \to 0.$$

A finitely generated R-module is torsion free if and only if its π-torsion vanishes.

(2) After a faithfully flat unramified base change, the smooth model $\mathcal{Y}$ has a section and is therefore isomorphic to $\mathcal{J}$. By flat base change and the description of τ_Y above, $\tau_{\mathcal{Y}}$ becomes an isomorphism; faithful flatness then gives $a(\mathcal{Y}) = b(\mathcal{Y}) = 0$.

(3) The scheme $\mathcal{Y} \otimes_R \widehat{R}$ is regular [Sta26, Tag 0BG6] and has the same closed fibre as $\mathcal{Y}$, so it is the minimal regular model over $\widehat{R}$, and similarly for $\mathcal{J}$. Flat base change identifies $\tau_{\mathcal{Y} \otimes_R \widehat{R}}$ with $\tau_{\mathcal{Y}} \otimes_R \widehat{R}$, whose kernel and cokernel have the same lengths over $\widehat{R}$ as those of $\tau_{\mathcal{Y}}$ over R.

(4) The perfect-residue-field case is [LLR04, Theorem 3.8]. By (3), the split multiplicative case reduces to [LLR04, Remark 8.13] over $\widehat{R}$.

(5) The map induced by $\tau_{\mathcal{Y}}$ on $H^1(\mathcal{Y},\mathcal{O}_{\mathcal{Y}})/\mathrm{tors}$ is injective, with cokernel of length $b(\mathcal{Y})$. Dualizing gives an inclusion of the Jacobian differential lattice $H^0(\mathcal{J},\omega_{\mathcal{J}/R})$ into $H^0(\mathcal{Y},\omega_{\mathcal{Y}/R})$ with the same cokernel length, which is (5.2) for the chosen generators. Since τ_Y is compatible with base change [LLR04, 1.4], the identity persists after flat base change. □

5.2 Good Jacobian reduction

Assume further that R is excellent of equal characteristic $p>0$ and that J has good reduction. The curve Y is a torsor under J. By [LLR04, Proposition 8.1 and Remark 8.2], this action extends to $\mathcal{J}\times_R\mathcal{Y}\to\mathcal{Y}$ and is transitive on the geometric points of the closed fibre. In particular, the closed fibre $\mathcal{Y}_s$ is geometrically irreducible. As an effective Cartier divisor, it has the form

$$\mathcal{Y}_s = rV,$$

where $r\geq 1$ is an integer and V is integral. By [LLR04, Proposition 8.1], V is smooth of genus one over $k_0=H^0(V,\mathcal{O}_V)$, and k_0/k is finite and purely inseparable.

Let $\eta_{\mathcal{Y}}$ be a generator of $H^0(\mathcal{Y},\omega_{\mathcal{Y}/R})$ as before. Its restriction to the smooth generic curve has no zeros, so its zero divisor is qV for a unique integer $q\geq 0$. If $q\geq r$, then $(1/\pi)\eta_{\mathcal{Y}}$ has divisor $(q-r)V\geq 0$ and hence is a global regular section, contradicting $H^0(\mathcal{Y},\omega_{\mathcal{Y}/R})=R\eta_{\mathcal{Y}}$. Thus

$$0\leq q<r, \qquad \mathrm{div}(\eta_{\mathcal{Y}}) = qV. \tag{5.3}$$

Via the identification of $H^0(Y,\omega)$ with $H^0(J,\omega)$ in Subsection 5.1 and Lemma 5.2(5), we regard the Jacobian differential η_J as a global section of $\omega_{\mathcal{Y}/R}$. By (5.2), $\eta_J=\pi^{b(\mathcal{Y})}\eta_{\mathcal{Y}}$ up to a unit; since $\mathrm{div}(\pi)=rV$, (5.3) gives $\mathrm{div}(\eta_J)=\chi V$, where

$$\chi = r\,b(\mathcal{Y})+q.$$

Since $0\leq q<r$, the integer χ determines both $b(\mathcal{Y})=\lfloor\chi/r\rfloor$ and q. The integer q plays the same role as the coefficient a in the canonical bundle formula [BM77, Theorem 2]. When the residue field is algebraically closed, χ is computed by the following result of Raynaud, whose proof is written out in [BT14].

Proposition 5.3 ([BT14, Proposition 3.17])**.** *Assume that R is complete and k is algebraically closed, and let ν be the order of $\mathcal{O}_{\mathcal{Y}}(V)|_V$ in* $\mathrm{Pic}(V)$*. Then $r=\nu p^s$ for an integer $s\geq 0$, and there are positive integers $k_0,\dots,k_{s-1}$ such that*

$$\chi = (r-1)+\sum_{i=0}^{s-1} k_i\,(r-\nu p^i). \tag{5.4}$$

Here s and the k_i record how the order of $\mathcal{O}_{\mathcal{Y}}(-V)|_{nV}$ in $\mathrm{Pic}(nV)$ grows from ν to r along the thickenings $nV\subset\mathcal{Y}$ defined by $\mathcal{O}_{\mathcal{Y}}(-nV)$ [BT14, Lemma 3.12]. In the tame case $s=0$ we get $\chi=r-1$, so $b(\mathcal{Y})=0$ and $q=r-1$; if $s\geq 1$, then $\chi\geq r$ and $b(\mathcal{Y})\geq 1$. Since $r-\nu p^i=\nu p^i(p^{s-i}-1)$ is divisible by $p-1$, (5.4) gives $\chi\equiv r-1 \pmod{p-1}$. After passing to the completion of the strict henselization of R, this implies Theorem 5.4 below when k is perfect; the rest of the section treats imperfect k by reduction to this case.

Theorem 5.4. *In this setting,*

$$r\,b(\mathcal{Y})+q\equiv r-1 \pmod{p-1}. \tag{5.5}$$

Corollary 5.5. *Assume that $r=1$, so that $\mathcal{Y}_s=V$ is reduced. Then $b(\mathcal{Y})$ is divisible by $p-1$. Moreover, $\mathcal{Y}$ is smooth over R if and only if $b(\mathcal{Y})=0$.*

If $\mathcal{Y}_s$ is reduced and $\mathcal{Y}/R$ is not smooth, then the image of η_J in $H^0(\mathcal{Y},\omega_{\mathcal{Y}/R})$ vanishes to order at least $p-1$ along $\mathcal{Y}_s$ by (5.2).

5.3 Normalization under extension of constants

To prove Theorem 5.4, we will compare $\mathcal{Y}$ with a regular model over a discrete valuation ring with algebraically closed residue field. Extending the coefficient field can destroy regularity, so we must normalize after base change. We first treat finite extensions and compute the changes in r, q, and b; these relations will then allow us to construct a finite regular normalization after extension to $\bar{k}$.

Lemma 5.6. *Let $\mathcal{Y}/R$ be as in Subsection 5.2, and assume moreover that R is henselian with a separably closed coefficient field k. Let ℓ/k be a finite purely inseparable extension and put $R_\ell = R \otimes_k \ell$.*

(1) *The normalization $\mathcal{Y}_\ell$ of $\mathcal{Y} \times_R \operatorname{Spec} R_\ell$ is its minimal regular model.*

(2) *Let ℓ'/ℓ be a further finite purely inseparable extension. Write r_ℓ, V_ℓ, q_ℓ for r, V, q of Subsection 5.2 applied to $\mathcal{Y}_\ell$, and similarly for ℓ'. There are nonnegative integers $e = e(\ell'/\ell)$, $N = N(\ell'/\ell)$ and $n = n(\ell'/\ell)$, with $(p-1) \mid N$, such that*

$$r_{\ell'} = p^e r_\ell, \tag{5.6}$$

$$n r_{\ell'} = N + q_{\ell'} - p^e q_\ell, \tag{5.7}$$

$$b(\mathcal{Y}_\ell) = n + b(\mathcal{Y}_{\ell'}). \tag{5.8}$$

Here p^e is the ramification index along $V_{\ell'}$ over V_ℓ, and the divisorial conductor of $\mathcal{Y}_{\ell'} \to (\mathcal{Y}_\ell)_{R_{\ell'}}$ is $NV_{\ell'}$.

Proof. (1) Since R_ℓ is finite free over R and $R_\ell/\pi R_\ell \cong \ell$, it is a one-dimensional Noetherian local ring with maximal ideal πR_ℓ, hence a discrete valuation ring with uniformizer π. The ring R_ℓ is excellent because it is finite over R, so normalization is finite. Since the generic curve is geometrically integral, the base change is integral. The composite $\mathcal{Y}_\ell \to \mathcal{Y}$ is finite and purely inseparable with normal target, hence a universal homeomorphism; so is the normalization map.

The action of $\mathcal{J}_{R_\ell}$ on $\mathcal{Y} \times_R \operatorname{Spec} R_\ell$ lifts to $\mathcal{Y}_\ell$. The product $\mathcal{J}_{R_\ell} \times_{R_\ell} \mathcal{Y}_\ell$ is normal because it is smooth over the normal scheme $\mathcal{Y}_\ell$. Hence the induced morphism to $\mathcal{Y} \times_R \operatorname{Spec} R_\ell$ factors through its normalization $\mathcal{Y}_\ell$. The identity and associativity axioms hold on the generic fibre and hence everywhere. The action on geometric points of the closed fibre is transitive because the normalization map is a universal homeomorphism.

Let S be the singular locus of the normal excellent surface $\mathcal{Y}_\ell$; it is finite, lies in the closed fibre, and is invariant under translations by $\mathcal{J}(R_\ell)$. Every ℓ-point of $\mathcal{J}_s$ lifts to an R_ℓ-point because R_ℓ is henselian and $\mathcal{J}$ is smooth. Since ℓ is separably closed, the group $G = \mathcal{J}_s(\ell)$ is infinite.

Fix a closed point $x \in (\mathcal{Y}_\ell)_s$ and a geometric point $\bar{x}$ over it. By transitivity, the geometric orbit map $g \mapsto g \cdot \bar{x}$ is a nonconstant morphism from an elliptic curve, so its fibres are finite. Hence the subgroup $H = \{g \in G \mid g \cdot \bar{x} = \bar{x}\}$ is finite, and the orbit $G \cdot \bar{x} \cong G/H$ is infinite. Since ℓ is separably closed, each closed point has a unique geometric point over it, so these orbit points lie over infinitely many distinct closed points. The lifting statement above places all these closed points in the $\mathcal{J}(R_\ell)$-orbit of x. If x were singular, its entire orbit would lie in the finite set S, a contradiction. Thus S is empty and $\mathcal{Y}_\ell$ is regular. The reduced closed fibre is a homogeneous curve under an elliptic curve. The argument of [LLR04, Proposition 8.1] shows that it is smooth of genus one over its purely inseparable constant field. It contains no (-1)-curve, so $\mathcal{Y}_\ell$ is minimal.

(2) Put $\mathcal{X} = \mathcal{Y}_\ell \times_{R_\ell} \operatorname{Spec} R_{\ell'}$.

$$\begin{array}{ccccc} \mathcal{Y}_{\ell'} & \xrightarrow{\nu} & \mathcal{X} & \longrightarrow & \mathcal{Y}_\ell \\ & & \downarrow & & \downarrow \\ & & \operatorname{Spec} R_{\ell'} & \longrightarrow & \operatorname{Spec} R_\ell. \end{array}$$

The right square is Cartesian, and ν is the normalization map. The composite $\mathcal{Y}_{\ell'} \to \mathcal{Y}_\ell$ is finite and purely inseparable, so its ramification index along $V_{\ell'}$ is p^e for some integer $e \geq 0$. Since π is a uniformizer of both R_ℓ and $R_{\ell'}$, the fibre multiplicities satisfy

$$r_{\ell'} = \operatorname{ord}_{V_{\ell'}}(\pi) = p^e \operatorname{ord}_{V_\ell}(\pi) = p^e r_\ell,$$

which proves (5.6). The generic fibre is smooth, so the divisorial conductor is $NV_{\ell'}$ for an integer $N \geq 0$. The regular model $\mathcal{Y}_\ell/R_\ell$ is Gorenstein. For every finite extension ℓ''/ℓ, the scheme $\mathcal{Y}_\ell \times_\ell \operatorname{Spec} \ell'' = \mathcal{Y}_\ell \times_{R_\ell} \operatorname{Spec} R_{\ell''}$ is integral because it is flat over the discrete valuation ring $R_{\ell''}$ with generic fibre a base change of the geometrically integral curve Y. Since ℓ is separably closed, $\mathcal{Y}_\ell$ is therefore geometrically integral over ℓ [Sta26, Tags 035X and 038K]. Applying Lemma 2.1 with $(k, A, X, k') = (\ell, R_\ell, \mathcal{Y}_\ell, \ell')$ gives

$$\omega_{\mathcal{Y}_{\ell'}/R_{\ell'}} \cong \nu^* \omega_{\mathcal{X}/R_{\ell'}} \otimes \mathcal{O}_{\mathcal{Y}_{\ell'}}(-NV_{\ell'}).$$

In particular, $(p-1) \mid N$.

Let $\eta_{\mathcal{Y}_\ell}$ and $\eta_{\mathcal{Y}_{\ell'}}$ be the generators of Subsection 5.2 for $\mathcal{Y}_\ell$ and $\mathcal{Y}_{\ell'}$, and write $\eta_{\mathcal{Y}_\ell} = \pi^{-n}\eta_{\mathcal{Y}_{\ell'}}$ up to a unit. By (5.3) applied to $\mathcal{Y}_\ell$, we have $\operatorname{div}(\eta_{\mathcal{Y}_\ell}) = q_\ell V_\ell$. Since $\omega_{\mathcal{X}/R_{\ell'}}$ is the pullback of $\omega_{\mathcal{Y}_\ell/R_\ell}$ and V_ℓ pulls back to $p^e V_{\ell'}$ on $\mathcal{Y}_{\ell'}$, the pullback of $\eta_{\mathcal{Y}_\ell}$ is a section of $\nu^*\omega_{\mathcal{X}/R_{\ell'}}$ with divisor $p^e q_\ell V_{\ell'}$. Via the isomorphism above, it is a rational section of $\omega_{\mathcal{Y}_{\ell'}/R_{\ell'}}$ with divisor $p^e q_\ell V_{\ell'} - NV_{\ell'}$. Taking divisors gives (5.7). Since $p^e q_\ell < p^e r_\ell = r_{\ell'}$, the right-hand side of (5.7) is greater than $-r_{\ell'}$. The integer n is therefore nonnegative. The Jacobian differential extends unchanged through constant extension because the Jacobian has good reduction. Equation (5.2) then gives (5.8). □

Proposition 5.7. *Let $\mathcal{Y}/R$ be as in Lemma 5.6. For a field extension ℓ/k, let $\nu_\ell\colon \mathcal{Y}_\ell \to \mathcal{Y}\times_R \operatorname{Spec}(R\otimes_k \ell)$ be the normalization. Then there exists a finite extension ℓ/k such that ν_ℓ is finite, $\mathcal{Y}_\ell$ is geometrically regular over ℓ, and V_ℓ is geometrically reduced over ℓ.*

Proof. Fix a closed point P of degree d of the generic curve Y over K. For a finite extension ℓ/k, let K_ℓ be the fraction field of R_ℓ, and let P_ℓ be the unique point of Y_{K_ℓ} over P; its degree d_ℓ over K_ℓ is at most d. Let H_ℓ be the closure of P_ℓ in the regular proper model $\mathcal{Y}_\ell$. Then H_ℓ is a horizontal Cartier divisor, finite and flat of degree d_ℓ over R_ℓ. Intersecting it with the closed fibre gives

$$d_\ell = (H_\ell \cdot (r_\ell V_\ell))_\ell = r_\ell (H_\ell \cdot V_\ell)_\ell.$$

The final intersection is a positive integer, so $r_\ell \le d$.

Choose ℓ_1 for which r_{ℓ_1} is maximal among all finite purely inseparable constant extensions. For every finite extension ℓ'/ℓ_1, maximality and (5.6) give $r_{\ell'} = r_{\ell_1}$; hence $e(\ell''/\ell') = 0$ whenever $\ell_1 \subseteq \ell' \subseteq \ell''$. Choose $\ell_2 \supseteq \ell_1$ for which $b(\mathcal{Y}_{\ell_2})$ is minimal among such extensions, which is possible because (5.8) makes these nonnegative integers nonincreasing. Hence $n(\ell''/\ell') = 0$ whenever $\ell_2 \subseteq \ell' \subseteq \ell''$. Finally choose $\ell_3 \supseteq \ell_2$ for which q_{ℓ_3} is minimal. For every $\ell' \supseteq \ell_3$, equation (5.7) now gives

$$N(\ell'/\ell_3) = q_{\ell_3} - q_{\ell'} = 0.$$

The scheme $(\mathcal{Y}_{\ell_3}) \times_{R_{\ell_3}} \operatorname{Spec} R_{\ell'}$ is S_2 and regular in codimension one since $N(\ell'/\ell_3) = 0$, thus it is normal by Serre's criterion. It consequently equals its normalization $\mathcal{Y}_{\ell'}$ and is regular by Lemma 5.6 (1). Since $(\mathcal{Y}_{\ell_3})_{R_{\ell'}} = \mathcal{Y}_{\ell_3} \times_{\ell_3} \operatorname{Spec} \ell'$ and every finite extension of ℓ_3 is purely inseparable, $\mathcal{Y}_{\ell_3}$ is geometrically regular over ℓ_3. Since $e(\ell'/\ell_3) = 0$ for every finite extension ℓ'/ℓ_3, the pullback of V_{ℓ_3} to $\mathcal{Y}_{\ell_3} \times_{\ell_3} \operatorname{Spec} \ell' = \mathcal{Y}_{\ell'}$ is the reduced divisor $V_{\ell'}$; hence V_{ℓ_3} is geometrically reduced over ℓ_3 [Sta26, Tag 035X]. Finally, ν_{ℓ_3} is finite because R_{ℓ_3} is excellent. □

5.4 Proof of Theorem 5.4 and Corollary 5.5

Proof. We may replace R by the completion of its strict henselization, which is isomorphic to $k^{\mathrm{s}}[[\pi]]$. Indeed, this base change preserves the minimal regular model, its closed fibre, and generators of $H^0(\mathcal{Y}, \omega_{\mathcal{Y}/R})$, hence r, q, and b: for the strict henselization because it is ind-étale over R, and for the completion by Lemma 5.2(3). After renaming k^{s} as k, Lemma 5.6 and Proposition 5.7 apply.

Put $R_\infty = R \otimes_k \bar{k}$. It is the union of the discrete valuation rings $R_{\ell'}$, for finite extensions ℓ'/k, with uniformizer π, so every nonzero element of R_∞ is a unit times a power of π. Thus R_∞ is a Noetherian discrete valuation ring. By Proposition 5.7, there is a finite extension ℓ/k such that ν_ℓ is finite, $\mathcal{Y}_\ell$ is geometrically regular over ℓ, and V_ℓ is geometrically reduced over ℓ.

Fix such an ℓ. Then $\mathcal{Z} = \mathcal{Y}_\ell \times_\ell \operatorname{Spec} \bar{k} = \mathcal{Y}_\ell \times_{R_\ell} \operatorname{Spec} R_\infty$ is finite and birational over $\mathcal{Y}_{R_\infty}$; in particular, $\mathcal{Z}$ is Noetherian, since $\mathcal{Y}_{R_\infty}$ is of finite type over the Noetherian ring R_∞. Since $\mathcal{Y}_\ell$ is geometrically regular over ℓ and $\mathcal{Z}$ is Noetherian, $\mathcal{Z}$ is regular [GD67, Proposition 5.13.7]. Being regular, finite and birational over $\mathcal{Y}_{R_\infty}$, the scheme $\mathcal{Z}$ is the normalization of $\mathcal{Y}_{R_\infty}$.

Since V_ℓ is geometrically reduced over ℓ, the closed fibre of $\mathcal{Z}$ is mE, where $m = r_\ell$ and $E = V_\ell \times_\ell \operatorname{Spec} \bar{k}$ is a smooth elliptic curve. By (5.6), $m = p^e r$ with $e = e(\ell/k)$. Let NE denote the divisorial conductor of $\mathcal{Z} \to \mathcal{Y}_{R_\infty}$. It is the base change of the divisorial conductor $N(\ell/k)V_\ell$ of ν_ℓ, so $N = N(\ell/k) \ge 0$ and $(p-1) \mid N$ by Lemma 5.6 (2). Let $\bar{q}$ be the integer in (5.3) for $\mathcal{Z}$, and put $\bar{a} = a(\mathcal{Z}) = b(\mathcal{Z})$, where the last equality holds by Lemma 5.2(4) since $\bar{k}$ is perfect. The relations of Lemma 5.6 (2) give

$$nm = N + \bar{q} - p^e q, \qquad b(\mathcal{Y}) = n + \bar{a}. \tag{5.9}$$

From now on, work over the completion $\bar{k}[[\pi]]$ of R_∞. The closed fibre mE contains no (-1)-curve, so $\mathcal{Z}$ is the minimal regular model of its generic fibre; by Lemma 5.2(3) and flat base change, its base change to $\bar{k}[[\pi]]$ is again the minimal regular model, with the same closed fibre mE and the same integers $\bar{q}$ and $\bar{a}$. We replace $\mathcal{Z}$ by this base change. By Proposition 5.3 for $\mathcal{Z}$, whose integer χ is $m\bar{a}+\bar{q}$, we have $m\bar{a}+\bar{q} \equiv m-1 \pmod{p-1}$. Combining this with (5.9) gives

$$mb(\mathcal{Y}) + p^e q = N + \bar{q} + m\bar{a} \equiv m - 1 \pmod{p-1}.$$

Since $m = p^e r$ and $p^e \equiv 1 \pmod{p-1}$, this is (5.5).

Suppose that $r = 1$. Then $q = 0$, so $(p-1) \mid b(\mathcal{Y})$. If $b(\mathcal{Y}) = 0$, equation (5.9) forces $n = \bar{a} = N = \bar{q} = 0$. The conductor vanishes, so $\mathcal{Z} = \mathcal{Y} \otimes_R \bar{k}[[\pi]]$. Moreover, the integer $\chi = m\bar{a} + \bar{q}$ of $\mathcal{Z}$ vanishes; since every summand in (5.4) for $\mathcal{Z}$ is nonnegative and the first one is $m-1$, we get $m = 1$. The closed fibre is smooth, and the generic fibre is smooth, so $\mathcal{Z}$ is smooth over $\bar{k}[[\pi]]$, and hence $\mathcal{Y}$ is smooth over R by faithfully flat descent. Conversely, if $\mathcal{Y}$ is smooth, then $b(\mathcal{Y}) = 0$ by Lemma 5.2(2). □

6 Genus-one fibrations on K-trivial surfaces

Under the Assumption 4.1, we have shown that (Proposition 4.4) either there is a fibration $f\colon X \to B$ with smooth generic fiber or $\rho(X) = 2$ and $N^1(X)_{\mathbb{Q}}$ is anisotropic. We deal with the former case in this section.

Assumption 6.1. In this section, k is a separably closed field of characteristic $p \geq 5$, and X is a regular projective geometrically integral surface over k with

$$H^0(X, \mathcal{O}_X) = k, \qquad H^1(X, \mathcal{O}_X) = 0, \qquad \omega_X \cong \mathcal{O}_X.$$

Moreover X admits a fibration $f\colon X \to B$ onto a regular projective curve whose generic fibre X_K, where $K = k(B)$, is a smooth curve of genus one.

Corollary 5.5 gives a lower bound for a nonzero value of the invariant b in Definition 5.1 after the Jacobian acquires good reduction. We now obtain an upper bound from the triviality of ω_X and compare the two bounds. The good and multiplicative reduction cases are treated directly, and the additive case is treated after a tame base change.

6.1 Weierstrass equations

We recall some standard facts on Weierstrass equations [Del75, §1, (2.6) and Proposition 5.1]. Let E be an elliptic curve over a field F, given by a Weierstrass equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in F,$$

with the origin of E at $(0:1:0)$. Let c_4, c_6 and Δ be the quantities attached to this equation in [Del75, (1.4) and (1.5)], where Δ is its discriminant, so that $1728\Delta = c_4^3 - c_6^2$ and $\Delta \neq 0$, and let $\omega_0 = dx/(2y + a_1x + a_3)$ be the invariant differential, a basis of $H^0(E, \omega_{E/F})$. Any two Weierstrass equations of E differ by a change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ with $u \in F^*$ and $r, s, t \in F$, which replaces $(c_4, c_6, \Delta, \omega_0)$ by $(u^{-4}c_4, u^{-6}c_6, u^{-12}\Delta, u\,\omega_0)$; in particular $\Delta\,\omega_0^{\otimes 12}$ and the j-invariant $j(E) = c_4^3/\Delta$ of E do not depend on the equation.

Now let $F = \mathrm{Frac}(R)$ for a discrete valuation ring R with valuation v. A Weierstrass equation is *integral* if all a_i lie in R, and *minimal* if it is integral and $v(\Delta)$ is minimal among integral Weierstrass equations. Any two minimal Weierstrass equations of E differ by a change of variables with $u \in R^*$, since their discriminants have the same valuation; hence $v(c_4)$, $v(c_6)$ and $v(\Delta)$ do not depend on the choice of the minimal equation.

If $6 \in R^*$, an integral Weierstrass equation can be transformed over R into one of the form $y^2 = x^3 - \frac{1}{48}c_4x - \frac{1}{864}c_6$ with the same c_4, c_6 and Δ [Del75, (2.1) and (2.2)]; together with the rescaling rule above, this shows that an integral Weierstrass equation is minimal if and only if $v(c_4) < 4$ or $v(c_6) < 6$. Let $\mathcal{E}$ be the minimal regular model of E over R, and let ω_0 be the invariant differential of a minimal Weierstrass equation of E. Then $\omega_{\mathcal{E}/R} = \omega_0\,\mathcal{O}_{\mathcal{E}}$ [Liu02, Corollary 9.4.37], so $H^0(\mathcal{E}, \omega_{\mathcal{E}/R}) = \omega_0\,H^0(\mathcal{E}, \mathcal{O}_{\mathcal{E}}) = \omega_0 R$.

Fix a minimal Weierstrass equation of E over R, with quantities c_4, c_6 and Δ, let $W \subset \mathbb{P}^2_R$ be its Weierstrass model, and let $\bar{\kappa}$ be an algebraic closure of the residue field of R. The geometric closed fibre

$W_{\bar\kappa}$ is the cubic defined by the image of the minimal equation over $\bar\kappa$; it is a curve of genus one in the sense of [Del75, §1], and the quantities c_4, c_6 and Δ attached to this cubic are the images of c_4, c_6 and Δ in $\bar\kappa$. Following [Del75, §1], $W_{\bar\kappa}$ is of *elliptic*, *multiplicative* or *additive type* if it is smooth, a nodal cubic or a cuspidal cubic, respectively. By [Del75, Proposition 5.1], these three cases occur if and only if $v(\Delta) = 0$, if and only if $v(\Delta) > 0$ and $v(c_4) = 0$, and if and only if $v(\Delta) > 0$ and $v(c_4) > 0$, respectively.

With this notation, E has good reduction if and only if $v(\Delta) = 0$. If $v(\Delta) > 0$, then E has bad reduction, which we call *multiplicative* or *additive* according as $W_{\bar\kappa}$ is of multiplicative or additive type. If E has multiplicative reduction and the residue field of R is separably closed, then E has split multiplicative reduction of type I_n in the sense of [LLR04, Section 8], where $n = -v(j(E))$ by loc. cit.; hence $n = v(\Delta)$, as $v(c_4) = 0$.

6.2 The base and relative differentials

We work under Assumption 6.1 and fix the following notation. For a closed point $P \in B$ let $R_P = \mathcal{O}_{B,P}$ be the local ring with maximal ideal $\mathfrak{m}_P$, let $t_P \in \mathfrak{m}_P$ be a uniformizer of R_P, let $\widehat{R}_P$ be the $\mathfrak{m}_P$-adic completion of R_P and $\widehat{K}_P$ its fraction field, and put $X_{(P)} = X \times_B \operatorname{Spec} R_P$. Let J_K be the Jacobian of X_K; the curve X_K is a torsor under J_K. Let $g\colon \mathcal{J} \to B$ be the minimal regular model of J_K over B. Put

$$L_X = f_*\omega_{X/B}, \qquad L_J = g_*\omega_{\mathcal{J}/B}.$$

Lemma 6.2. (1) *$B \cong \mathbb{P}^1_k$ and the morphism f is flat.*

(2) *For every closed point $P \in B$, $X_{(P)}$ and $X_{(P)} \otimes_{R_P} \widehat{R}_P$ are the minimal regular models of X_K over R_P and of $X_K \otimes_K \widehat{K}_P$ over $\widehat{R}_P$.*

(3) *$L_X \cong \mathcal{O}_{\mathbb{P}^1}(2)$ and $\omega_{X/B} \cong f^*L_X$. In particular $\omega_{X_{(P)}/R_P}$ is generated by a global section.*

(4) *$R^1f_*\mathcal{O}_X$ is invertible. For every closed point $P \in B$ we have $a(X_{(P)}) = 0$ and $H^0(X_P, \mathcal{O}_{X_P}) = k(P)$.*

(5) *For every closed point $P \in B$, the multiplicities of the components of X_P have greatest common divisor one.*

Proof. (1) We have $H^0(B, \mathcal{O}_B) = k$, and the Leray spectral sequence gives an exact sequence

$$0 \to H^1(B, \mathcal{O}_B) \to H^1(X, \mathcal{O}_X) \to H^0(B, R^1f_*\mathcal{O}_X) \to H^2(B, \mathcal{O}_B) = 0. \tag{6.1}$$

As $H^1(X, \mathcal{O}_X) = 0$, we get $H^1(B, \mathcal{O}_B) = 0$. Hence $\deg \omega_B = -2$, and B is a conic over k [Kol13, Lemma 10.6(3)]. Since B is regular, k is separably closed and $p \neq 2$, the conic B is smooth and split, so $B \cong \mathbb{P}^1_k$. The morphism f is flat by [Har77, III, Proposition 9.7], as X is integral and B is a regular curve.

(2) Since $K_X \equiv 0$, the regular proper model $X_{(P)}$ of the genus-one curve X_K contains no (-1)-curve, so it is the minimal regular model. The same holds for $X_{(P)} \otimes_{R_P} \widehat{R}_P$, which is regular by [Sta26, Tag 0BG6] and has the same closed fibre.

(3) Since X and B are regular, f is a local complete intersection morphism, and $\omega_{X/B} \cong \omega_X \otimes f^*\omega_B^{-1} \cong f^*\omega_B^{-1}$ [Liu02, Theorem 6.4.9]. By the projection formula and $f_*\mathcal{O}_X = \mathcal{O}_B$, we get $L_X \cong \omega_B^{-1} \cong \mathcal{O}_{\mathbb{P}^1}(2)$ and $\omega_{X/B} \cong f^*L_X$.

(4) We have $H^0(B, R^1f_*\mathcal{O}_X) = 0$ since $H^1(X, \mathcal{O}_X) = 0$. The torsion subsheaf of $R^1f_*\mathcal{O}_X$ has finite support and no nonzero global sections, hence vanishes. Thus $R^1f_*\mathcal{O}_X$ is locally free, of rank $h^1(X_K, \mathcal{O}) = 1$. By flat base change $H^1(X_{(P)}, \mathcal{O}) = (R^1f_*\mathcal{O}_X)_P$ is free, and $X_{(P)}$ is the minimal regular model by (2); so $a(X_{(P)}) = 0$ and $H^0(X_P, \mathcal{O}_{X_P}) = k(P)$ by Lemma 5.2 (1).

(5) Write $X_P = mD$, where m is the greatest common divisor of the multiplicities and D is an effective Cartier divisor on $X_{(P)}$, and suppose that $m > 1$. The ideal $\mathcal{O}_{X_{(P)}}(-(m-1)D)/\mathcal{O}_{X_{(P)}}(-mD)$ of $\mathcal{O}_{X_P}$ is isomorphic to $\mathcal{O}_D(-(m-1)D) \cong \mathcal{O}_D(D)$, because $mD = X_P$ is principal, and its square is zero because $2(m-1) \geq m$. By adjunction [Liu02, Theorem 9.1.37] and $\omega_{X_{(P)}/R_P} \cong \mathcal{O}_{X_{(P)}}$, which holds by (3), we have $\omega_{D/k(P)} \cong \mathcal{O}_D(D)$. As $(D^2)_{k(P)} = m^{-1}(D \cdot X_P)_{k(P)} = 0$, the degree of $\omega_{D/k(P)}$ over $k(P)$ is zero, and Riemann–Roch on the projective Gorenstein curve D gives $\chi(\mathcal{O}_D) = -\frac{1}{2}\deg_{k(P)} \omega_{D/k(P)} = 0$ [Liu02, Theorem 7.3.17 and Corollary 7.3.31]. By Serre duality $h^0(D, \omega_{D/k(P)}) = h^1(D, \mathcal{O}_D) = h^0(D, \mathcal{O}_D) > 0$. Hence $H^0(X_P, \mathcal{O}_{X_P})$ contains nonzero elements of a square-zero ideal, which is impossible since $H^0(X_P, \mathcal{O}_{X_P}) = k(P)$ is a field by (4). □

Over an algebraically closed field, Lemma 6.2 (3)–(5) are the case $\omega_X \cong \mathcal{O}_X$, $h^1(\mathcal{O}_X) = 0$ of the canonical bundle formula [BM77, Theorem 2 and Proposition 4]: there are no wild fibres, hence no multiple fibres. We give direct proofs because X need not be smooth and k may be imperfect.

For a closed point $P \in B$, let v_P be the valuation of R_P, put $b_P = b(X_{(P)})$, and let $c_{4,P}$, $c_{6,P}$ and Δ_P be the quantities attached to a minimal Weierstrass equation of J_K over R_P.

Lemma 6.3. (1) *There is an injective homomorphism $L_J \to L_X$ whose cokernel has length b_P at every closed point P. Consequently* $\deg L_J + \sum_P b_P[k(P) : k] = 2$.

(2) $12 \deg L_J = \sum_P v_P(\Delta_P)[k(P) : k]$.

In particular $0 \le \deg L_J \le 2$, and if J_K has bad reduction at some point, then $\sum_P b_P[k(P) : k] \le 1$.

Proof. (1) The stalks of L_X and L_J at P are $H^0(X_{(P)}, \omega)$ and $H^0(\mathcal{J} \times_B \operatorname{Spec} R_P, \omega)$ by flat base change, and their generic fibres are identified by the dual of the isomorphism τ_{X_K} of (5.1). By (5.2) this identification maps $(L_J)_P$ into $(L_X)_P$ with cokernel of length b_P. Since $\deg L_X = 2$ by Lemma 6.2 (3), the formula follows.

(2) Let $\Delta_K = \Delta\, \omega_0^{\otimes 12}$, where Δ and ω_0 are the discriminant and the invariant differential of a Weierstrass equation of J_K; the element Δ_K of $H^0(J_K, \omega_{J_K/K})^{\otimes 12}$ is nonzero and does not depend on the equation. By flat base change $H^0(J_K, \omega_{J_K/K})^{\otimes 12}$ is the stalk of $L_J^{\otimes 12}$ at the generic point of B, so Δ_K is a nonzero rational section of $L_J^{\otimes 12}$. Fix a closed point $P \in B$. Put $\mathcal{J}_{(P)} = \mathcal{J} \times_B \operatorname{Spec} R_P$. Then $\mathcal{J}_{(P)}$ is the minimal regular model of J_K over R_P. By flat base change, $\omega_{\mathcal{J}_{(P)}/R_P}$ is the pullback of $\omega_{\mathcal{J}/B}$, and $H^0(\mathcal{J}_{(P)}, \omega_{\mathcal{J}_{(P)}/R_P}) = (g_* \omega_{\mathcal{J}/B})_P = (L_J)_P$. Choose a minimal Weierstrass equation of J_K over R_P, with discriminant Δ_P and invariant differential $\omega_{0,P}$. As recalled above, $\omega_{0,P}$ generates $(L_J)_P$, and $\Delta_K = \Delta_P\, \omega_{0,P}^{\otimes 12}$ since Δ_K does not depend on the equation. Hence Δ_K lies in $(L_J^{\otimes 12})_P$ and vanishes to order $v_P(\Delta_P)$ at P. As P is arbitrary, Δ_K is a global section of $L_J^{\otimes 12}$ with divisor $\sum_P v_P(\Delta_P) P$, a finite sum since $\Delta_K \neq 0$. Taking degrees over k, where a closed point P has degree $[k(P) : k]$, gives $12 \deg L_J = \sum_P v_P(\Delta_P)[k(P) : k]$. □

6.3 Good and multiplicative reduction

Lemma 6.4. *Assume $p \ge 13$, and let $P \in B$ be a closed point at which J_K has bad reduction. If P is not k-rational, then $[k(P) : k] = p \le 23$, and J_K has multiplicative reduction at P with $v_P(\Delta_P) = 1$. In particular, every point at which J_K has additive reduction is k-rational.*

Proof. By Lemma 6.3, $v_P(\Delta_P)[k(P) : k] \le 12 \deg L_J \le 24$. If P is not k-rational, then $[k(P) : k] > 0$ is a power of p, hence at least 13. Therefore $v_P(\Delta_P) = 1$, and $[k(P) : k] = p$ and $p \le 23$. If $v_P(c_{4,P}) > 0$, then $c_{6,P}^2 = c_{4,P}^3 - 1728\Delta_P$ would have valuation one, which is odd; hence $v_P(c_{4,P}) = 0$ and the reduction is multiplicative. □

Lemma 6.5. *If J_K has good reduction at a closed point P, then X_P is smooth over $k(P)$, and X is smooth over k at every point of X_P.*

Proof. By Subsection 5.2 and Lemma 6.2 (2), $X_P = rV$ with V integral and smooth over $k_0 = H^0(V, \mathcal{O}_V)$. By Lemma 6.2 (5), $r = 1$. Then $k_0 = H^0(X_P, \mathcal{O}_{X_P}) = k(P)$ by Lemma 6.2 (4), so X_P is smooth over $k(P)$. Since f is flat and $B \cong \mathbb{P}^1_k$ by Lemma 6.2 (1), f is smooth along X_P, and hence X is smooth over k along X_P. □

Let $\mathcal{O}_L$ be a complete discrete valuation ring with fraction field L and residue field κ, and let L'/L be a finite separable extension. The integral closure $\mathcal{O}_{L'}$ of $\mathcal{O}_L$ in L' is a discrete valuation ring finite over $\mathcal{O}_L$, because $\mathcal{O}_L$ is complete and L'/L is separable; let $\mathfrak{m}_{L'}$ be its maximal ideal and κ' its residue field. Since $\operatorname{Tr}_{L'/L}(\mathcal{O}_{L'}) \subseteq \mathcal{O}_L$ and the trace form of L'/L is nondegenerate, the set $\{\beta \in L' : \operatorname{Tr}_{L'/L}(\beta \mathcal{O}_{L'}) \subseteq \mathcal{O}_L\}$ is a fractional ideal of $\mathcal{O}_{L'}$ containing $\mathcal{O}_{L'}$. Its inverse $\mathfrak{D}_{L'/L}$ is therefore an ideal of $\mathcal{O}_{L'}$, the *different* of L'/L, and the *different exponent* of L'/L is the integer $\delta_{L'/L} \ge 0$ with $\mathfrak{D}_{L'/L} = \mathfrak{m}_{L'}^{\delta_{L'/L}}$.

Lemma 6.6. *Assume that L'/L is cyclic of degree p with ramification index one, and that κ'/κ is purely inseparable with $[\kappa' : \kappa] = p$. Then $\delta_{L'/L} \ge p - 1$.*

Proof. Since the ramification index is one, we have $\mathcal{O}_{L'} \otimes_{\mathcal{O}_L} \kappa = \kappa'$. Let σ be a generator of the Galois group of L'/L, and let $z \in \mathcal{O}_{L'}$ be a lift of a generator of κ' over κ. As $[\kappa' : \kappa] = p$, the images of $1, z, \dots, z^{p-1}$

form a basis of κ' over κ, so $\mathcal{O}_{L'} = \mathcal{O}_L[z]$ by Nakayama's lemma. Let $g(X) = \prod_{j=0}^{p-1}(X - \sigma^j z) \in \mathcal{O}_L[X]$ be the minimal polynomial of z over L, and let $g'(X)$ be its derivative. By [Ser79, III, §6, Corollary 2 to Proposition 11], the equality $\mathcal{O}_{L'} = \mathcal{O}_L[z]$ implies that $\mathfrak{D}_{L'/L}$ is generated by $g'(z) = \prod_{j=1}^{p-1}(z - \sigma^j z)$. The automorphism σ acts trivially on κ' because κ'/κ is purely inseparable, so each factor lies in $\mathfrak{m}_{L'}$; hence $g'(z) \in \mathfrak{m}_{L'}^{p-1}$ and $\delta_{L'/L} \geq p-1$. □

Lemma 6.7. *Let C be a smooth projective geometrically connected curve of genus one over L whose Jacobian has multiplicative reduction of type I_n, and let $\mathcal{Y}$ be the minimal regular model of C over $\mathcal{O}_L$. Assume that $\mathcal{O}_L$ contains k, that κ is separably closed, and that $\mathcal{Y}_s$ is not multiple, that is, the multiplicities of its components have greatest common divisor one. Then exactly one of the following holds.*

(1) *$a(\mathcal{Y}) \geq p-1$.*

(2) *$a(\mathcal{Y}) = 0$ and C has a rational point. Moreover:*

- *if $n = 1$, then $\mathcal{Y}_s$ is irreducible and reduced, with a κ-rational double point, and its normalization is $\mathbb{P}^1_\kappa$;*
- *if $n \geq 2$, then $\mathcal{Y}_s$ is a cycle of n reduced components isomorphic to $\mathbb{P}^1_\kappa$, and every intersection point of two components is κ-rational, with the two components meeting transversally there.*

Proof. The two cases exclude each other, so we assume $a(\mathcal{Y}) < p-1$ and prove (2). Let M be the cyclic extension of L attached to C in [LLR04, Proposition 8.3(a)], so that $C(M) \neq \emptyset$, with ramification index e and residue degree f. The multiplicities of the components of $\mathcal{Y}_s$ have greatest common divisor e [LLR04, before Proposition 8.11], so $e = 1$, and [LLR04, Proposition 8.11(b)] gives $\delta_{M/L} = a(\mathcal{Y}) < p-1$. As κ is separably closed, f is a power of p. Suppose $f > 1$, and let L' be the subextension of M/L of degree p. The extension L'/L is cyclic with ramification index dividing $e = 1$, so $[\kappa' : \kappa] = p$, and κ'/κ is purely inseparable because κ is separably closed. Since $\mathfrak{D}_{M/L} = \mathfrak{D}_{M/L'}\mathfrak{D}_{L'/L}\mathcal{O}_M$ [Ser79, III, §4, Proposition 8], Lemma 6.6 gives $\delta_{M/L} \geq \delta_{L'/L} \geq p-1$, a contradiction. Hence $f = 1$ and $M = L$, so C has a rational point and $a(\mathcal{Y}) = \delta_{M/L} = 0$. The description of $\mathcal{Y}_s$ follows from [LLR04, Proposition 8.3(b)]. □

Lemma 6.8. *Let $P \in B$ be a closed point at which J_K has multiplicative reduction, and put $q = [k(P) : k]$. Then X is smooth over k at every point of X_P that is not a singular point of X_P. Every singular point z of X_P has $\kappa(z) = k(P)$, and there is an isomorphism*

$$\widehat{\mathcal{O}}_{X_{\bar{k}},\bar{z}} \cong \overline{k}[[s, Y_1, Y_2]]/(Y_1Y_2 - s^q),$$

where $\bar{z}$ is the point of $X_{\bar{k}}$ over z. In particular $X_{\bar{k}}$ is normal at every point over X_P, and X is smooth over k along X_P if P is k-rational.

Proof. We first apply Lemma 6.7 over $\widehat{R}_P$, then describe X_P and the completed local rings of X at the singular points of X_P, and finally pass to $\overline{k}$. Choose an affine coordinate u on $B = \mathbb{P}^1_k$ with $P \neq \infty$, so that P lies in $\operatorname{Spec} k[u] = B \setminus \{\infty\}$; let $\mathfrak{p} \subset k[u]$ be the maximal ideal of P, so that $R_P = k[u]_{\mathfrak{p}}$, and let $\bar{u} \in k(P)$ be the image of u. Since k is separably closed, the residue field $k(P) = k(\bar{u})$ is purely inseparable of degree q over k, so $\bar{u}$ has minimal polynomial $T^q - c$ over k for some $c \in k$. Hence $\mathfrak{p} = (u^q - c)$, and we take $t_P = u^q - c$, which generates $\mathfrak{m}_P = \mathfrak{p}R_P$. The pullback of t_P to $X_{(P)}$ is still denoted by t_P.

Step 1: a rational point over $\widehat{K}_P$. Let $R = \widehat{R}_P$. By Lemma 6.2 (2), $Y = X_{(P)} \otimes_{R_P} R$ is the minimal regular model of $X_{\widehat{K}_P}$ over R, with closed fibre $Y_s = X_P$. By Lemma 6.2 (4) and Lemma 5.2 (3), $a(Y) = 0$, and by Lemma 6.2 (5) the multiplicities of the components of Y_s have greatest common divisor one. The Jacobian $J_K \otimes_K \widehat{K}_P$ of $X_{\widehat{K}_P}$ still has multiplicative reduction over R, since the minimality criterion $v(c_4) < 4$ or $v(c_6) < 6$ of Section 6.1 is unchanged by completion. Since k is separably closed, so is $k(P)$, and thus this reduction is split multiplicative of type I_n with $n = v_P(\Delta_P)$ by Section 6.1. Hence Lemma 6.7 (2) applies to Y, and $X_{\widehat{K}_P}$ has a rational point.

Step 2: the fibre X_P and its singular points. Suppose first $n \geq 2$. By Lemma 6.7 (2), $Y_s = X_P$ is a cycle of n reduced components isomorphic to $\mathbb{P}^1_{k(P)}$, and every intersection point of two components is $k(P)$-rational, with the two components meeting transversally there. Every other point of X_P lies on exactly one component, where X_P is smooth over $k(P)$, so the singular points of X_P are the intersection points of components, and they are $k(P)$-rational. At such a point z, let Y_1 and Y_2 be local equations of the

two components through z. Since these components meet transversally at z, the maximal ideal of $\mathcal{O}_{X,z}$ is (Y_1, Y_2), and since X_P is reduced, $t_P = \lambda Y_1 Y_2$ with $\lambda \in \mathcal{O}_{X,z}^*$.

Suppose $n = 1$. Since $X_{\widehat{K}_P}$ has a rational point, the curve $X_{\widehat{K}_P}$ is isomorphic to its Jacobian, so by uniqueness of minimal regular models, Y is isomorphic to the minimal regular model of the Jacobian. Choose a minimal Weierstrass equation $y^2 = x^3 - \frac{1}{48}c_4 x - \frac{1}{864}c_6$ over R. Here $v(c_4) = 0$ and $v(\Delta) = n = 1$, so $v(c_6) = 0$ because $1728\Delta = c_4^3 - c_6^2$. Replacing x by $x - c_6/(12c_4)$ gives $y^2 = x^3 + a_2 x^2 + a_4 x + a_6$ with

$$a_2 = -\frac{c_6}{4c_4}, \qquad a_4 = -\frac{36\Delta}{c_4^2}, \qquad a_6 = \frac{c_6\Delta}{c_4^3},$$

so $v(a_2) = 0$ and $v(a_4) = v(a_6) = 1$; thus the reduced cubic is $y^2 = x^2(x + \bar{a}_2)$ with $\bar{a}_2 \neq 0$, whose unique singular point is the node $x = y = 0$. As the equation is congruent to $-a_6$ modulo the square of the maximal ideal (t_P, x, y) of $R[x, y]$ at the node and $v(a_6) = 1$, the equation does not lie in this square, so the Weierstrass model is regular at the node; it is regular elsewhere because its closed fibre is smooth there. The closed fibre of the Weierstrass model is irreducible and not a (-1)-curve, so the Weierstrass model is the minimal regular model Y. The closed fibre $X_P = Y_s$ is a nodal cubic, smooth over $k(P)$ away from the node z, which is $k(P)$-rational. Since 2 is invertible and the residue of a_2 is a nonzero square in the separably closed field $k(P)$, Hensel's lemma gives $\gamma \in R[[x, y]]$ with $\gamma^2 = a_2 + x$. The elements $Y_1 = y - \gamma x$ and $Y_2 = y + \gamma x$ satisfy $(Y_1, Y_2) = (x, y)$ and $Y_1 Y_2 = y^2 - x^2(a_2 + x)$. Writing $a_4 = t_P a_4'$ and $a_6 = t_P a_6'$ with $a_4', a_6' \in R^*$, the equation becomes $Y_1 Y_2 = t_P(a_4' x + a_6')$, so $t_P = \lambda Y_1 Y_2$ with the unit $\lambda = (a_4' x + a_6')^{-1}$.

In both cases, f is flat and $B \cong \mathbb{P}^1_k$ by Lemma 6.2 (1), so X is smooth over k at every point of X_P other than the singular points, where X_P is smooth over $k(P)$. At a singular point z of X_P we have $\kappa(z) = k(P)$, and the maximal ideal of $\widehat{\mathcal{O}}_{X,z}$ is generated by Y_1 and Y_2, and $t_P = \lambda Y_1 Y_2$ for a unit λ. The local R-algebra map $R[[Y_1, Y_2]] \to \widehat{\mathcal{O}}_{X,z}$ is surjective, because it is surjective on residue fields and on cotangent spaces. The kernel of this map is a height-one prime of the factorial ring $R[[Y_1, Y_2]]$ containing $t_P - \tilde{\lambda} Y_1 Y_2$, for a lift $\tilde{\lambda}$ of λ; since this element is a regular parameter, it generates the kernel. Hence

$$\widehat{\mathcal{O}}_{X,z} \cong R[[Y_1, Y_2]]/(t_P - \tilde{\lambda} Y_1 Y_2). \tag{6.2}$$

Step 3: base change to $\overline{k}$. Let $k' = k(\alpha)$ with $\alpha^q = c$ and put $s = u - \alpha$. The scheme $B_{k'}$ has a unique point over P, namely $s = 0$, and $\widehat{R}_P \otimes_k k' = k'[[s]]$ with $t_P = s^q$. Tensoring (6.2) with k' gives the completed local ring of $X_{k'}$ at the unique point over z:

$$k'[[s]][[Y_1, Y_2]]/(s^q - \tilde{\lambda} Y_1 Y_2) \cong k'[[s, Y_1, Y_2]]/(Y_1 Y_2 - s^q),$$

where we replaced Y_1 by $\tilde{\lambda} Y_1$. Completing after base change to $\overline{k}$ gives the asserted isomorphism. The ring $\overline{k}[[s, Y_1, Y_2]]/(Y_1 Y_2 - s^q)$ is a two-dimensional hypersurface with an isolated singular point, hence normal by Serre's criterion, so $X_{\bar{k}}$ is normal at $\bar{z}$. At the other points over X_P, the scheme $X_{\bar{k}}$ is smooth over $\overline{k}$ by Step 2, hence normal. Finally, if $q = 1$, then every singular point z of X_P has $\kappa(z) = k$, so the regular surface X is smooth over k at z [Sta26, Tag 00TV]; together with Step 2, X is smooth over k along X_P. □

6.4 Additive reduction: tame base change

Let $P \in B(k)$, let $U \subseteq B$ be an affine open neighbourhood of P and $t \in \mathcal{O}(U)$ with divisor P on U, and write $R = R_P$ and $Y = X_{(P)}$. For an integer $d \geq 1$ prime to p put

$$B_d = \operatorname{Spec} \mathcal{O}(U)[s]/(s^d - t), \qquad R_d = \mathcal{O}_{B_d, P_d} = R[s]/(s^d - t), \qquad K_d = k(B_d),$$

where P_d is the unique point of B_d over P. The curve B_d is smooth and geometrically integral over k and finite over U, the point P_d is k-rational, and R_d is a discrete valuation ring with uniformizer s. The group $G = \mu_d(k)$, of order d, acts on B_d over U by $s \mapsto \zeta s$, and K_d is a cyclic extension of K with group G.

Let Y_d be the minimal regular model of $X_K \otimes_K K_d$ over R_d. For $g \in G$, the twist of Y_d by the automorphism g of $\operatorname{Spec} R_d$ is again a minimal regular model of $X_K \otimes_K K_d$, since $X_K \otimes_K K_d$ is defined over K. Since a birational map between minimal regular models of a curve of genus one is an isomorphism [Liu02, Definition 9.3.12 and Corollary 9.3.24], the action of G on $X_K \otimes_K K_d$ through K_d extends to an action on Y_d over its action on R_d.

Let $\mathcal{Z}$ be the normalization of $f^{-1}(U) \times_U B_d$, and let $\widetilde{\mathcal{Z}} \to \mathcal{Z}$ be a resolution of singularities; it is projective over the affine curve B_d [Sta26, Tag 0C5P]. Contracting successively the (-1)-curves in the fibre over P_d gives a regular surface $\mathcal{Y}_d$, proper and flat over B_d, whose localization at P_d is minimal and hence isomorphic to Y_d. The surface $\mathcal{Y}_d$ is geometrically integral over k: its base change to $\overline{k}$ is flat over the integral curve $(B_d)_{\overline{k}}$, with generic fibre $X_K \otimes_K \overline{k}(B_d)$, which is integral because X_K is geometrically integral.

Lemma 6.9. *Let R be a discrete valuation ring with $6 \in R^*$ and uniformizer π, and let J be an elliptic curve over $\mathrm{Frac}(R)$ with additive reduction. Let c_4, c_6 and Δ be the quantities attached to a minimal Weierstrass equation, and put $\delta = v(\Delta)$. For an integer $d \geq 1$ with $d \in R^*$, let $R' = R[s]/(s^d - \pi)$, and let η_J and $\eta_{J'}$ be generators of $H^0(\omega)$ of the minimal regular models of J over R and of $J \otimes \mathrm{Frac}(R')$ over R'. Then exactly one of the following holds.*

(1) *$3v(c_4) \geq \delta$. Then $\delta \in \{2,3,4,6,8,9,10\}$, and for $d = 12/\gcd(12,\delta) \in \{2,3,4,6\}$ the curve $J \otimes \mathrm{Frac}(R')$ has good reduction, and $\eta_{J'} = s^{w_J}\eta_J$ up to a unit, with $w_J = d\delta/12 \leq d-1$.*

(2) *$3v(c_4) < \delta$. Then $v(c_4) = 2$, $v(c_6) = 3$ and $\delta = n+6$ with $n \geq 1$; for $d = 2$ the curve $J \otimes \mathrm{Frac}(R')$ has multiplicative reduction with $v_{R'}(\Delta') = 2n$ for the discriminant Δ' of a minimal equation, and $\eta_{J'} = s\eta_J$ up to a unit.*

Proof. (1) Assume that $3v(c_4) \geq \delta$. If $\delta \geq 12$, then $v(c_4) \geq 4$ and $2v(c_6) = v(c_4^3 - 1728\Delta) \geq 12$, contradicting minimality; so $0 < \delta < 12$. If $3v(c_4) > \delta$, then $2v(c_6) = \delta$, so δ is even; otherwise $3v(c_4) = \delta$. This gives the list. Put $u = s^{d\delta/12} \in R'$; the exponent is an integer by the choice of d. With $v' = v_{R'}$, so that $v' = dv$ on $\mathrm{Frac}(R)$, we have $v'(u^{-4}c_4) \geq d\delta/3 - d\delta/3 = 0$, $v'(u^{-6}c_6) \geq d\delta/2 - d\delta/2 = 0$ and $v'(u^{-12}\Delta) = 0$, using $2v(c_6) \geq \delta$. Hence the rescaled equation, given by $x = u^2x'$ and $y = u^3y'$, has coefficients in R' and unit discriminant, so J has good reduction over R'. Its invariant differential is u times that of the original equation, and both generate the respective modules $H^0(\omega)$ by Section 6.1, as both equations are minimal. Thus $w_J = v'(u) = d\delta/12$, which is at most $d-1$ as $\delta < 12$.

(2) Assume that $3v(c_4) < \delta$. Then $2v(c_6) = v(c_4^3 - 1728\Delta) = 3v(c_4)$, so $v(c_4) = 2m$ and $v(c_6) = 3m$ with $m \geq 1$, and minimality forces $m = 1$. Then $\delta > 6$. Rescaling with $u = s$ over $R' = R[s]/(s^2 - \pi)$ gives quantities c_4, c_6 and Δ of valuations 0, 0 and $2\delta - 12 = 2n$, so the new equation is minimal with multiplicative reduction, and its invariant differential is s times the original one. □

Lemma 6.10. *Let R be a discrete valuation ring whose residue field has characteristic p, with uniformizer t, let $d \geq 1$ be prime to p, and let $R' = R[s]/(s^d - t)$. Let Y and Y' be the minimal regular models of a smooth genus-one curve X_{K_R} over $K_R = \mathrm{Frac}(R)$ and of $X_{K_R} \otimes_{K_R} \mathrm{Frac}(R')$ over R'. If R is excellent and the multiplicities of the components of Y_s have greatest common divisor one, then the same holds for Y'.*

Proof. Let Γ be a component of Y_s of multiplicity r, and let $A = \mathcal{O}_{Y,\Gamma}$, with uniformizer ϖ, so that $t = \varpi^r w$ with $w \in A^*$. Put $g = \gcd(r,d)$, $r = gr'$ and $d = gd'$. In the function field $K(Y) \otimes_{K_R} \mathrm{Frac}(R')$, which is a field because X_{K_R} is geometrically integral, the element $\varepsilon = s^{d'}\varpi^{-r'}$ satisfies $\varepsilon^g = w$. The extension generated by ε is unramified over A, because g is prime to p and w is a unit. Over it, $s^{d'} = \varepsilon\varpi^{r'}$ with $\gcd(d',r') = 1$, so every valuation ν extending the valuation of A has ramification index d' and satisfies $\nu(s) = r\nu(\varpi)/d = r/g$. Let N be the normalization of $Y \otimes_R R'$, which is finite since R is excellent. Its closed fibre is the divisor of s, so every component of N_s over Γ has multiplicity $r/\gcd(r,d)$. If a prime number ℓ divided all these multiplicities, it would divide every r, which is impossible.

Let $\widetilde{N} \to N$ be a resolution of singularities. The strict transforms of the components of N_s keep their multiplicities, so the multiplicities of $\widetilde{N}_s$ have greatest common divisor one. The morphism $\widetilde{N} \to Y'$ provided by minimality is a sequence of blowups in closed points [Sta26, Tag 0C5R]. If $Z_1 \to Z_2$ is the blowup of a closed point z of a regular surface over R', then the exceptional curve has multiplicity $\sum_i r_i \,\mathrm{mult}_z(\Gamma_i)$, where the Γ_i are the components of the closed fibre of Z_2 with multiplicities r_i; hence Z_1 and Z_2 have the same greatest common divisor. Therefore the greatest common divisor for Y' is one. □

Lemma 6.11. *In the setting above, let η_X and η_J be generators of $H^0(\omega)$ of Y and of the minimal regular model of J_K over R, and let η_{Y_d} and η_{J_d} be those of Y_d and of the minimal regular model of $J_K \otimes_K K_d$ over R_d. Write $\eta_{Y_d} = s^{w_X}\eta_X$ and $\eta_{J_d} = s^{w_J}\eta_J$ up to units. Then $w_X \geq 0$ and*

$$b(Y_d) = d\,b_P - w_X + w_J.$$

Proof. By (5.2), $\eta_J = t^{b_P}\eta_X = s^{db_P}\eta_X$, and hence $\eta_{J_d} = s^{db_P - w_X + w_J}\eta_{Y_d}$ up to units. Comparing with (5.2) for Y_d gives the formula.

The scheme $M = Y \otimes_R R_d$ is flat over R_d with integral generic fibre, hence integral, and it is Gorenstein with ω_{M/R_d} generated by η_X. Let $Z \to M$ be a resolution of singularities of the normalization of M; it is an isomorphism over the points of codimension one of the normalization. At a point ξ of codimension one of M, the semilocal ring $\widetilde{A}$ of Z over ξ is the normalization of $A = \mathcal{O}_{M,\xi}$, and $\omega_{\widetilde{A}} = \mathrm{Hom}_A(\widetilde{A}, \omega_A)$ embeds into ω_A by evaluation at 1. Since M satisfies S_2 and ω_{M/R_d} is invertible, a rational section of ω_{M/R_d} is regular if it is regular at all points of codimension one. Hence $H^0(Z, \omega_{Z/R_d}) \subseteq H^0(M, \omega_{M/R_d}) = R_d\eta_X$. The morphism $Z \to Y_d$ given by minimality is a sequence of blowups in closed points [Sta26, Tag 0C5R]. The canonical bundle formula for these blowups gives $H^0(Y_d, \omega_{Y_d/R_d}) \subseteq H^0(Z, \omega_{Z/R_d}) \subseteq R_d\eta_X$, that is, $w_X \geq 0$. □

6.5 Additive reduction: descent of smoothness

Lemma 6.12. *In the setting of the tame base change, assume that Y_d is smooth over k at every point of its closed fibre. Then X is smooth over k at every point of X_P.*

Proof. Step 1: the quotient. The scheme Y_d is projective over R_d [Sta26, Tag 0C5P], so the quotient $W = Y_d/G$ exists [Gro71, Exp. V, Proposition 1.8]. It is a normal surface, proper over R, with generic fibre $(X_K \otimes_K K_d)/G = X_K$, and $Y_d \to W$ is finite. Since $|G|$ is invertible, formation of invariants commutes with flat base change; in particular $W_{k'} = (Y_d)_{k'}/G$ for every field extension k'/k. Let y be a closed point of the closed fibre of Y_d, with stabilizer $H \subseteq G$ and image $w \in W$. Then $\widehat{\mathcal{O}}_{W,w} = (\widehat{\mathcal{O}}_{Y_d,y})^H$, since G permutes the points over w transitively.

Step 2: local structure. Let k' be a finite purely inseparable extension of k containing a copy of $\kappa(y)$, and let y' and w' be the points over y and w. Then $\kappa(y') = k'$, and $\mathcal{O}_{(Y_d)_{k'},y'}$ is regular because Y_d is geometrically regular at y. Hence $\widehat{\mathcal{O}}_{(Y_d)_{k'},y'} \cong k'[[y_1, y_2]]$. The cyclic group H has order prime to p, and k contains the $|H|$-th roots of unity, so H acts on $\mathfrak{m}/\mathfrak{m}^2$ through two characters χ_1, χ_2. Applying the projectors $|H|^{-1}\sum_{h\in H}\chi_i(h)^{-1}h$ to lifts of an eigenbasis, we may choose y_1, y_2 to be eigenvectors with characters χ_1, χ_2. The group H acts faithfully on $k'[[y_1, y_2]]$, because it acts faithfully on the function field, so (χ_1, χ_2) is injective. We say that w is of the *first kind* if χ_1 or χ_2 is trivial, and of the *second kind* otherwise. If χ_2 is trivial, then χ_1 is injective, and $\widehat{\mathcal{O}}_{W_{k'},w'} = k'[[y_1^{|H|}, y_2]]$ is regular; similarly if χ_1 is trivial. So if w is of the first kind, since this holds for all sufficiently large k' and regularity descends along flat local homomorphisms [Sta26, Tag 00OF], W is geometrically regular at w.

Suppose now that w is of the second kind, and let h_0 generate the cyclic group H. Let $\mathcal{F} \subseteq Y_d$ be the fixed-point scheme of H, defined near y by the ideal generated by all $h^*a - a$. Its formation commutes with base change, and near y' the ideal of $\mathcal{F}_{k'}$ contains $(\chi_1(h_0) - 1)y_1$ and $(\chi_2(h_0) - 1)y_2$ and is contained in (y_1, y_2). So $\mathcal{O}_{\mathcal{F},y} \otimes_k k' = k'$, hence $\mathcal{O}_{\mathcal{F},y} = k$ and $\kappa(y) = k$. Then $\widehat{\mathcal{O}}_{Y_d,y} \cong k[[y_1, y_2]]$, and applying the projectors in $\mathcal{O}_{Y_d,y}$ we find eigenvectors $y_1, y_2 \in \mathcal{O}_{Y_d,y}$ forming a regular system of parameters. Let $\mathcal{U} = \operatorname{Spec} k[Y_1, Y_2]^H$, where H acts on Y_i through χ_i; it is the affine toric surface over k attached to the positive quadrant and the lattice generated by $\mathbb{Z}^2$ and the weights of H. The invariant monomials in y_1, y_2 lie in $\mathcal{O}_{Y_d,y}^H = \mathcal{O}_{W,w}$, and they define a morphism $\psi\colon V \to \mathcal{U}$ from an open neighbourhood V of w, with $\psi(w)$ the torus-fixed point 0. On completions, ψ induces the isomorphism $k[[Y_1, Y_2]]^H \to k[[y_1, y_2]]^H$. Hence $\mathcal{O}_{\mathcal{U},0} \to \mathcal{O}_{W,w}$ is flat, because $\widehat{\mathcal{O}}_{W,w}$ is faithfully flat over $\mathcal{O}_{W,w}$ and flat over $\mathcal{O}_{\mathcal{U},0}$, and $\mathfrak{m}_0\mathcal{O}_{W,w} = \mathfrak{m}_w$; also $\kappa(w) = \kappa(0) = k$. The fibre $\psi^{-1}(0)$ has local ring $\mathcal{O}_{W,w}/\mathfrak{m}_0\mathcal{O}_{W,w} = \kappa(w)$ at w, so w is an isolated point of it; after shrinking V, it is the only point of V over 0. The points w of this second kind are isolated points of fixed-point schemes of subgroups of G, hence there are finitely many of them.

Step 3: a smooth model. Let $\rho\colon \widetilde{\mathcal{U}} \to \mathcal{U}$ be the toric resolution [Ful93, Section 2.6]; the construction is defined over any field. It is proper and birational, it is an isomorphism over $\mathcal{U} \setminus \{0\}$, and the surface $\widetilde{\mathcal{U}}$ is covered by affine planes over k; in particular $\widetilde{\mathcal{U}}$ is smooth over k. Put $\widetilde{V} = V \times_{\mathcal{U}} \widetilde{\mathcal{U}}$. Then $\widetilde{V} \to V$ is proper and birational and an isomorphism over $V \setminus \{w\}$. Let $\tilde{v} \in \widetilde{V}$ lie over w, with image $\tilde{u} \in \widetilde{\mathcal{U}}$. The homomorphism $\mathcal{O}_{\widetilde{\mathcal{U}},\tilde{u}} \to \mathcal{O}_{\widetilde{V},\tilde{v}}$ is flat, with closed fibre a localization of $\mathcal{O}_{W,w} \otimes_{\mathcal{O}_{\mathcal{U},0}} \kappa(\tilde{u}) = \kappa(\tilde{u})$, a field. The same holds after base change to every finite purely inseparable extension k'/k. Since $\widetilde{\mathcal{U}}$ is smooth over k, [Sta26, Tags 031E and 038U] show that $\widetilde{V}$ is geometrically regular at $\tilde{v}$. Performing this construction at each of the finitely many points w of the second kind in Step 2 and gluing with the complement, we obtain a regular surface $\widetilde{W}$, proper over R with generic fibre X_K, which is geometrically regular at every point of

its closed fibre.

Step 4: contraction to Y. By minimality there is a morphism $\widetilde{W} \to Y$, and it is a sequence of blowups in closed points [Sta26, Tag 0C5R]: $\widetilde{W} = Z_0 \to Z_1 \to \cdots \to Z_n = Y$. Suppose that Z_i is geometrically regular at every point of its closed fibre, and let $Z_i \to Z_{i+1}$ be the blowup of the closed point z, with exceptional curve $E \cong \mathbb{P}^1_{\kappa(z)}$ and $q = [\kappa(z) : k]$. Then $(E^2)_k = -q$. Let $k' \supseteq \kappa(z)$ be a finite purely inseparable extension of k. The surface $(Z_i)_{k'}$ is regular along $E_{k'}$, and $E_{k'} = qE'$ as Cartier divisors, where $E' = (E_{k'})_{\mathrm{red}}$, because $\kappa(z) \otimes_k k'$ is local of length q with residue field k'. Intersection numbers are preserved by the flat base change to k', and $R \otimes_k k'$ is a discrete valuation ring because P is k-rational; so $-q = (E^2)_k = ((E_{k'})^2)_{k'} = q^2(E'^2)_{k'}$ with $(E'^2)_{k'} \in \mathbb{Z}$. Hence $q = 1$ and $\kappa(z) = k$. Since $\kappa(z) = k$, the surface Z_{i+1} is geometrically regular at z [Sta26, Tag 031E], and it is so at the other points of its closed fibre because $Z_{i+1} \setminus \{z\} \cong Z_i \setminus E$. By induction, $Y = X_{(P)}$ is geometrically regular along its closed fibre, that is, X is smooth over k along X_P. □

6.6 Conclusion

Proposition 6.13. *Assume $p \geq 13$, and let $P \in B$ be a closed point at which J_K has additive reduction. Then P is k-rational and X is smooth over k at every point of X_P.*

Proof. The point P is k-rational by Lemma 6.4, and $b_P \leq 1$ by Lemma 6.3. The multiplicities of X_P have greatest common divisor one by Lemma 6.2 (5), and hence so do those of Y_d for every d prime to p, by Lemma 6.10.

Suppose that we are in case (1) of Lemma 6.9, and take $d = 12/\gcd(12, \delta)$. The Jacobian has good reduction at P_d. By Lemma 6.11 and Lemma 6.9 (1),

$$0 \leq b(Y_d) = d\, b_P - w_X + w_J \leq d + (d-1) \leq 11 < p - 1.$$

The closed fibre of Y_d has greatest common divisor one, so Corollary 5.5 gives $b(Y_d) = 0$ and smoothness over R_d. Since B_d is smooth over k, Y_d is smooth over k along its closed fibre, and Lemma 6.12 applies.

Suppose that we are in case (2), and take $d = 2$. The Jacobian has split multiplicative reduction of type I_{2n} at P_2, so $a(Y_2) = b(Y_2)$ by Lemma 5.2(4). By Lemma 6.11 and Lemma 6.9(2), $b(Y_2) = 2b_P - w_X + 1 \leq 3$. By Lemma 5.2 (3), the minimal regular model $Y_2 \otimes_{R_2} \widehat{R}_2$ has the same closed fibre as Y_2, whose multiplicities have greatest common divisor one by the first paragraph, and $a(Y_2 \otimes_{R_2} \widehat{R}_2) = a(Y_2) \leq 3 < p - 1$. Since the residue field k of $\widehat{R}_2$ is separably closed, Lemma 6.7 (2) shows that the closed fibre of Y_2 is a cycle of $2n$ reduced copies of $\mathbb{P}^1_k$ meeting transversally in k-rational points. Every point of this fibre is either a smooth point of the fibre or a k-rational point, so Y_2 is smooth over k along its closed fibre [Sta26, Tag 031E]. Lemma 6.12 applies again. □

Proposition 6.14. *Under Assumption 6.1, assume $p \geq 13$. Then X is geometrically normal. More precisely, X is smooth over k except at the singular points of fibres X_P over closed points P which are not k-rational and at which J_K has multiplicative reduction; such points exist only if $p \leq 23$, and at them $X_{\bar{k}}$ has singularities of type A_{p-1}.*

Proof. Every closed point $P \in B$ is of good, multiplicative or additive reduction for J_K. The three cases are treated by Lemma 6.5, Lemma 6.8 together with Lemma 6.4, and Proposition 6.13. Since $X_{\bar{k}}$ is normal along every fibre, it is normal. □

7 The anisotropic case and geometric normality

Proposition 4.4 gives two alternatives: X admits a genus-one fibration, or $\rho(X) = 2$, the intersection form on $N^1(X)_{\mathbb{Q}}$ is anisotropic, and X contains a negative curve. In this section, we deal with the latter case. For $p \geq 13$, we first construct a contraction $X \to X'$ with $\rho(X') = 1$ and X' not geometrically normal (Subsection 7.1). We then derive a contradiction by studying the exceptional curves on the minimal resolution of the normalization of $X'_{\bar{k}}$ (Subsection 7.2).

7.1 Reduction to Picard number one

Throughout this subsection, Assumption 4.1 holds, and we use the notation S and ϕ from Section 4.

Lemma 7.1. *Let k be an F-finite separably closed field of characteristic $p \geq 7$, and let X be a regular projective geometrically integral surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$ which is not geometrically normal. Assume that $\rho(X) = 2$ and that the intersection form on $N^1(X)_{\mathbb{Q}}$ is anisotropic. Let $C \subset X$ be a negative curve with contraction $X \to X'$ as in Lemma 2.6, and assume that X' is geometrically normal. Then there is a negative curve $\Gamma \neq C$ on X whose contraction X_Γ is not geometrically normal.*

Proof. The morphism $X_{\bar{k}} \to X'_{\bar{k}}$ is an isomorphism away from $C_{\bar{k}}$, and $X'_{\bar{k}}$ is normal. Hence $X_{\bar{k}}$ is normal away from $C_{\bar{k}}$. Since $X_{\bar{k}}$ is Gorenstein, hence S_2, its non-normal locus is a nonempty union of curves, so it equals $C_{\bar{k}}$ as a set. Let $\overline{C}$ be the reduced curve $\phi^{-1}(C)$ in S; it is irreducible because ϕ is a homeomorphism. By Proposition 4.2(2), $D = \delta\overline{C}$ for an integer $\delta \geq 1$, so

$$K_S \equiv -(p-1)\delta\,\overline{C}.$$

The surface S is $\mathbb{Q}$-factorial with $\rho(S) = 2$, and ϕ^* is an isometry $N^1(X)_{\mathbb{Q}} \cong N^1(S)_{\mathbb{Q}}$ for the intersection pairings over k and $\bar{k}$, respectively; in particular the intersection form on $N^1(S)_{\mathbb{Q}}$ is anisotropic. As ϕ^*C is a positive multiple of $\overline{C}$, we have $\overline{C}^2 < 0$.

Let $V = N^1(S)_{\mathbb{R}}$, a plane, and let $\overline{\mathrm{NE}}(S) \subset V$ be the closed cone of curves. It contains an ample class in its interior and contains no line. Every curve other than $\overline{C}$ has nonnegative intersection with $\overline{C}$, while $\overline{C}^2 < 0$; hence $\mathbb{R}_{\geq 0}[\overline{C}]$ is one of the two boundary rays of $\overline{\mathrm{NE}}(S)$. Let R_2 be the other boundary ray, and let r_2 span it. Write an ample class as $h = \alpha[\overline{C}] + \beta r_2$ with $\alpha, \beta > 0$. Then $0 < h \cdot \overline{C} = \alpha\overline{C}^2 + \beta\, r_2 \cdot \overline{C}$, so $r_2 \cdot \overline{C} > 0$ and

$$K_S \cdot r_2 = -(p-1)\delta\,\overline{C} \cdot r_2 < 0.$$

Since S is a normal projective surface over $\bar{k}$ with K_S $\mathbb{Q}$-Cartier, the cone theorem [Tan14, Theorem 3.13 and Proposition 3.15] shows that R_2 is spanned by the class of a rational curve Γ_S. Its class is rational and nonzero, so $\Gamma_S^2 \neq 0$ by anisotropy. If $\Gamma_S^2 > 0$, then $[\Gamma_S]$ would lie in the open positive cone, which is contained in the interior of $\overline{\mathrm{NE}}(S)$; this contradicts $[\Gamma_S] \in R_2$. Hence $\Gamma_S^2 < 0$, and $\Gamma_S \neq \overline{C}$ because $R_2 \neq \mathbb{R}_{\geq 0}[\overline{C}]$.

Let $\Gamma = \phi(\Gamma_S)$, a curve on X different from C. As $\phi^*\Gamma$ is a positive multiple of Γ_S, we get $\Gamma^2 < 0$. Let $X \to X_\Gamma$ be the contraction of Γ given by Lemma 2.6. It is an isomorphism over the complement of the image of Γ, and this complement contains the image of $C \setminus \Gamma$, which is nonempty. Since $X_{\bar{k}}$ is not normal at any point of $C_{\bar{k}}$, the surface $(X_\Gamma)_{\bar{k}}$ is not normal. □

Corollary 7.2. *Let k be an F-finite separably closed field of characteristic $p \geq 13$, and let X be a regular projective geometrically integral surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$ which is not geometrically normal. Assume that $\rho(X) = 2$ and that the intersection form on $N^1(X)_{\mathbb{Q}}$ is anisotropic. Then X contains a negative curve whose contraction is not geometrically normal.*

Proof. By Proposition 4.4(2), there is a negative curve C. If its contraction is not geometrically normal, we are done; otherwise Lemma 7.1 gives the required curve. □

7.2 Exclusion of the contracted surface

Throughout this subsection, k is an F-finite separably closed field of characteristic $p \geq 7$, and X is a regular projective geometrically integral surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$ which is not geometrically normal; we assume moreover that $\rho(X) = 2$, that the intersection form on $N^1(X)_{\mathbb{Q}}$ is anisotropic, and that $\Gamma \subset X$ is a negative curve whose contraction $h\colon X \to X'$ of Lemma 2.6 is not geometrically normal. By Corollary 7.2, such a curve exists if $p \geq 13$. Let $P = h(\Gamma)$. We write $\nu'\colon S' \to X'_{\bar{k}}$ for the normalization and $\phi'\colon S' \to X'$ for its composite with the projection. Let $s_0 \in S'$ be the unique point over P, and let $\mu\colon T' \to S'$ be the minimal resolution. Let Y' be the normalization of $X'_{k^{1/p}}$ and $C'_1 = C_{k^{1/p}}$ the divisor obtained by applying (2.1) to X', and write the divisorial conductor of $S' \to X'_{\bar{k}}$ as $(p-1)D'$.

Lemma 7.3. (1) *The surface X' is normal, projective and geometrically integral, with $H^0(X', \mathcal{O}_{X'}) = k$, $\rho(X') = 1$ and $\omega_{X'} \cong \mathcal{O}_{X'}$; it is a local complete intersection, regular away from P, and $2D$ is Cartier for every Weil divisor D on X'.*

(2) *$C'_1 \neq 0$ and $K_{Y'} \sim -(p-1)C'_1$, and there is an ample Cartier divisor D'_1 on X' whose pullback to Y' is $2pC'_1$.*

(3) *S' is a rational surface with rational singularities; it is $\mathbb{Q}$-factorial with $\rho(S') = 1$; D' is a nonzero effective $\mathbb{Z}$-divisor with $-K_{S'} \sim (p-1)D'$; and $-K_{S'}$ is ample.*

(4) *Let H be the pullback of D'_1 to T' and $H_{S'}$ its pullback to S', and define $\lambda \in \mathbb{Q}$ by $-K_{S'} \equiv \lambda H_{S'}$. Then H is nef and big, $H \cdot E = 0$ for every μ-exceptional curve E, H is positive on every other curve,*

$$H^2 \in 2p\mathbb{Z}_{>0}, \qquad \lambda \geq r := \frac{p-1}{2p},$$

and

$$-K_{T'} \equiv \lambda H + Z \tag{7.1}$$

with Z an effective μ-exceptional $\mathbb{Q}$-divisor.

(5) *The point s_0 is a singular point of S'. The exceptional curves of μ are smooth rational curves meeting transversally, and the absolute value of the determinant of their intersection matrix is $2p^e$ for some $e \geq 0$. Moreover $\pi_1^{\mathrm{et}}(U^\circ_{s_0}) \cong \mathbb{Z}/2$, and the connected double cover of $U^\circ_{s_0}$ is simply connected.*

Proof. (1) This follows from Lemma 2.6. Since $h^*\omega_{X'} \cong \omega_X \cong \mathcal{O}_X$ and $h_*\mathcal{O}_X = \mathcal{O}_{X'}$, the projection formula gives $\omega_{X'} \cong \mathcal{O}_{X'}$. The surface $X'_{\bar{k}}$ is the image of the integral surface $X_{\bar{k}}$, hence irreducible, and it is Cohen–Macaulay and generically reduced, hence integral.

(2) By [Tan21, Proposition 2.10(3)], $X'_{k^{1/p}}$ is not normal, so $C'_1 \neq 0$. Equation (2.1) gives $K_{Y'} \sim -(p-1)C'_1$. The ramification indices of $Y' \to X'$ likewise divide p, so pC'_1 descends to an effective nonzero Weil divisor on X'. By Lemma 2.6(4), its double is a Cartier divisor D'_1 whose pullback to Y' is $2pC'_1$. As $\rho(X') = 1$, D'_1 is numerically a positive multiple of an ample divisor, hence ample by the Nakai–Moishezon criterion.

(3) The surface S' is birational to the normalization S of $X_{\bar{k}}$, which is rational by Proposition 4.2(3). By [Tan18a, Proposition 2.4 and Lemma 2.5], the surface S' is $\mathbb{Q}$-factorial and $\rho(S') = \rho(X') = 1$. As in the proof of Proposition 4.2(2), applying (2.1) after a finite extension shows that D' is a nonzero effective $\mathbb{Z}$-divisor and that $-K_{S'} \sim (p-1)D'$. Since D' is nonzero and effective, $-K_{S'}$ is numerically a positive multiple of an ample divisor, hence ample. The argument of Proposition 4.2(3) shows that S' has rational singularities.

(4) The morphism $T' \to X'_{\bar{k}}$ is birational and contracts exactly the μ-exceptional curves, since $S' \to X'_{\bar{k}}$ is finite; this gives the first assertions. Intersection numbers of Cartier divisors are preserved by the pullbacks $T' \to X'$ and $Y' \to X'$, so

$$(H^2)_{\bar{k}} = ((D'_1)^2)_k = ((2pC'_1)^2)_{k^{1/p}} = 2p\left((2pC'_1)\cdot C'_1\right)_{k^{1/p}},$$

where $\left((2pC'_1)\cdot C'_1\right)_{k^{1/p}}$ is a positive integer, as in the proof of Proposition 4.3. Comparing conductor ideals as in the proof of Proposition 4.3 gives $D' \geq \mu'^*_1 C'_1$, where $\mu'_1 \colon S' \to Y'$ is the natural morphism, and $H_{S'} = 2p\,\mu'^*_1 C'_1$. Hence

$$\lambda = \frac{(p-1)\,D' \cdot H_{S'}}{H^2} \geq \frac{(p-1)\,\mu'^*_1 C'_1 \cdot H_{S'}}{H^2} = \frac{p-1}{2p}.$$

By [Kol13, Theorem 10.5 and Claim 10.3.5], $K_{T'} = \mu^* K_{S'} - Z$ with $Z \geq 0$ exceptional, which gives (7.1).

(5) By Lemmas 3.1 and 3.2, $\pi_1^{\mathrm{et}}(U^\circ_s) = 1$ for $s \neq s_0$, while $\pi_1^{\mathrm{et}}(U^\circ_{s_0}) \cong \mathbb{Z}/2$ and the connected double cover of $U^\circ_{s_0}$ is simply connected. If s_0 were a regular point, $U^\circ_{s_0}$ would be simply connected by purity [Gro71, Exp. X, Corollaire 3.3]; so s_0 is singular. The remaining assertions follow from Corollary 3.4, because the intersection matrix is the direct sum of the matrices at the singular points. □

Lemma 7.4. *Assume $p \geq 13$. Let $\varphi \colon T' \to \mathbb{P}^1$ be a morphism whose general fibre F is isomorphic to $\mathbb{P}^1$, and put $a = H \cdot F$. Then exactly one μ-exceptional curve A is not contained in a fibre of φ, and it satisfies $A \cdot F = 1$. Moreover $a \leq 2$, unless either $p \in \{13, 17\}$, $a = 3$ and $H^2 = 2p$, or $p = 13$, $a = 4$, $H^2 = 26$ and $A^2 = -2$.*

Proof. Since H is nef and big and F is a nonzero nef class with $F^2 = 0$, the Hodge index theorem gives $a \geq 1$. Intersecting (7.1) with F gives

$$2 = -K_{T'} \cdot F = \lambda a + Z \cdot F. \tag{7.2}$$

As $Z \cdot F \geq 0$ and $\lambda \geq r$, we get $a \leq 2/r = 4p/(p-1) < 5$, so $a \leq 4$.

Call a μ-exceptional curve horizontal if it is not contained in a fibre. At least one horizontal curve exists: otherwise a general fibre F is disjoint from the exceptional locus, so its image in S' has self-intersection $F^2 = 0$, whereas every curve on S' has positive self-intersection since $\rho(S') = 1$. Let $A_1, \ldots, A_m$ be the horizontal curves, and put $d_i = A_i \cdot F \geq 1$ and $w_i = -A_i^2 \geq 2$. Let β_i be the coefficient of A_i in Z; since the vertical exceptional curves do not meet F, we have $Z \cdot F = \sum_i \beta_i d_i$.

Lower bound for β_i. The curve A_i is a smooth rational curve by Lemma 7.3(5), and $H \cdot A_i = 0$, so (7.1) and adjunction give $Z \cdot A_i = -K_{T'} \cdot A_i = 2 - w_i$. On the other hand $Z \cdot A_i \geq -\beta_i w_i$, because $Z - \beta_i A_i$ is effective and does not contain A_i. Hence

$$\beta_i \geq 1 - \frac{2}{w_i}.$$

Lower bound for w_i. The class $w_i F + d_i A_i$ has square $2w_i d_i^2 - w_i d_i^2 = w_i d_i^2 > 0$ and satisfies $H \cdot (w_i F + d_i A_i) = w_i a$. By the Hodge index theorem, $w_i^2 a^2 \geq H^2 \cdot w_i d_i^2$, that is, $w_i \geq H^2 d_i^2 / a^2$. Since w_i is an integer and $H^2 \geq 2p$,

$$w_i \geq \left\lceil \frac{2p\, d_i^2}{a^2} \right\rceil.$$

Case analysis. If $a = 4$, then $Z \cdot F = 2 - 4\lambda \leq 2 - 4r = 2/p < 1/3$, so $\beta_i < 1/3$ and hence $w_i = 2$ for all i. As $w_i \geq \lceil p\, d_i^2/8 \rceil$, this forces $d_i = 1$ and $p = 13$, and the inequality $w_i \geq H^2 d_i^2/a^2$ gives $H^2 \leq 32$, so $H^2 = 26$. If there were two horizontal curves $A_1 \neq A_2$, the class $D = 2F + A_1 + A_2$ would satisfy $D^2 = 4 + 2A_1 \cdot A_2 \geq 4$ and $H \cdot D = 8$, and the Hodge index theorem would give $64 = (H \cdot D)^2 \geq H^2 D^2 \geq 104$, which is impossible. If $a = 3$ and $p \geq 19$, then $w_i \geq \lceil 2p/9 \rceil \geq 5$, so $\beta_i \geq 3/5$, while $Z \cdot F \leq 2 - 3r = (p+3)/(2p) < 3/5$ since $p > 15$; this is impossible. If $a = 3$ and $p \in \{13, 17\}$, then $w_i \geq \lceil 2p/9 \rceil \geq 3$, so $\beta_i \geq 1/3$ and $\frac{1}{3} \sum_i d_i \leq Z \cdot F \leq 2 - 3r = (p+3)/(2p) < 2/3$; hence $\sum_i d_i = 1$, that is, there is exactly one horizontal curve A_1, and $d_1 = 1$. If moreover $H^2 \geq 4p$, then $w_1 \geq \lceil 4p/9 \rceil \geq 6$, so $\beta_1 \geq 2/3 > Z \cdot F$, which is impossible; hence $H^2 = 2p$. If $a = 2$, then $w_i \geq \lceil p/2 \rceil \geq 7$, so $\beta_i \geq 5/7$ and $\frac{5}{7} \sum_i d_i \leq Z \cdot F \leq 2 - 2r = 1 + 1/p < 10/7$; hence $\sum_i d_i = 1$. If $a = 1$, then $w_i \geq 2p$, so $\beta_i \geq 1 - 1/p$ and $(1 - 1/p) \sum_i d_i \leq 2 - r = (3p+1)/(2p) < 2(1 - 1/p)$; hence $\sum_i d_i = 1$. In all remaining cases there is exactly one horizontal curve, with $d = 1$, and either $a \leq 2$, or $p \in \{13, 17\}$, $a = 3$ and $H^2 = 2p$, or $p = 13$, $a = 4$, $H^2 = 26$ and $A^2 = -2$. □

Proposition 7.5. *Assume $p \geq 7$. Then T' admits no morphism $\varphi\colon T' \to \mathbb{P}^1$ with general fibre $\mathbb{P}^1$ for which the conclusion of Lemma 7.4 holds. In particular, the situation of this subsection does not occur if $p \geq 13$.*

Proof. Suppose that φ is such a morphism, with general fibre F. Let A be the horizontal exceptional curve, a section of φ, and put $b = -A^2 \geq 2$ and $a = H \cdot F$.

Step 1: shape of the fibres. Since S' is $\mathbb{Q}$-factorial with $\rho(S') = 1$, $\rho(T')$ is one plus the number of μ-exceptional curves, that is, $2 + v$, where v is the number of vertical exceptional curves. On the other hand $\rho(T') = 2 + \sum_{F_0} (n_{F_0} - 1)$, the sum over the fibres F_0 with n_{F_0} components. No fibre consists of exceptional curves only, since $H \cdot F = a > 0$. Hence $n_{F_0} - 1$ is at least the number of exceptional curves in F_0, and comparing the two formulas shows that every fibre has exactly one component which is not exceptional. Let $F_0 = mG + \sum_j m_j E_j$ be a reducible fibre, with G the non-exceptional component. A reducible fibre of a ruled surface contains a (-1)-curve, and the E_j have self-intersection at most -2; so G is a (-1)-curve. Since $H \cdot G$ is a positive integer, $a = H \cdot F_0 = m\, H \cdot G$ is divisible by m. Adjunction gives

$$-2 = K_{T'} \cdot F_0 = -m + \sum_j m_j(-E_j^2 - 2),$$

so $m \geq 2$. Since $A \cdot F_0 = 1$, the section A meets exactly one component of F_0, transversally, and this component has multiplicity one; as $m \geq 2$, it is one of the E_j.

Until Step 7 we assume $a \leq 2$. As $m \geq 2$ divides $a \leq 2$, we get $m = a = 2$, and then $E_j^2 = -2$ for all j. The equations $F_0 \cdot G = 0$ and $F_0 \cdot E_j = 0$ say that the multiplicities of the neighbours of G add up to 2 and that twice the multiplicity of each E_j is the sum of the multiplicities of its neighbours. It follows that either G meets two curves u, v of multiplicity one, each meeting nothing else, or G is the end of a chain $c_1, \ldots, c_l$ of curves of multiplicity two, with c_1 meeting G and c_l meeting two curves u, v of multiplicity one, which meet nothing else. Removing G leaves the configuration $A_1 \sqcup A_1$ or D_n with $n = l + 2 \geq 3$, where $D_3 = A_3$.

The components of multiplicity one are u and v, so by the above A meets exactly one of them, and no other component of F_0; we label them so that A meets u.

Step 2: at most one reducible fibre. Let t be the number of reducible fibres, and let $\Phi_1, \dots, \Phi_t$ be the corresponding configurations of type $A_1 \sqcup A_1$ or D_{n_i}. Each has determinant 4, and the diagonal entry of the inverse of its (positive definite) negated intersection matrix at the curve met by A is $q_i = 1/2$ for $A_1 \sqcup A_1$ and $q_i = n_i/4$ for D_{n_i}. Taking the Schur complement at A, the absolute value of the determinant of all exceptional curves is

$$4^t\Big(b - \sum_i q_i\Big) = 4^{t-1}\Big(4b - \sum_i 4q_i\Big), \qquad 4q_i \in \mathbb{Z}.$$

By Lemma 7.3(5) it equals $2p^e$ with p odd, so $t \le 1$.

Step 3: the numerical invariants. Suppose $t = 0$. Then φ is a $\mathbb{P}^1$-bundle, $T' \cong \mathbb{F}_b$, and A is the negative section and the only exceptional curve; so $b = 2p^e$. Suppose $t = 1$, with reducible fibre F_0. Contracting G, then successively $c_1, \dots, c_l$ (each becomes a (-1)-curve after the previous contraction), and finally v, turns F_0 into a smooth fibre, and none of these curves meets A. Hence T' is obtained from $\mathbb{F}_b$, in which A is the negative section, by blowing up n points on a fibre f_0 ($n = 2$ for $A_1 \sqcup A_1$): first a point $x_1 \in f_0 \setminus A$, then the intersection point of the strict transform of f_0 with the first exceptional curve, and then each time a general point of the last exceptional curve. Let E_i^{tot} be the total transform of the i-th exceptional curve and $\pi\colon T' \to \mathbb{F}_b$ the blowdown, and write $H = \pi^*(\alpha A + \beta f) - \sum_i c_i E_i^{\mathrm{tot}}$. Then $H \cdot A = 0$ and $H \cdot F = a$ give $H = a\,\pi^*(A + bf) - \sum_i c_i E_i^{\mathrm{tot}}$. The curves $u = \pi^* f_0 - E_1^{\mathrm{tot}} - E_2^{\mathrm{tot}}$, $v = E_1^{\mathrm{tot}} - E_2^{\mathrm{tot}}$ and $E_i^{\mathrm{tot}} - E_{i+1}^{\mathrm{tot}}$ for $2 \le i < n$ are exceptional, so H is orthogonal to them; hence $c_1 = c_2 = \cdots = c_n = a/2 = 1$. Using $-K_{T'} = \pi^*(2A + (b+2)f) - \sum_i E_i^{\mathrm{tot}}$ we obtain

$$H^2 = 4b - n, \qquad -K_{T'} \cdot H = 2b + 4 - n. \tag{7.3}$$

In both cases, $-K_{T'} \cdot H = -K_{S'} \cdot H_{S'} = (p-1)\, D' \cdot H_{S'}$ with $D' \cdot H_{S'} \in \mathbb{Z}$, because D' is a $\mathbb{Z}$-divisor and $H_{S'}$ is Cartier. So $-K_{T'} \cdot H$ is divisible by $p - 1$, and it is positive because $-K_{S'}$ is ample.

Step 4: $t = 0$. Here $H \equiv a(A + bf)$ and $-K_{T'} \cdot H = a(b+2) = a(2p^e + 2) \equiv 4a \pmod{p-1}$. So $(p-1) \mid 4a$ with $a \le 2$, which is impossible for $p \ge 7$.

Step 5: $A_1 \sqcup A_1$. The exceptional curves form the chain A, u, with determinant $2b - 1$, and the isolated curve v, with determinant 2; so $2(2b-1) = 2p^e$. By (7.3), $-K_{T'} \cdot H = 2b + 2 = p^e + 3 \equiv 4 \pmod{p-1}$, which is impossible for $p \ge 7$.

Step 6: D_n. Label the configuration as follows: u and v are the two tips of multiplicity one, $c = c_l$ is the curve meeting them, and $w_j = c_{l-j}$ for $1 \le j \le n-3$ form the long arm, so that $w_{n-3} = c_1$. All exceptional curves form one connected configuration, of determinant $4b - n = 2p^e$; so they lie over one singular point, which is s_0 by Lemma 7.3(5), and n is even, hence $n \ge 4$. Let $T_0 = T' \times_{S'} \operatorname{Spec} \mathcal{O}^{\mathrm{sh}}_{S',s_0}$, and let $\mathcal{B} = v + w_1 + w_3 + \cdots + w_{n-3}$. These components are disjoint, every component of $\mathcal{B}$ has even self-intersection, and every other exceptional curve meets $\mathcal{B}$ in an even number of points: c meets v and w_1, each w_{2j} meets w_{2j-1} and w_{2j+1}, and A and u meet no component of $\mathcal{B}$. Hence $\mathcal{O}_{T_0}(\mathcal{B})$ has even degree on every exceptional curve, so Lemma 3.3(1) gives $\mathcal{O}_{T_0}(\mathcal{B}) \cong N^{\otimes 2}$ for a line bundle N. The double cover $\varpi\colon V_0 = \mathbf{Spec}(\mathcal{O}_{T_0} \oplus N^{-1}) \to T_0$ defined by the section of $N^{\otimes 2}$ vanishing on $\mathcal{B}$ is branched exactly along the smooth divisor $\mathcal{B}$, so V_0 is regular. Over $U^\circ_{s_0}$ it is the μ_2-torsor attached to the nonzero class $\mathcal{B}$ in Lemma 3.3(1), hence the connected double cover. Let $B_0 = H^0(V_0, \mathcal{O}_{V_0})$; it is a strictly henselian normal local ring, and $V_0 \to \operatorname{Spec} B_0$ is a resolution whose punctured spectrum is the connected double cover of $U^\circ_{s_0}$.

The exceptional curves of V_0 are as follows. Each component of $\mathcal{B}$ lifts to a smooth rational curve R with $2R = \varpi^*(\text{component})$, so $R^2 = -1$. The curves A and u meet no branch curve, so each splits into two disjoint copies, A', A'' and u', u'', with $A'^2 = A''^2 = -b$ and $u'^2 = u''^2 = -2$. The curve c and every w_{2j} meet $\mathcal{B}$ in two points, so their preimages $\tilde{c}$ and $\tilde{w}_{2j}$ are smooth rational curves of self-intersection -4. Contracting the (-1)-curves R raises the self-intersections of $\tilde{c}$ and of each $\tilde{w}_{2j}$ to -2, since each of these meets exactly two of the curves R. The resulting minimal configuration is a tree of smooth rational curves meeting transversally: a central (-2)-curve $\tilde{c}$ with two arms u', A' and u'', A'' of self-intersections $(-2, -b)$, and a third arm of $(n-4)/2$ curves $\tilde{w}_2, \tilde{w}_4, \dots$ of self-intersection -2. Put $Q = 2b-1$ and $q = (n-2)/2$. Each short arm has determinant Q and inverse entry b/Q at its end next to $\tilde{c}$, and the third arm has determinant q and inverse entry $(q-1)/q$ there. Hence the absolute value of the determinant is

$$Q^2 q\Big(2 - \frac{2b}{Q} - \frac{q-1}{q}\Big) = Q(Q-q) = (2b-1)\,\frac{4b-n}{2} = (2b-1)\,p^e.$$

The punctured spectrum of B_0 is simply connected by Lemma 7.3(5), so Lemma 3.3(2) shows that $(2b-1)p^e$ is a power of p. Write $2b-1=p^f$. Then $n=2(Q-p^e)+2=2p^f-2p^e+2\geq 4$, so $f>e$. By (7.3),

$$-K_{T'}\cdot H=2b+4-n=2p^e-p^f+3\leq (2-p)p^e+3\leq 5-p<0,$$

contradicting the positivity in Step 3.

Step 7: $a\geq 3$. By Lemma 7.4, either $p\in\{13,17\}$, $a=3$ and $H^2=2p$, or $p=13$, $a=4$, $H^2=26$ and $b=2$. As at the end of Step 3, $-K_{T'}\cdot H$ is a positive multiple of $p-1$. By (7.1) and (7.2), $-K_{T'}\cdot H=\lambda H^2$ with $\lambda\leq 2/a\leq 2/3$, so $-K_{T'}\cdot H\leq 4p/3<2(p-1)$; hence $-K_{T'}\cdot H=p-1$.

We claim that every reducible fibre of φ contains a (-1)-curve which does not meet A. More generally, let F_1 be a reducible fibre of a morphism with general fibre $\mathbb{P}^1$ from a smooth projective surface to $\mathbb{P}^1$, and let A_1 be a section; we show by induction on the number of components of F_1 that F_1 contains a (-1)-curve not meeting A_1. The section A_1 meets exactly one component u_1 of F_1, and u_1 has multiplicity one. Every (-1)-curve in F_1 other than u_1 is disjoint from A_1, and F_1 contains a (-1)-curve; so we may suppose that u_1 is the only one. Then $F_1\cdot u_1=0$ shows that u_1 meets exactly one other component C, transversally, and that C has multiplicity one. Let σ be the contraction of u_1. If $\sigma(F_1)$ is irreducible, then it is a smooth fibre $\sigma(C)$, so $C^2=\sigma(C)^2-1=-1$, a contradiction. Otherwise, by induction, $\sigma(F_1)$ contains a (-1)-curve not meeting $\sigma(A_1)$; it does not pass through the point $\sigma(u_1)\in\sigma(A_1)$, so its strict transform is a (-1)-curve in F_1 other than u_1, again a contradiction.

Contracting such curves successively, we obtain a birational morphism from T' to a $\mathbb{P}^1$-bundle over $\mathbb{P}^1$ which is an isomorphism near A. As $A^2=-b$, this bundle is $\mathbb{F}_b$ with negative section A, and the morphism $\pi\colon T'\to\mathbb{F}_b$ is a composite of N blowups at points not on the images of A. Let E_i^{tot} be the total transform of the i-th exceptional curve, and put $c_i=H\cdot E_i^{\mathrm{tot}}\in\mathbb{Z}$. As in Step 3, $H=a\,\pi^*(A+bf)-\sum_i c_iE_i^{\mathrm{tot}}$ and $-K_{T'}=\pi^*(2A+(b+2)f)-\sum_i E_i^{\mathrm{tot}}$, so

$$H^2=a^2b-\sum_i c_i^2\quad\text{and}\quad -K_{T'}\cdot H=a(b+2)-\sum_i c_i. \tag{7.4}$$

Fix i, let $F_0=mG+\sum_j m_jE_j$ be the fibre of φ containing E_i^{tot}, as in Step 1, and let e_i be the coefficient of G in E_i^{tot}. Since H vanishes on the E_j and $H\cdot G=a/m$, we have $c_i=e_ia/m$. Let the i'-th centre be the last one lying over the i-th centre, possibly $i'=i$. No later centre lies on the i'-th exceptional curve, so its strict transform is a (-1)-curve on T' contained in E_i^{tot}; it is not μ-exceptional, as μ is minimal, so it is G, and $e_i\geq 1$. Moreover $F_0=\pi^*f_0$ for a fibre f_0 of $\mathbb{F}_b$, and the i-th centre lies on the total transform of f_0 on the surface being blown up; hence $F_0-E_i^{\mathrm{tot}}$ is effective. If $e_i=m$, this divisor would be supported on the E_j, on which $K_{T'}$ is nonnegative by adjunction, whereas $K_{T'}\cdot(F_0-E_i^{\mathrm{tot}})=-2+1=-1$. Hence

$$1\leq e_i\leq m-1.$$

Finally, the exceptional curve of a blowup at a point of a fibre has as multiplicity the sum of the multiplicities of the components of the fibre through that point, and there are at most two of them, as the reduced fibres are trees of smooth rational curves meeting transversally. Hence the first three exceptional curves over a fibre f_0 of $\mathbb{F}_b$ have multiplicity at most 1, 2 and 3, respectively; so if at most $\ell\in\{1,2,3\}$ centres lie over f_0, then the components of π^*f_0 have multiplicity at most ℓ.

Suppose that $a=4$, so that $p=13$ and $b=2$. Then (7.4) gives $\sum_i c_i=4$ and $\sum_i c_i^2=6$. As the c_i are positive integers, $\sum_i c_i(c_i-1)=2$ shows that one c_i equals 2 and the others equal 1, so $N=3$. Choose i with $c_i=1$; then $m=4$ for the fibre containing E_i^{tot}, since e_i and a/m are positive integers with product 1. But at most three centres lie over the image of this fibre in $\mathbb{F}_b$, so its components have multiplicity at most 3, a contradiction.

Suppose that $a=3$. Since $m\geq 2$ divides a by Step 1, $m=3$ for every reducible fibre; hence $c_i=e_i\in\{1,2\}$ and $c_i^2=3c_i-2$ for all i. Now (7.4) gives

$$p-3=3(-K_{T'}\cdot H)-H^2=18-3\sum_i c_i+\sum_i c_i^2=18-2N,$$

so $N=(21-p)/2$. Every reducible fibre contains the component G of multiplicity 3, so at least three centres lie over its image in $\mathbb{F}_b$. If $p=17$, then $N=2$, so there is no reducible fibre and $N=0$, a contradiction. If

$p = 13$, then $N = 4$, and all centres lie over the image f_0 of a single reducible fibre $F_0 = 3G + \sum_j m_j E_j$ with four components E_j. By the adjunction formula of Step 1, $\sum_j m_j(-E_j^2 - 2) = 1$; so one of the E_j, which we call x, has multiplicity one and $x^2 = -3$, and the others are (-2)-curves. The equations $F_0 \cdot C = 0$ say that the multiplicities of the neighbours of a component C add up to $-C^2$ times the multiplicity of C. In particular, a component of multiplicity one which meets a component of multiplicity three is x, and it meets nothing else. Put $y_0 = G$. If $y_0, \dots, y_l$ is a chain of components of multiplicity three, each meeting the next, then the neighbours of y_l other than y_{l-1} (if $l \geq 1$) have multiplicities adding up to 3, as $G^2 = -1$ and $y_l^2 = -2$ for $l \geq 1$, and they are not three curves of multiplicity one; so they are either a single curve y_{l+1} of multiplicity three, or x and a curve z of multiplicity two. In the second case, as z is a (-2)-curve, it has exactly one further neighbour t, of multiplicity one; then $t \neq x$ is a (-2)-curve, so it meets nothing else. Since F_0 has finitely many components, the second case occurs for some l, and then $y_0, \dots, y_l, x, z, t$ are all the components of F_0, as F_0 is connected. As there are five of them, $l = 1$ and $F_0 = 3G + 3y_1 + x + 2z + t$, where y_1 meets G, x and z, and z meets t. The curve A meets x or t, the components of multiplicity one. Contracting G, y_1, z, and then the one of x and t which does not meet A, turns F_0 into a smooth fibre; each of these curves is a (-1)-curve when it is contracted, and none of them meets A, so we may assume that π is the composite of these contractions. If A meets t, then $t = \pi^* f_0 - E_1^{\mathrm{tot}} - E_2^{\mathrm{tot}}$, $x = E_1^{\mathrm{tot}} - E_2^{\mathrm{tot}} - E_3^{\mathrm{tot}}$, $z = E_2^{\mathrm{tot}} - E_3^{\mathrm{tot}}$, $y_1 = E_3^{\mathrm{tot}} - E_4^{\mathrm{tot}}$ and $G = E_4^{\mathrm{tot}}$, so $H \cdot G = 1$ and $H \cdot x = H \cdot z = H \cdot y_1 = 0$ give $(c_i) = (2, 1, 1, 1)$. If A meets x, then $x = \pi^* f_0 - E_1^{\mathrm{tot}} - E_2^{\mathrm{tot}} - E_3^{\mathrm{tot}}$, $t = E_1^{\mathrm{tot}} - E_2^{\mathrm{tot}}$, $z = E_2^{\mathrm{tot}} - E_3^{\mathrm{tot}}$, $y_1 = E_3^{\mathrm{tot}} - E_4^{\mathrm{tot}}$ and $G = E_4^{\mathrm{tot}}$, which gives $(c_i) = (1, 1, 1, 1)$. Hence $\sum_i c_i^2 \in \{4, 7\}$, whereas $\sum_i c_i^2 = 9b - 26 \equiv 1 \pmod 9$ by (7.4); this is impossible.

Finally, if $p \geq 13$, then $T' \ncong \mathbb{P}^2$, because $\mathbb{P}^2$ contains no curve of negative self-intersection while μ has exceptional curves by Lemma 7.3(5). A smooth projective rational surface other than $\mathbb{P}^2$ admits a birational morphism to a Hirzebruch surface, hence a morphism to $\mathbb{P}^1$ with general fibre $\mathbb{P}^1$, and Lemma 7.4 applies to it. □

7.3 Geometric normality

Theorem 7.6. *Let k be a field of characteristic $p \geq 13$, and let X be a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. Then X is geometrically normal over k.*

Proof. We may assume that k is F-finite and separably closed. Suppose that X is not geometrically normal. Since $p \geq 13$, the surface X is geometrically integral by [WZ26, Theorem 1.2], so Assumption 4.1 holds. By Proposition 4.4, either there is a fibration $f\colon X \to B$ onto a regular projective curve whose generic fibre is a smooth curve of genus one, or $\rho(X) = 2$ and the intersection form on $N^1(X)_{\mathbb{Q}}$ is anisotropic.

In the first case, Assumption 6.1 holds by Proposition 4.2, and Proposition 6.14 shows that X is geometrically normal, a contradiction. In the second case, Corollary 7.2 gives a negative curve $\Gamma \subset X$ whose contraction is not geometrically normal, so the situation of Section 7.2 occurs; this contradicts Proposition 7.5. Hence X is geometrically normal. □

Remark 7.7. The hypothesis $p \geq 13$ is used in Proposition 6.14 and Lemma 7.4, and the rank-two case does not occur when $p \equiv 3 \pmod 4$ (Proposition 4.4(2)). For the primes $p \leq 11$ the argument gives the following information.

(1) For $p = 5$ and $p = 7$ the conclusion of Theorem 1.1 fails: [WZ26, Example 6.2 and Remark 1.5(2)] constructs, over the function field of a curve over an algebraically closed field, regular projective geometrically integral surfaces X with $\omega_X \cong \mathcal{O}_X$ and $H^0(X, \mathcal{O}_X)$ equal to the base field which are not geometrically normal.

(2) Let $p = 11$. Since $11 \equiv 3 \pmod 4$, a counterexample would admit a genus-one fibration as in Assumption 6.1. The arguments of Section 6 then apply, except at the following closed points $P \in B$ of additive reduction for J_K: (a) points with $[k(P) : k] = 11$, where necessarily $v_P(\Delta_P) = 2$ and the reduction is of type II; (b) k-rational points in case (1) of Lemma 6.9 with $b(Y_d) = 10$, which forces $\delta = 10$ (type II*), $d = 6$, $b_P = 1$ and $w_X = 1$ in Lemma 6.11.

Configuration (b) occurs, by the example of Section 10 (see Remark 10.9); we do not know whether configuration (a) occurs.

8 Smoothness for $p \geq 23$

We now prove that the surfaces of Theorem 1.1 are smooth over k for $p \geq 23$. If X admits a genus-one fibration, Proposition 6.14 shows that X is smooth except possibly at nodes of multiplicative fibres over points which are not k-rational, which occur only for $p \leq 23$, and at which $X_{\bar{k}}$ has singularities of type A_{p-1}. In general, we study the minimal resolution of the normal surface $X_{\bar{k}}$. By Lemma 3.5, its rational double points have type A_{p^e-1} with $e \geq 1$. We restrict the remaining exceptional configurations using the Picard lattice of X, a bound for the determinant of the exceptional lattice, and an analysis of the tangent cone of the non-rational singular point.

Throughout this section, k is an F-finite separably closed field of characteristic $p \geq 13$, and X is a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. Put $V = X_{\bar{k}}$ and write $\phi\colon V \to X$ for the projection. By Theorem 1.1, the surface V is normal. It is Gorenstein, because X is regular and ω_V is the pullback of ω_X; so K_V is Cartier and $K_V \equiv 0$. It is also a local complete intersection over $\bar{k}$: a closed immersion of the regular scheme X into a smooth k-scheme is a regular immersion, and this property is preserved by the base change to $\bar{k}$. As V is normal, the geometric normalization of X is V itself, so Corollary 3.4(1) applies, with $W = X$ and $S = V$, at every singular point of V. Moreover, X is smooth over k if and only if V is smooth, because smoothness is compatible with, and descends along, the extension $\bar{k}/k$. Let $\mu\colon T \to V$ be the minimal resolution, and write

$$K_T = \mu^* K_V - Z$$

with Z a μ-exceptional divisor. By [Kol13, Theorem 10.5 and Claim 10.3.5], $Z \geq 0$, and Z is a $\mathbb{Z}$-divisor because K_V is Cartier. In particular $K_T \equiv -Z$.

Lemma 8.1. (1) *If $Z = 0$, then T is a smooth projective surface with $K_T \equiv 0$, and every singular point of V is a rational double point whose exceptional curves are smooth rational curves of self-intersection -2.*

(2) *If $Z \neq 0$, then T is rational, $H^1(X, \mathcal{O}_X) = 0$, $\omega_X \cong \mathcal{O}_X$ and $\chi(\mathcal{O}_X) = 2$. Moreover exactly one singular point y_0 of V is not rational; it has geometric genus $p_g(y_0) = 1$, and all other singular points of V are rational double points.*

Proof. (1) If $Z = 0$, then $K_T \equiv 0$. Every μ-exceptional curve E satisfies $K_T \cdot E = 0$ and $E^2 < 0$, so adjunction gives $p_a(E) = 0$ and $E^2 = -2$. Two exceptional curves meet in at most one point, transversally, since otherwise their sum would have nonnegative square; hence the exceptional configurations are negative definite configurations of (-2)-curves, that is, Dynkin diagrams of types A, D and E, and the singular points are rational double points [Art77].

(2) Since Z is effective and nonzero, $h^0(T, nK_T) = h^0(T, -nZ) = 0$ for $n \geq 1$, so T has Kodaira dimension $-\infty$ and is ruled over a smooth projective curve C [Mum69]. If C had positive genus, then every rational curve on T would be contracted to a point of C, by Lüroth's theorem. By Corollary 3.4(1), all μ-exceptional curves are rational, so Z would be vertical, and a general fibre $F \cong \mathbb{P}^1$ would satisfy $-2 = K_T \cdot F = -Z \cdot F = 0$, which is absurd. Hence T is rational, and $H^1(T, \mathcal{O}_T) = H^2(T, \mathcal{O}_T) = 0$. The Leray spectral sequence for μ gives $H^1(V, \mathcal{O}_V) = 0$ and $h^2(V, \mathcal{O}_V) = \sum_y p_g(y)$. By Serre duality $h^2(V, \mathcal{O}_V) = h^0(V, \omega_V)$, and $h^0(V, \omega_V) \leq 1$, because a nonzero section of ω_V has an effective numerically trivial divisor, which is zero; in that case $\omega_V \cong \mathcal{O}_V$. If $h^0(V, \omega_V) = 0$, then all singular points of V are rational, hence rational double points since V is Gorenstein [Art77], so the minimal resolution is crepant and $Z = 0$, a contradiction. Therefore $\omega_V \cong \mathcal{O}_V$ and $\sum_y p_g(y) = 1$, which gives the assertion about the singular points, since rational Gorenstein singularities are rational double points. By flat base change, $h^0(X, \omega_X) = 1$ and $H^1(X, \mathcal{O}_X) = 0$; as above $\omega_X \cong \mathcal{O}_X$, and $\chi(\mathcal{O}_X) = 1 - 0 + 1 = 2$. □

Proposition 8.2. *If $p \geq 23$ and $Z = 0$, then X is smooth over k.*

Proof. By Lemma 8.1(1) and the classification of surfaces [BM77], T is a K3, Enriques, abelian or bielliptic surface, so $b_2(T) \leq 22$. The μ-exceptional curves span a negative definite sublattice of $\mathrm{NS}(T)$, so there are at most $\rho(T) - 1 \leq 21$ of them. By Lemma 3.5, a singular point of V would have type A_{p^e-1} for some $e \geq 1$, requiring at least $p - 1 \geq 22$ exceptional curves, a contradiction. Hence V is smooth, and so is X. □

Proposition 8.3. *If $Z \neq 0$, then $\rho(X) = 2$, the intersection form on $N^1(X)_{\mathbb{Q}}$ is anisotropic,* $\det \mathrm{Pic}(X) = -p^{2a+1}$ *for some $a \geq 0$, and $p \equiv 1 \pmod 4$.*

Proof. By Lemma 8.1(2), $\omega_X \cong \mathcal{O}_X$, $H^1(X, \mathcal{O}_X) = 0$ and $\chi(\mathcal{O}_X) = 2$. Every numerically trivial line bundle $\mathcal{L}$ on X is trivial: its pullback to the rational surface T is numerically trivial, hence trivial, so $\mathcal{L}_{\bar{k}} = \mu_*\mu^*\mathcal{L}_{\bar{k}}$ is trivial, and a nonzero section of $\mathcal{L}$ has an effective numerically trivial divisor. The proofs of Proposition 4.2(6)–(7) now apply verbatim, with S replaced by V: T is rational, each exceptional lattice has determinant $\pm$ a power of p by Corollary 3.4(1), and V is the geometric normalization of X. So $\mathrm{Pic}(X)$ is an even lattice of signature $(1, \rho(X) - 1)$ and determinant $\pm p^a$. By Lemma 2.7, either the assertion holds or the intersection form on $N^1(X)_{\mathbb{Q}}$ is isotropic. In the latter case, Lemma 2.8 gives a fibration $f\colon X \to B$ onto a regular projective curve, whose generic fibre is a smooth curve of genus one by the argument in the proof of Proposition 4.4. Then Assumption 6.1 holds, and by Proposition 6.14 every singular point of V is a rational double point of type A_{p-1}; this contradicts the existence of the non-rational singular point y_0 of Lemma 8.1(2). □

Lemma 8.4. *Suppose that $Z \neq 0$, let y_0 be as in Lemma 8.1(2), and let $d = -Z^2$. Then y_0 is the only singular point of V and $1 \leq d \leq 4$. The exceptional locus of μ is* $\mathrm{Supp}\, Z = E_1 \cup \cdots \cup E_r$*, where $r = 8 + d$. Moreover, if $Z = \sum_i z_i E_i$, $a_i = K_T \cdot E_i$ and $w_i = -E_i^2$, then $z_i \geq 1$, $a_i \geq 0$ and*

$$w_i = a_i + 2 - 2p_a(E_i) \geq 1, \qquad w_i z_i - a_i = \sum_{j \neq i} z_j\, E_i \cdot E_j, \qquad \sum_i z_i a_i = d.$$

Proof. By Lemma 8.1(2), the singular points of V other than y_0 are rational double points, whose minimal resolutions are crepant; so Z is supported over y_0. If some exceptional curve over y_0 were not contained in $\mathrm{Supp}\, Z$, then, the exceptional fibre over y_0 being connected, some such curve E would meet $\mathrm{Supp}\, Z$; then $K_T \cdot E = -Z \cdot E < 0$ and E would be a (-1)-curve, contradicting minimality. Hence $\mathrm{Supp}\, Z$ is the exceptional fibre over y_0, and $z_i \geq 1$. Since μ is minimal, $a_i \geq 0$; adjunction gives $w_i = a_i + 2 - 2p_a(E_i)$, and $w_i \geq 1$ because the intersection matrix of the E_i is negative definite, which also gives $d \geq 1$. Since $K_T \equiv -Z$, the middle identity is $Z \cdot E_i = -a_i$, and the last one is $K_T \cdot Z = -Z^2$.

Let $U \subseteq V$ be an affine open neighbourhood of y_0 containing no other singular point of V. Since $\omega_V \cong \mathcal{O}_V$ and $p_g(y_0) = 1$, the point $y_0 \in U$ is an elliptic singularity in the sense of [Tan25, Definition 5.1], and $\mu^{-1}(U) \to U$ is its minimal resolution. If $d \geq 3$, then by [Tan25, Lemma 5.2, Notation 5.15 and the proof of Theorem 5.19], applied to U, the powers of the maximal ideal of $\widehat{\mathcal{O}}_{V,y_0}$ are the completions of the ideals $\mu_*\mathcal{O}_T(-jZ)$, and the associated graded ring is $\bigoplus_{j \geq 0} H^0(Z, \mathcal{O}_Z(-jZ))$. For $j \geq 1$, the group $H^1(Z, \mathcal{O}_Z(-jZ))$ is a quotient of $H^1(\mu^{-1}(U), \mathcal{O}_T(-jZ))$, because U is affine and the fibres of μ have dimension at most one; and the latter group vanishes by [Tan25, Notation 5.15]. Since $h^0(Z, \mathcal{O}_Z) = h^1(Z, \mathcal{O}_Z) = 1$ [Tan25, Lemma 5.2], the Riemann–Roch theorem on Z shows that the degree-j part has dimension $-jZ^2 = jd$ for $j \geq 1$. So $\widehat{\mathcal{O}}_{V,y_0}$ has embedding dimension d and multiplicity d. As V is a local complete intersection, $\widehat{\mathcal{O}}_{V,y_0}$ is the quotient of a regular local ring of dimension d by a regular sequence of $d - 2$ elements of the square of its maximal ideal, so its multiplicity is at least 2^{d-2}; hence $d \leq 4$.

Since T is rational, Noether's formula gives $\rho(T) = 10 - K_T^2 = 10 + d$. The surface V is $\mathbb{Q}$-factorial and $\rho(V) = \rho(X)$, which is 2 by Proposition 8.3. For a divisor D on T, some multiple $m\mu_*D$ is Cartier, and $mD - \mu^*(m\mu_*D)$ is μ-exceptional; so $N^1(T)_{\mathbb{Q}}$ is spanned by $\mu^*N^1(V)_{\mathbb{Q}}$ and the classes of the μ-exceptional curves. These two parts are orthogonal, the second is negative definite, and μ^* is injective on $N^1(V)_{\mathbb{Q}}$ by the projection formula. Hence μ has exactly $\rho(T) - \rho(V) = 8 + d$ exceptional curves. At least one lies over y_0, so at most eleven could lie over another singular point. Such a point would be a rational double point and would require at least $p - 1 \geq 12$ exceptional curves by Lemma 3.5, a contradiction. Thus y_0 is the only singular point of V and $r = 8 + d$. □

Proposition 8.5. *Suppose that $Z \neq 0$, and let $E \subset \mathrm{Pic}(T)$ be the sublattice spanned by the classes of the μ-exceptional curves. Then $|\det E|$ is a power of p, and*

$$p \leq |\det E| \leq 2^8 \cdot 3^d \leq 20736.$$

Proof. By Corollary 3.4(1), $|\det E|$ is a power of p. By the proof of Proposition 8.3, the proof of Proposition 4.2(7) applies with S replaced by V. In the notation of that proof, $|\det \mathrm{Pic}(X)| = [Q : \mu^*\phi^* \mathrm{Pic}(X)]^2\, |\det Q|$,

where the index is a power of p, and $|\det Q| = |\det E^{\mathrm{sat}}|$ divides $|\det E|$. Since $\det \mathrm{Pic}(X) = -p^{2a+1}$ by Proposition 8.3, $|\det Q|$ is an odd power of p; hence $p \le |\det Q| \le |\det E|$.

By Lemma 8.4, the exceptional curves are $E_1, \dots, E_r$ with $r = 8 + d$, and $w_i \le a_i + 2$ and $\sum_i a_i \le \sum_i z_i a_i = d$. Since the matrix $(-E_i \cdot E_j)$ is positive definite, Hadamard's inequality and Bernoulli's inequality $1 + a/2 \le (3/2)^a$ for integers $a \ge 0$ give

$$|\det E| \le \prod_i w_i \le \prod_i 2\Big(1 + \frac{a_i}{2}\Big) \le 2^{8+d}\Big(\frac{3}{2}\Big)^d = 2^8 \cdot 3^d \le 2^8 \cdot 3^4 = 20736.$$

□

Lemma 8.6. *Suppose that $Z \neq 0$ and $p \ge 13$, and let $E_1, \dots, E_r$, z_i, a_i, w_i and d be as in Lemma 8.4.*

(1) *We have $r = 8 + d$, and the determinant of the matrix $A = (-E_i \cdot E_j)_{1 \le i,j \le r}$ is an odd power of p; if $p \ge 29$, then $\det A = p$.*

(2) *Every E_i is a rational curve which is unibranch at every point. Let Γ be the graph whose vertices are the curves E_i and the points lying on at least two of them, with an edge joining each such point to each E_i through it. Then Γ is a tree; in particular, two of the curves E_i meet in at most one point.*

(3) *A point lies on at most three of the curves E_i, and on at most two with $a_i = 0$. If E_i and E_j meet, then $(E_i \cdot E_j)^2 < w_i w_j$.*

Proof. (1) By Lemma 8.4, all exceptional curves lie over y_0 and $r = 8 + d$, so $\det A = |\det E|$ in the notation of Proposition 8.5. By the proof of that proposition, $|\det E| = [E^{\mathrm{sat}} : E]^2 \, |\det Q|$ with $|\det Q|$ an odd power of p, and the index is a power of p because E^{sat}/E embeds in the dual $E^\vee/E$, a group of order $|\det E|$. So $|\det E|$ is an odd power of p; if $p \ge 29$, it is smaller than p^3 because $20736 < 29^3$, and hence $\det A = p$.

(2) Let $G = E_1 \cup \cdots \cup E_r$. By Corollary 3.4(1), the normalization $\pi \colon \widetilde{G} \to G$ is a disjoint union of r smooth rational curves. Since $V = X_{\bar{k}}$ descends to the regular surface X, the group scheme $\mathrm{Pic}^0_{G/\bar{k}}$ is unipotent and every E_i is unibranch by [Sch08, Corollaries 2.6 and 2.7]. Thus $H^1_{\mathrm{et}}(G, \mu_n) = 0$ for every integer $n > 1$ prime to p. For $x \in G$, let $\beta(x) = |\pi^{-1}(x)|$. Since G is connected, the normalization exact sequence for μ_n gives

$$\sum_x \big(\beta(x) - 1\big) = r - 1.$$

As the components are unibranch, $\beta(x)$ is the number of components through x, so this identity says that Γ has one fewer edge than vertices, including when three components pass through one point. The graph Γ is connected, so it is a tree.

(3) Suppose that k of the curves pass through a point, and let D be their sum. Any two of them meet, so $D^2 \ge -\sum w_i + k(k-1)$, the sum running over these k curves; and $\sum w_i \le \sum (a_i + 2) \le d + 2k \le 4 + 2k$ by Lemma 8.4. As $D^2 < 0$, we obtain $k(k-1) < 4 + 2k$, that is, $k \le 3$. A curve with $a_i = 0$ has $p_a(E_i) = 0$ and $w_i = 2$, since $w_i \ge 1$; if three such curves pass through a point, then $D^2 \ge -6 + 6 = 0$, which is absurd. The last assertion holds because the 2×2 principal minors of the positive definite matrix A are positive. □

Lemma 8.7. *If $Z \neq 0$ and $p \ge 13$, then $d \le 3$.*

Proof. Suppose that $d = 4$, and use the notation of Lemmas 8.4 and 8.6. Let $x \in X$ be the image of y_0. Since k is separably closed, $\kappa(x)$ is purely inseparable over k, and y_0 is the only point of V over x, so Lemma 2.4 applies to X and x, with $\bar{x} = y_0$. Let $\mathfrak{m}_{y_0}$ be the maximal ideal of $\mathcal{O}_{V,y_0}$. By the proof of Lemma 8.4, $\mathcal{O}_{V,y_0}$ has embedding dimension 4 and $\dim \mathfrak{m}_{y_0}^j/\mathfrak{m}_{y_0}^{j+1} = 4j$ for $j \ge 1$. Write $\mathrm{gr}_{\mathfrak{m}_{y_0}} \mathcal{O}_{V,y_0} = S/I^*$ with $S = \mathrm{Sym}_{\bar{k}}(\mathfrak{m}_{y_0}/\mathfrak{m}_{y_0}^2)$, so that I^* is the ideal of the tangent cone of V at y_0; then $I_1^* = 0$, $\dim I_2^* = 2$ and $\dim I_3^* = 8$.

Let L be the space of vertical directions in the Zariski tangent space of V at y_0, as in Lemma 2.4, and let u be its codimension, which is at most 2 by Lemma 2.4(1). Choose a basis $y_1, \dots, y_4$ of $\mathfrak{m}_{y_0}/\mathfrak{m}_{y_0}^2$ such that L consists of the tangent vectors v with $v(y_1) = \cdots = v(y_u) = 0$. By Lemma 2.4(2), I^* is stable under $\partial/\partial y_j$ for $j > u$. As $I_1^* = 0 \neq I_2^*$ and $p > 2$, Lemma 2.4(3) gives $I_2^* \subseteq \bar{k}[y_1, \dots, y_u]$. Since $\dim I_2^* = 2$, we get $u = 2$ and $I_2^* \subseteq \bar{k}[y_1, y_2]$.

We claim that the zero set of I^* in $\mathbb{P}^3_k = \operatorname{Proj} S$ is contained in the line $y_1 = y_2 = 0$. If the two quadrics spanning I_2^* have no common factor, they have no common zero with $(y_1, y_2) \neq 0$. Otherwise, after a linear change of y_1, y_2, we have $I_2^* = \langle y_1^2, y_1 y_2 \rangle$. The space $I_2^* S_1$ has dimension 7 and is spanned by the monomials divisible by y_1^2 or $y_1 y_2$, so I_3^* contains an element $G = y_1 G_1(y_3, y_4) + G_2(y_2, y_3, y_4)$, with G_1 and G_2 homogeneous of degrees 2 and 3, not both zero. Now $\partial G/\partial y_3 = y_1\, \partial G_1/\partial y_3 + \partial G_2/\partial y_3$ lies in I_2^*, and comparing monomials gives $\partial G_1/\partial y_3 = \partial G_2/\partial y_3 = 0$; similarly for y_4. Since $p > 3$, it follows that $G_1 = 0$ and $G_2 = c y_2^3$ with $c \neq 0$. So $y_2^3 \in I^*$, and the claim follows.

By [Tan25, Notation 5.3 and Theorem 5.19], applied to the affine neighbourhood U of y_0 chosen in the proof of Lemma 8.4, the blow-up $\widetilde{U}$ of U at y_0 is normal, and $\mu|_{\mu^{-1}(U)}$ factors through a projective birational morphism $g\colon \mu^{-1}(U) \to \widetilde{U}$ whose exceptional curves are exactly the curves E_i with $a_i = K_T \cdot E_i = 0$. The fibre of $\widetilde{U}$ over y_0 is $\operatorname{Proj}(\operatorname{gr}_{\mathfrak{m}_{y_0}} \mathcal{O}_{V,y_0})$, whose underlying set is the zero set of I^*, which is contained in a line by the claim. Since $\widetilde{U}$ is normal, g has connected fibres, so the curves E_i with $a_i > 0$ have pairwise distinct images; each of them is an irreducible curve contained in that line, hence equal to it. As $d > 0$, it follows that exactly one curve E_h has $a_h > 0$.

The other curves E_i have $a_i = 0$, so they are (-2)-curves, and by Lemma 8.6(2) and (3) they meet transversally along a forest. As in the proof of Lemma 8.1(1), their intersection matrix $-A_0$ is a sum of Dynkin diagrams of types A_n, D_n and E_n, of total rank $r - 1 \leq 11$; so the prime factors of $\det A_0$ are at most 11. Write z_0 for the vector of the z_i with $i \neq h$ and $e = (E_i \cdot E_h)_{i \neq h}$. The identity $Az = a$ of Lemma 8.4 gives $A_0 z_0 = z_h e$ and $w_h z_h - e \cdot z_0 = a_h$, so the Schur complement $w_h - e \cdot A_0^{-1} e$ of A_0 in A equals a_h/z_h, and

$$z_h \det A = a_h \det A_0.$$

Since $a_h \leq z_h a_h = 4$ and $p \geq 13$, the right-hand side is prime to p, whereas $\det A$ is divisible by p by Lemma 8.6(1). This contradiction proves that $d \neq 4$. □

Proposition 8.8. *If $Z \neq 0$ and $p \geq 29$, then $d = 4$.*

Proof. The proof is a computer enumeration, carried out by `smooth/configs.py` in Appendix A.1. By Lemmas 8.4 and 8.6, A is a positive definite symmetric integral matrix of size $r = 8 + d$ with $1 \leq d \leq 4$ and the following properties. Its diagonal entries are the w_i, where (w_i, a_i) is either $(2, 0)$ or one of the ten pairs with $1 \leq a_i \leq 4$, $1 \leq w_i \leq a_i + 2$ and $w_i \equiv a_i \pmod 2$. Its off-diagonal entries are the numbers $-E_i \cdot E_j$, which are described by a tree Γ and local intersection numbers as in Lemma 8.6(2) and (3). Finally $Az = a$ for a vector z of positive integers with $z \cdot a = d$. The script lists all such matrices. It builds Γ from a curve with $a_i > 0$ by breadth-first search: the balance $w_i z_i - a_i = \sum_{j \neq i} z_j\, E_i \cdot E_j$ of each curve E_i is split between the point joining E_i to its predecessor and the other points on E_i, each of which carries one or two new curves by Lemma 8.6(3). Every partial configuration is required to have a positive definite matrix. The search is finite, because $r \leq 12$, $z_i a_i \leq 4$, every term $z_j\, E_i \cdot E_j$ is at most $w_i z_i - a_i$, and $(E_i \cdot E_j)^2 < w_i w_j \leq (a_i + 2)(a_j + 2) \leq 16$. The script finds that the only primes $p \geq 29$ occurring as $\det A$ are 53 and 113, both with $d = 4$ and $r = 12$. Every determinant it finds already occurs for a configuration in which any two curves meet transversally and no three pass through one point; the determinants of these configurations are recomputed independently by `smooth/crosscheck.py` in Appendix A.1, which runs over all trees with r vertices. Since $\det A = p$ by Lemma 8.6(1), we obtain $d = 4$. □

Theorem 8.9. *Let k be a field of characteristic $p \geq 23$, and let X be a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. Then X is smooth over k.*

The proof is computer-assisted, through Proposition 8.8, only when $p \equiv 1 \pmod 4$ and $p \leq 6912$.

Proof. We may assume that k is F-finite and separably closed. Then the notation of this section applies. If $Z = 0$, then X is smooth by Proposition 8.2. Suppose that $Z \neq 0$. Then $p \equiv 1 \pmod 4$ by Proposition 8.3, so $p \geq 29$, and $d \leq 3$ by Lemma 8.7. Hence $p \leq 2^8 \cdot 3^3 = 6912$ by Proposition 8.5, which is a contradiction if $p > 6912$. In general, Proposition 8.8 gives $d = 4$, contradicting $d \leq 3$. □

9 Classification of geometric singularities for $p = 13, 17, 19$

We now determine the geometric singularities in the three remaining prime characteristics $p \geq 13$.

Proposition 9.1. *Let k be a field of characteristic $p \in \{13, 17, 19\}$, and let X be a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. Then $X_{\bar{k}}$ is normal and is either smooth or has exactly one singular point, which is a rational double point of type A_{p-1}. In the latter case its minimal resolution is a K3 surface and $\omega_X \cong \mathcal{O}_X$. In particular $X_{\bar{k}}$ is strongly F-regular.*

Proof. We may assume that k is F-finite and separably closed. The surface $V = X_{\bar{k}}$ is normal by Theorem 1.1. We use the notation of Section 8: let $\mu\colon T \to V$ be the minimal resolution and write $K_T = \mu^* K_V - Z$ with $Z \geq 0$.

Suppose first that $Z = 0$. By Lemma 8.1(1) and [BM77], T is a K3, Enriques, abelian or bielliptic surface, and every singular point of V is a rational double point. The μ-exceptional curves span a negative definite sublattice of $\mathrm{NS}(T)$ orthogonal to the pullback of an ample class, so there are at most $\rho(T)-1 \leq b_2(T)-1 \leq 21$ of them. By Lemma 3.5, every singular point of V has type A_{p^e-1} for some $e \geq 1$. The bound $p^e - 1 \leq 21$ forces $e = 1$, since $p \geq 13$. There is at most one such point because $2(p-1) > 21$. If V is singular, then $b_2(T) \geq \rho(T) \geq p \geq 13$, so T is a K3 surface. Then $K_T \sim 0$ and $\omega_V \cong \mathcal{O}_V$. Flat base change gives $h^0(X, \omega_X) = 1$; since ω_X is numerically trivial, its nonzero section has no zeros, so $\omega_X \cong \mathcal{O}_X$. Rational double points of type A are strongly F-regular in every characteristic [Har98, Theorem 1.1]. Thus all assertions hold when $Z = 0$.

Suppose now that $Z \neq 0$. By Proposition 8.3, $p \equiv 1 \pmod 4$, so $p \in \{13, 17\}$. By Lemma 8.1(2) and Proposition 8.3, T is rational, $H^1(V, \mathcal{O}_V) = 0$, $\omega_V \cong \mathcal{O}_V$ and $\rho(X) = 2$. By Lemma 8.4, V has a unique singular point P, which is non-rational, and its exceptional fibre consists of $r = 8 + d \leq 12$ curves $E_1, \ldots, E_r$, where $d = -Z^2 \in \{1, 2, 3, 4\}$.

Put $A = (-E_i \cdot E_j)$, $z_i = \mathrm{mult}_{E_i} Z$, $a_i = K_T \cdot E_i$ and $w_i = -E_i^2$. By Lemma 8.6, $\det A$ is an odd power of p, the incidence graph is a tree, and its intersection numbers satisfy the conditions used in the enumeration. By Proposition 8.5,

$$\det A \leq 2^8 \cdot 3^d \leq 20736 < 13^5 \leq p^5,$$

so $\det A \in \{p, p^3\}$.

By Lemma 8.7, $1 \leq d \leq 3$.

The scripts `smooth/p13.py` and `smooth/p17.py` in Appendix A.2 run the enumeration of Proposition 8.8 and keep the matrices with $\det A \in \{p, p^3\}$. Among the configurations with $r = 8 + d$ and $1 \leq d \leq 3$, there are none for $p = 13$, so this case is impossible. For $p = 17$ there is exactly one configuration up to isomorphism, with $d = 2$ and $r = 10$. Its exceptional curves are $F_1, \ldots, F_6$, G_1, G_1', G_6 and G_6', with $G_1^2 = G_6^2 = -3$ and all other self-intersections -2. The intersecting pairs are $\{F_i, F_{i+1}\}$ for $1 \leq i \leq 5$ and $\{F_1, G_1\}$, $\{F_1, G_1'\}$, $\{F_6, G_6\}$ and $\{F_6, G_6'\}$, and $Z = G_1 + G_1' + G_6 + G_6' + 2F_1 + \cdots + 2F_6$. Its curves are smooth and rational because $w_i = a_i + 2$, that is $p_a(E_i) = 0$, for all i, and they meet transversally because distinct curves have intersection number 0 or 1. We have $K_T \sim -Z$, since $K_T = \mu^* K_V - Z$ and $\omega_V \cong \mathcal{O}_V$.

Put $F = \bar{k}$. Since T is rational, $H^1(V, \mathcal{O}_V) = 0$ and $\omega_V \cong \mathcal{O}_V$, the Leray spectral sequence gives $p_g(P) = 1$. Thus P is an elliptic singularity in the sense of [Tan25, Definition 5.1]. By [Tan25, Lemma 5.2], Z is its fundamental cycle, and direct computation gives

$$Z^2 = -2, \qquad Z \cdot G_1 = Z \cdot G_6 = -1, \qquad Z \cdot C = 0$$

for each of the other eight exceptional curves C.

We first recover the part of the local equation needed for the argument. Write $R = \widehat{\mathcal{O}}_{V,P}$ and $I_j = (\mu_* \mathcal{O}_T(-jZ))^{\wedge}_P$. For the local constructions below, replace μ by its base change to $\operatorname{Spec} R$ and retain the notation T and Z. By [Tan25, Notation 5.15, Lemmas 5.12 and 5.21, and Theorem 5.23], I_1 is the maximal ideal of R, and the associated graded algebra for the I_j is generated by elements $\bar{x}, \bar{y}$ of degree 1 and $\bar{z}$ of degree 2. Its degree-j part has dimension $2j$ for $j \geq 1$; in particular the monomials in $\bar{x}, \bar{y}, \bar{z}$ of weighted degree less than 4 are linearly independent, where the weights are $(1, 1, 2)$. Choose lifts $x, y \in I_1$ and $z \in I_2$. The sections x, y generate $\mathcal{O}_T(-Z)$ near the exceptional fibre, by [Tan25, Theorem 5.7] and Nakayama's lemma. Consequently (x, y) is a reduction of the maximal ideal, and $\operatorname{length} R/(x, y) = 2$, since R is Cohen–Macaulay of multiplicity 2 [Tan25, Theorem 5.23]. It follows that R is free of rank two over $A = F[[x, y]]$.

The independence of $\bar{z}$ from $\bar{x}^2, \bar{x}\bar{y}, \bar{y}^2$ gives $z \notin (x,y)R$, so $1, z$ form an A-basis by Nakayama's lemma. Write its relation as $z^2 = a(x,y)z + b(x,y)$. The independence of the monomials of weighted degree less than 4 gives $a \in (x,y)^2$ and $b \in (x,y)^4$. Replacing z by $z - a/2$, we therefore obtain

$$R \cong F[[x,y,z]]/(z^2 - h(x,y)), \qquad h \in (x,y)^4,$$

and the class of z in I_2/I_3 is still independent of $\bar{x}^2, \bar{x}\bar{y}, \bar{y}^2$.

Let B be the blowup of $\operatorname{Spec} A$ at the origin, and let W be its normalization in $\operatorname{Frac} R$. The generation of $\mathcal{O}_T(-Z)$ by x, y gives a factorization $T \to W \to B$ in the completed local setting. The morphism $g\colon T \to W$ contracts precisely the curves of degree zero for $\mathcal{O}_T(-Z)$, namely the chain

$$G_1', F_1, \dots, F_6, G_6'.$$

Hence W has a single rational double point Q of type A_8 [Art77]. The two remaining exceptional curves on W are the images $\overline{G}_1, \overline{G}_6$ of G_1, G_6, and they meet only at Q. The pullback $\overline{Z}$ of the exceptional divisor of B is Cartier, satisfies $g^*\overline{Z} = Z$, and equals $\overline{G}_1 + \overline{G}_6$. In particular the ideals I_j are determined by order at least j along these two curves.

Let h_j denote the homogeneous degree-j part of h. We have $h_4 \neq 0$. Indeed, otherwise $z^2 = h \in (x,y)^5$ would give order at least 3 for z along both $\overline{G}_1$ and $\overline{G}_6$, since x, y have order at least 1 there; hence $z \in I_3$, a contradiction. On the chart $y = xv$ of B, putting $w = z/x^2$ gives the equation

$$w^2 = h_4(1,v) + xh_5(1,v) + x^2h_6(1,v) + \cdots.$$

This double cover is normal: h is reduced by the normality of R, and $h_4 \neq 0$ means that removing the factor x^4 from its pullback leaves a reduced branch divisor. Thus this is a chart of W. Each of $\overline{G}_1$ and $\overline{G}_6$ maps with degree one to the exceptional projective line of B, since $-Z \cdot G_1 = -Z \cdot G_6 = 1$. The exceptional double cover therefore splits, so h_4 is the square of a homogeneous quadratic. Its zeros are exactly the points where the two components meet. Since the components meet only at Q, a linear change of x, y and a scaling of z give $h_4 = y^4$, with Q at $x = v = w = 0$.

Put $P(v) = h_5(1,v)$ and $S(v) = h_6(1,v)$. The equation of W at Q is

$$w^2 = v^4 + xP(v) + x^2S(v) + O(x^3).$$

Since Q is singular, $P(0) = 0$. The quadratic part at an A_8 point has rank two, so $P'(0) = 0$ and $b := S(0) \neq 0$. Write $P(v) = av^2 + O(v^3)$. Completing the square in x by the formal Morse lemma gives a remaining series in v whose degree-four coefficient is

$$1 - \frac{a^2}{4b}.$$

More explicitly, the critical section is $x = -av^2/(2b) + O(v^3)$, and substitution gives this coefficient. If it were nonzero, Q would be of type A_3, rather than A_8. Therefore $a^2 = 4b$ and $a \neq 0$. We have proved that

$$h = y^4 + ax^3y^2 + cx^2y^3 + dxy^4 + ey^5 + O((x,y)^6), \qquad a \neq 0.$$

Put $f = z^2 - h$. Since V descends to the regular surface X, [Sch08, Corollary 5.4] implies that one of f_x, f_y, f_z belongs to the ideal generated by f and the other two partial derivatives. This is impossible for $f_z = 2z$, since f, f_x, f_y have order at least two. Modulo z, the remaining possibilities are $h_x \in (h, h_y)$ and $h_y \in (h, h_x)$ in $F[[x,y]]$. The latter is impossible because h_y has order three, whereas h and h_x have order four. For the former, the degree-four part of every element of (h, h_y) of order at least four lies in the span of xy^3 and y^4. But the degree-four part of h_x contains $3ax^2y^2$ with $3a \neq 0$. Thus neither possibility occurs, so $Z \neq 0$ is impossible. □

10 Counterexample in characteristic 11

In this section we show that Theorem 1.1 fails for $p = 11$. Let $k = \mathbb{F}_{11}(c)$ with c transcendental over $\mathbb{F}_{11}$; it is F-finite with p-basis $\{c\}$. Let γ be the 11th root of c in an algebraic closure, and $k' = k(\gamma)$, a purely inseparable extension of degree 11. Let $B = \mathbb{P}^1_k$ with affine coordinate u, and put $K = k(u)$ and $K' = k'(u)$.

Proposition 10.1. *There are a regular projective surface X over k and a morphism $f\colon X \to B$ whose generic fibre is a smooth curve of genus one, such that $H^0(X, \mathcal{O}_X) = k$ and $\omega_X \cong \mathcal{O}_X$, but $X_{k'}$ is not normal. In particular X is not geometrically normal. The same holds with k replaced by a separable closure k^{s} of k and k' by $k^{\mathrm{s}}(\gamma)$.*

Thus the bound $p \geq 13$ in Theorem 1.1 is optimal. Verification scripts for the polynomial identities are described in Appendix A.3.

10.1 The curve

Let E be the elliptic curve $y^2 = x^3 + u^5$ over K, and let $\omega_E = dx/2y$. Let ∂ be the K-derivation of $K(E)$ with $\partial x = 2y$ and $\partial y = 3x^2$, which is well defined because $\partial(y^2 - x^3 - u^5) = 0$. Then $df = \partial(f)\,\omega_E$ for $f \in K(E)$, since both sides are K-derivations of $K(E)$ which agree on x and y. Using $y^2 = x^3 + u^5$, one computes

$$\begin{aligned} &\partial x = 2y, && \partial^2 x = 6x^2, && \partial^3 x = 2xy, && \partial^4 x = 10x^3 + 4u^5, \\ &\partial^5 x = 5x^2 y, && \partial^6 x = 2x^4 + 9u^5 x, && \partial^7 x = 5x^3 y + 7u^5 y, && \partial^8 x = x^5 + 7u^5 x^2, \\ &\partial^9 x = 10x^4 y + 6u^5 xy, && \partial^{10} x = u^{10}. \end{aligned} \tag{10.1}$$

Since ∂^{11} is a derivation, $\partial^{11} x = \partial(u^{10}) = 0$ and $2y\,\partial^{11} y = \partial^{11}(x^3 + u^5) = 0$, we have $\partial^{11} = 0$.

The ring $K'(E) = K(E) \otimes_k k' = k'(u, x, y)$ is a field, because $K(E)$ is separable over k. We extend ∂ k'-linearly to $K'(E)$, let ∂_γ be the derivation of $K'(E) = K(E)[T]/(T^{11} - c)$ which vanishes on $K(E)$ and maps γ to 1, and put

$$\delta = u^{-1}\partial + \partial_\gamma.$$

Lemma 10.2. *Let Y be a scheme over $\mathbb{F}_{11}$, let D be a derivation of $\mathcal{O}_Y$ with $D^{11} = 0$, and let $g \in \Gamma(Y, \mathcal{O}_Y)$ with $D(g) = 1$. Then the sheaf $\mathcal{O}_Y^D$ of D-constants is the structure sheaf of a scheme Y^D with the same underlying space as Y, and $\mathcal{O}_Y = \bigoplus_{i=0}^{10} \mathcal{O}_{Y^D} g^i$. In particular $\mathcal{O}_Y \cong \mathcal{O}_{Y^D}[T]/(T^{11} - g^{11})$.*

Proof. The derivation D defines an α_{11}-action. Writing s for the coordinate on α_{11}, the coaction sends g to $g + s$ because $D(g) = 1$, so every stabilizer is contained in $s = 0$ and the action is free. By [Sch07, Proposition 1.1 and Section 2], the quotient Y^D exists with the same underlying space as Y and structure sheaf $\mathcal{O}_Y^D$, and $Y \to Y^D$ is an α_{11}-torsor. The invariant function g^{11} descends to Y^D, so

$$T = \mathbf{Spec}_{Y^D}\big(\mathcal{O}_{Y^D}[t]/(t^{11} - g^{11})\big)$$

is another α_{11}-torsor over Y^D. The function g defines an equivariant morphism $Y \to T$, which is an isomorphism because both schemes are torsors. The asserted decomposition follows from the basis $1, t, \ldots, t^{10}$ of $\mathcal{O}_T$ over $\mathcal{O}_{Y^D}$. □

Lemma 10.3. (1) *The derivation δ commutes with ∂, vanishes on K, and satisfies $\delta(\gamma) = 1$ and $\delta^{11} = 0$.*

(2) *The field $L = \{f \in K'(E) \mid \delta f = 0\}$ contains K, and $K'(E) = \bigoplus_{i=0}^{10} L\gamma^i$. Moreover L is the function field of a smooth projective geometrically integral curve X_K of genus one over K, and there is an isomorphism $X_K \otimes_K K' \cong E \otimes_K K'$ inducing $L \otimes_K K' = K'(E)$.*

(3) *We have $\partial(L) \subseteq L$, and there is a regular nowhere vanishing $\eta \in H^0(X_K, \Omega^1_{X_K/K})$ whose image in $\Omega^1_{K'(E)/K'}$ is ω_E; explicitly, $\eta = df/\partial(f)$ for every $f \in L$ with $\partial(f) \neq 0$.*

Proof. (1) The derivations ∂ and ∂_γ commute and vanish on u, so $u^{-1}\partial$, ∂_γ and ∂ commute pairwise. Hence $\delta^{11} = u^{-11}\partial^{11} + \partial_\gamma^{11}$, which vanishes, because $\partial^{11} = 0$ and ∂_γ^{11} vanishes on $K(E)$ and on γ^j for $0 \leq j \leq 10$. The remaining assertions are clear.

(2) The derivation ∂ is dual to ω_E, which is regular and nowhere vanishing on E, so ∂ is a derivation of $\mathcal{O}_E$. Hence δ is a derivation of the structure sheaf of $Y = E \otimes_K K' = E \otimes_k k'$, and Lemma 10.2 with $g = \gamma$ gives a scheme $X_K = Y^\delta$ over K with $\mathcal{O}_Y = \mathcal{O}_{X_K}[T]/(T^{11} - c) = \mathcal{O}_{X_K} \otimes_K K'$. Since properness, smoothness and geometric integrality descend along $K \to K'$, the scheme X_K is a smooth projective geometrically integral curve of genus one. Its function field is $K'(E)^\delta = L$, and $K'(E) = \bigoplus_i L\gamma^i$ by Lemma 10.2 at the generic point.

(3) As ∂ commutes with δ, it preserves L. If ∂ vanished on L, it would vanish on $L(\gamma) = K'(E)$; so there is $f \in L$ with $\partial(f) \neq 0$. The image of $df \in \Omega^1_{L/K}$ in $\Omega^1_{K'(E)/K'}$ is $\partial(f)\,\omega_E$, and $\partial(f) \in L$; so $\eta = df/\partial(f)$ maps to ω_E, and it does not depend on f because $\Omega^1_{L/K} \to \Omega^1_{K'(E)/K'}$ is injective. Since $\Omega^1_{X_K/K} \otimes_K K' = \Omega^1_{Y/K'}$ and ω_E is regular and nowhere vanishing on Y, so is η on X_K. □

10.2 Explicit equations

Put

$$a = \frac{x^{11}}{u^{18}}, \qquad b = \frac{y^{11}}{u^{27}}, \qquad \xi = \frac{a}{b} \qquad \text{and} \qquad \theta = \sum_{i=0}^{10} \Big(-\frac{\gamma}{u}\Big)^i \frac{\partial^i x}{i!}.$$

Lemma 10.4. *The elements a, b and θ lie in L, and $b^2 = a^3 + u$. Moreover $\theta^{11} = Q$, where*

$$\begin{aligned} Q = {} & -c^{10} + c^9a^4b + 5c^9uab - 2c^8u^2a^5 - 3c^8u^3a^2 + 3c^7u^4a^3b + 2c^7u^5b - 4c^6u^6a^4 \\ & + 4c^6u^7a + 5c^5u^8a^2b + 5c^4u^{10}a^3 + 2c^4u^{11} - 4c^3u^{12}ab + 3c^2u^{14}a^2 - 2cu^{16}b + u^{18}a. \end{aligned} \tag{10.2}$$

The field $K(a,b)$ is the function field of the elliptic curve $E'\colon b^2 = a^3 + u$ over K, and $L = K(a,b)(\theta)$ is an extension of degree 11 of $K(a,b)$.

Proof. Since ∂ kills 11th powers and ∂_γ kills $K(E)$, we have $\delta(a) = \delta(b) = 0$. Raising $y^2 = x^3 + u^5$ to the 11th power and dividing by u^{54} gives $b^2 = a^3 + u$. Using $\partial(u) = 0$, $\partial_\gamma(\gamma) = 1$ and $\partial^{11}x = 0$, we find

$$\delta(\theta) = \sum_{i=0}^{9} \frac{(-\gamma)^i}{i!\,u^{i+1}}\,\partial^{i+1}x - \sum_{i=1}^{10} \frac{(-\gamma)^{i-1}}{(i-1)!\,u^i}\,\partial^i x = 0.$$

Since the Frobenius is additive and $(i!)^{11} = i!$ in $\mathbb{F}_{11}$,

$$\theta^{11} = \sum_{i=0}^{10} \frac{(-c)^i}{i!\,u^{11i}}\,(\partial^i x)^{11}.$$

If $\partial^i x = P_i(u,x,y)$ with P_i as in (10.1), then $(\partial^i x)^{11} = P_i(u^{11}, u^{18}a, u^{27}b)$, and substituting gives (10.2). For instance, since $10! \equiv -1$ and $9! \equiv 1$ modulo 11, the terms with $i = 10$ and $i = 9$ contribute $-c^{10}$ and $c^9a^4b + 5c^9uab$.

We have $K'(a,b) = K'(x^{11}, y^{11})$, and $K'(E) = K'(x^{11}, y^{11})(x)$ because $y = y^{11}/(x^3 + u^5)^5$. As $[K'(E) : K'(x^{11})] = 22$, $[K'(E) : K'(x^{11}, y^{11})] \leq 11$ and $[K'(x^{11}, y^{11}) : K'(x^{11})] \leq 2$, both inequalities are equalities. The field $K(a,b)$ is separable over k, so $K(a,b) \otimes_k k' = K'(a,b)$ is a field, and $[K'(E) : K(a,b)] = 121$. Since $K(a,b) \subseteq L$ and $[K'(E) : L] = 11$ by Lemma 10.3(2), we get $[L : K(a,b)] = 11$.

Let $\mathcal{O}_0$ be the local ring of the plane $\operatorname{Spec} k[a,b]$ at the generic point of the curve $u = b^2 - a^3 = 0$. It is a discrete valuation ring with uniformizer u and residue field $k(s)$, where $s = b/a$, and $a \mapsto s^2$, $b \mapsto s^3$. By (10.2), $Q \in \mathcal{O}_0$ has residue $c^9s^{11} - c^{10}$, which does not lie in $k(s)^{11} = k^{11}(s^{11})$ because $c^9 \notin k^{11}$. If θ were in $K(a,b)$, it would lie in the normal ring $\mathcal{O}_0$, and the residue of $Q = \theta^{11}$ would be an 11th power. Hence $\theta \notin K(a,b)$, and $L = K(a,b)(\theta)$ because $[L : K(a,b)]$ is prime. □

Gradient identities. Put $\widetilde{Q}(c,a,b) = Q(b^2 - a^3, a, b, c)$, $u = b^2 - a^3$ and $h = 5\widetilde{Q}_c$. Then

$$\widetilde{Q}_c = -2h, \qquad \widetilde{Q}_a = ubh, \qquad \widetilde{Q}_b = 3uah. \tag{10.3}$$

These identities follow from weighted homogeneity and the truncated exponential defining θ. For weights $(6,2,3,11)$ on (u,a,b,c), the polynomial Q is homogeneous of degree 110. The substitution $u = b^2 - a^3$ preserves these weights, so Euler's identity in characteristic 11 gives

$$2a\widetilde{Q}_a + 3b\widetilde{Q}_b = 0.$$

Write $P_i = \partial^i x$ and $\Phi(P) = P(u^{11}, u^{18}a, u^{27}b)$, and put $\partial' = 2b\partial_a + 3a^2\partial_b$. Since $\partial' u = 0$, we have $\Phi(\partial P) = u^9\partial'\Phi(P)$. Differentiating the truncated sum for Q in c, and using $P_{11} = 0$, therefore gives

$$(\partial\theta)^{11} = \sum_{i=0}^{9} \frac{(-c)^i}{i!\,u^{11i}}\Phi(P_{i+1}) = -u^{11}\widetilde{Q}_c = u^9\partial'\widetilde{Q}.$$

Consequently $2b\widetilde{Q}_a + 3a^2\widetilde{Q}_b = -u^2\widetilde{Q}_c$. Combining this with Euler's identity and cancelling u in $\mathbb{F}_{11}[c,a,b]$ proves (10.3). The coefficient of c^0 in h is $u^{16}b$, so $h \neq 0$. Thus h is a greatest common divisor of the three derivatives, and their normalized quotients generate the unit ideal because $5\widetilde{Q}_c/h = 1$. The same identities give

$$\partial' Q = 2u^2 h, \qquad (\partial\theta)^{11} = 2u^{11} h. \tag{10.4}$$

Reducing $h = 5\widetilde{Q}_c$ modulo u and using $\overline{Q} = c^9 s^{11} - c^{10}$ gives

$$\overline{h} = c^8 s^{11} + 5c^9 \neq 0 \tag{10.5}$$

in the residue field $k(s)$.

10.3 The fibre over $u = 0$

Let $R = \mathcal{O}_{B,0}$, and let $\mathcal{W}$ be the closure of $b^2 = a^3 + u$ in $\mathbb{P}^2_R$. Its affine part $\operatorname{Spec} R[a,b]/(b^2 - a^3 - u)$ is a localization of the plane $\operatorname{Spec} A$, where $A = k[a,b]$. Near the point O at infinity of the special fibre, the functions $\xi = a/b$ and $\zeta = 1/b$ satisfy $\zeta = \xi^3 + u\zeta^3$. Hence $\mathcal{W}$ is smooth over k, the point O is k-rational with local coordinates u and ξ, and $\zeta = \xi^3\varepsilon$ for the unit $\varepsilon = (1 - u\zeta^2)^{-1}$ of $\mathcal{O}_{\mathcal{W},O}$. The special fibre $\mathcal{W}_0$ is the cuspidal cubic $b^2 = a^3$, and the local ring of $\mathcal{W}$ at its generic point is $\mathcal{O}_0$. Let $\mathcal{Z}$ be the normalization of $\mathcal{W}$ in L. It is finite over $\mathcal{W}$, proper over R with generic fibre X_K, and $\mathcal{Z} \to \mathcal{W}$ is a homeomorphism because $L/K(a,b)$ is purely inseparable.

Lemma 10.5. *The scheme $\mathcal{Z}$ is regular, and its special fibre $\mathcal{Z}_0$ is integral. The element u is a uniformizer of the local ring $\mathcal{O}_{\mathcal{Z},z_0}$ at the generic point z_0 of $\mathcal{Z}_0$, whose residue field is $k(s)(\bar\theta)$, where $\bar\theta$ is the residue of θ and $\bar\theta^{11} = c^9 s^{11} - c^{10}$. In particular $\mathcal{Z}$ is the minimal regular model of X_K over R.*

Proof. We first show that the normalization N of A in L is regular; the affine part of $\mathcal{Z}$ is a localization of $\operatorname{Spec} N$. Every element of L has its 11th power in $\operatorname{Frac}(A)$, so $N = L \cap A^{1/11}$, where $A^{1/11} = k^{1/11}[a^{1/11}, b^{1/11}]$. This is a regular ring with p-basis $\{c^{1/11}, a^{1/11}, b^{1/11}\}$ over $\mathbb{F}_{11}$, so the module $\mathcal{T} = \operatorname{Der}_{\mathbb{F}_{11}}(A^{1/11})$ is free on the dual basis. Then $\theta = \widetilde{Q}(c^{1/11}, a^{1/11}, b^{1/11}) \in A^{1/11}$, since both sides are 11th roots of Q. Let $\mathcal{F} = \{D \in \mathcal{T} \mid D(\theta) = 0\}$; it is saturated in $\mathcal{T}$ and closed under brackets and pth powers, so it is a foliation on $\operatorname{Spec} A^{1/11}$. The derivations of $\operatorname{Frac}(A)^{1/11}$ which vanish on $L = \operatorname{Frac}(A)(\theta)$ are those which vanish on θ, and L is the field of their constants by Jacobson's correspondence [Jac44]; hence $N = (A^{1/11})^{\mathcal{F}}$. By (10.3), the quotients $\widetilde{Q}_c/h$, $\widetilde{Q}_a/h$ and $\widetilde{Q}_b/h$ generate the unit ideal of $\mathbb{F}_{11}[c,a,b]$. For $D \in \mathcal{T}$ the chain rule gives $D(\theta) = h^{1/11}\lambda(D)$, where

$$\lambda(D) = \sum_{z \in \{c,a,b\}} (\widetilde{Q}_z/h)(c^{1/11}, a^{1/11}, b^{1/11})\, D(z^{1/11}),$$

and the $A^{1/11}$-linear map $\lambda\colon \mathcal{T} \to A^{1/11}$ is surjective. Hence $\mathcal{F} = \ker\lambda$ is a direct summand of $\mathcal{T}$ of rank two, that is, a subbundle of $\mathcal{T}$, and N is regular by [Eke87, Proposition 2.4].

Next we consider the point O. Every monomial $u^m a^i b^j$ in (10.2) satisfies $2i + 3j \leq 11$, with equality only for $c^9 a^4 b$. Near O we have $a^i b^j = \xi^i \zeta^{-i-j} = \xi^{-2i-3j}\varepsilon^{-i-j}$, so $\theta_\infty = \xi\theta$ satisfies $\theta_\infty^{11} = \xi^{11} Q \in \mathcal{O}_{\mathcal{W},O}$, and $\xi^{11}Q$ has value c^9 at O. Since $c^9 \notin k^{11}$, the ring $\mathcal{O}_{\mathcal{W},O}[T]/(T^{11} - \xi^{11}Q)$ is flat over $\mathcal{O}_{\mathcal{W},O}$ with closed fibre the field $k(c^{9/11})$, hence a regular local ring [Sta26, Tag 031E]. Being finite over $\mathcal{O}_{\mathcal{W},O}$ with fraction field L and normal, it is the local ring of $\mathcal{Z}$ at the point over O. As the special fibre $\mathcal{W}_0$ lies in the union of the affine part and $\{O\}$, and the generic fibre X_K is smooth, $\mathcal{Z}$ is regular.

Since $\mathcal{Z} \to \mathcal{W}$ is a homeomorphism, $\mathcal{Z}_0$ is irreducible. The ring $\mathcal{O}_{\mathcal{Z},z_0}$ is the integral closure of $\mathcal{O}_0$ in L, a discrete valuation ring finite over $\mathcal{O}_0$, so its ramification index e and residue degree f over $\mathcal{O}_0$ satisfy $ef = [L : K(a,b)] = 11$. Its residue field contains $k(s)$ and $\bar\theta$, where $\bar\theta^{11} = c^9 s^{11} - c^{10}$ is not an 11th power in $k(s)$ by the proof of Lemma 10.4. Hence $f = 11$ and $e = 1$, so u is a uniformizer, the residue field is $k(s)(\bar\theta)$, and the special fibre $\mathcal{Z}_0 = \operatorname{div}(u)$ is integral. Finally, $\mathcal{Z}_0^2 = 0$, so $\mathcal{Z}_0$ is not an exceptional curve, and $\mathcal{Z}$ is the minimal regular model [Liu02, Section 9.3]. □

Lemma 10.6. *The rational section $u^{-1}\eta$ of $\omega_{\mathcal{Z}/R}$ generates $\omega_{\mathcal{Z}/R}$; equivalently, $\operatorname{div}_{\mathcal{Z}}(\eta) = \mathcal{Z}_0$.*

Proof. Since $\mathcal{Z}$ is regular, $\omega_{\mathcal{Z}/R}$ is invertible, and since η is regular and nowhere vanishing on X_K, it suffices to show that η/τ has valuation one at z_0 for a generator τ of $\omega_{\mathcal{Z}/R}$ at z_0. The ring $\mathcal{O}_0[T]/(T^{11}-Q)$ is local with residue field $k(s)[T]/(T^{11}-c^9s^{11}+c^{10})$, so it is a discrete valuation ring, equal to $\mathcal{O}_{\mathcal{Z},z_0}$. Thus near z_0 the scheme $\mathcal{Z}$ is the hypersurface $g=0$, where $g=T^{11}-Q$, in $\mathcal{P}=\operatorname{Spec}A[T]$, which is smooth over R at the point over z_0. There b is a unit, and $\Omega^1_{\mathcal{P}/R}$ is free on da and dT, because $2b\,db=3a^2\,da$ in $\Omega^1_{\mathcal{P}/R}$. By the adjunction formula [Liu02, Section 6.4], $\omega_{\mathcal{Z}/R}$ is generated at z_0 by the rational differential τ on X_K determined by $dg\wedge\tau=da\wedge dT$ on the generic fibre of $\mathcal{P}$. Here $dg=-Q_a\,da-Q_b\,db=-\frac{\partial' Q}{2b}\,da$, as $11T^{10}=0$, where Q_a and Q_b are the partial derivatives of $Q(u,a,b)$ with u fixed and $\partial' Q=2bQ_a+3a^2Q_b$; so $\tau=-2b\,d\theta/\partial' Q$. On the other hand $\partial(\theta)\in L$ is nonzero and $\eta=d\theta/\partial(\theta)$ by Lemma 10.3(3), so

$$\frac{\eta}{\tau}=-\frac{\partial' Q}{2b\,\partial(\theta)}.$$

Let v be the valuation of $\mathcal{O}_{\mathcal{Z},z_0}$; by Lemma 10.5, it restricts to the valuation of $\mathcal{O}_0$ on $K(a,b)$. The residue (10.5) shows that h is a unit at z_0. By (10.4), we obtain $v(\partial' Q)=2$ and $v(\partial(\theta))=1$. Since $v(b)=0$, we have $v(\eta/\tau)=1$. □

Lemma 10.7. *Let $P\neq 0$ be a closed point of B, let $R_P=\mathcal{O}_{B,P}$, and let $X_{(P)}$ be the minimal regular model of X_K over R_P. Then $\omega_{X_{(P)}/R_P}$ is generated by η if $P\neq\infty$, and by $u\eta$ if $P=\infty$.*

Proof. If $P\neq\infty$, let $\mathcal{E}_P$ be the closure of $y^2=x^3+u^5$ in $\mathbb{P}^2_{R_P}$, and put $\omega_P=\omega_E$, $\partial_P=\partial$ and $D_P=u\delta=\partial_P+u\partial_\gamma$. If $P=\infty$, let $w=u^{-1}$, $x'=w^2x$ and $y'=w^3y$, let $\mathcal{E}_P$ be the closure of $y'^2=x'^3+w$ in $\mathbb{P}^2_{R_P}$, and put $\omega_P=dx'/2y'=u\,\omega_E$, $\partial_P=w\partial$ and $D_P=\delta=\partial_P+\partial_\gamma$; thus $\partial_P(x')=2y'$ and $\partial_P(y')=3x'^2$. In both cases $\mathcal{E}_P$ is regular with irreducible special fibre, $\omega_{\mathcal{E}_P/R_P}$ is generated by ω_P [LLR04, 9.1], and ∂_P is a derivation of $\mathcal{O}_{\mathcal{E}_P}$, as one checks on the two standard affine charts. Hence D_P is an R_P-linear derivation of the structure sheaf of $Y_P=\mathcal{E}_P\otimes_k k'$, with $D_P^{11}=0$ by Lemma 10.3(1), and $\varepsilon_P=D_P(\gamma)\in\{u,1\}$ is a unit of R_P. Lemma 10.2 with $g=\gamma/\varepsilon_P$ gives a scheme $Y_P^{D_P}$ with

$$\mathcal{O}_{Y_P}=\mathcal{O}_{Y_P^{D_P}}[T]/(T^{11}-c\,\varepsilon_P^{-11})\cong\mathcal{O}_{Y_P^{D_P}}\otimes_k k',$$

the last isomorphism mapping T to γ/ε_P. So $Y_P^{D_P}$ is proper and flat over R_P with generic fibre X_K, as $K'(E)^{D_P}=L$, and $Y_P^{D_P}\otimes_k k'\cong Y_P$, compatibly with $L\otimes_K K'=K'(E)$. The scheme Y_P is regular: for $P\neq\infty$ it is smooth over the regular ring $R_P\otimes_k k'$, a localization of $k'[u]$; for $P=\infty$ its affine part is a localization of $k'[x',y']$, and it is smooth over k' near the point at infinity. By faithfully flat descent [Sta26, Tag 00OF], $Y_P^{D_P}$ is regular. Its special fibre is irreducible, being homeomorphic to that of $\mathcal{E}_P$, so it contains no exceptional curve and $Y_P^{D_P}=X_{(P)}$. Finally, $\omega_{X_{(P)}/R_P}\otimes_k k'\cong\omega_{Y_P/R_P\otimes_k k'}$ [Liu02, Theorem 6.4.9], and this isomorphism maps η to ω_E by Lemma 10.3(3). As ω_P generates $\omega_{Y_P/R_P\otimes_k k'}$, the section $(\omega_P/\omega_E)\,\eta$, which is η or $u\eta$, generates $\omega_{X_{(P)}/R_P}$. □

10.4 Geometric non-normality

Lemma 10.8. *The ring $\mathcal{O}_{\mathcal{Z},z_0}\otimes_k k'$ is not normal.*

Proof. Put $t=y/x$, $r=x/t^2$ and $\beta=t^5/u^4$ in $K'(E)$. A direct computation using $y^2=x^3+u^5$ gives

$$\xi=\frac{u^9}{t^{11}},\quad u=\xi^5\beta^{11},\quad t=\xi^4\beta^9,\quad a=\xi^{-2}r^{11},\quad b=\xi^{-3}r^{11}\quad\text{and}\quad r^3-r^2+\xi\beta=0.$$

The function t on $E_{K'}$ has degree three, with poles at the origin and at the two points with $x=0$, and $K'(t)=k'(\xi,\beta)$. So $K'(E)=K'(t,x)=k'(\xi,\beta,r)$ is a cubic extension of $k'(\xi,\beta)$ in which r has minimal polynomial $T^3-T^2+\xi\beta$, and the subring $C=k'[\xi,\beta,r]$ is isomorphic to $k'[\xi,\beta][T]/(T^3-T^2+\xi\beta)$. Let $\mathcal{O}'$ be the localization of C at the height-one prime ideal $(\beta,r-1)$. Since r is a unit of $\mathcal{O}'$ and $r-1=-\xi\beta/r^2$, the ring $\mathcal{O}'$ is a discrete valuation ring with uniformizer β and residue field $k'(\xi)$; let v' be its valuation. Then $v'(u)=11$, $v'(a)=v'(b)=0$, and a has residue ξ^{-2}, which is transcendental over k. So the centre of $v'|_{K(a,b)}$ on $\operatorname{Spec}k[a,b]$ is a non-closed point of the curve $u=0$, that is, its generic point, and $v'|_{K(a,b)}$ is 11 times the valuation of $\mathcal{O}_0$.

Let $\mathcal{O} = \mathcal{O}_{\mathcal{Z},z_0}$ with valuation v. By Lemma 10.5, v restricts to the valuation of $\mathcal{O}_0$, and since $L/K(a,b)$ is purely inseparable, v is its only extension to L; hence $v'|_L = 11v$ and $\mathcal{O} \subseteq \mathcal{O}'$. For every $h \in \mathcal{O}'$, we have $h^{11} \in L$ and $v(h^{11}) = v'(h) \geq 0$, so $h^{11} \in \mathcal{O}$ and h is integral over $\mathcal{O}$. Since $\mathcal{O}'$ is normal and contains $\mathcal{O}[\gamma]$, it is the normalization of $\mathcal{O} \otimes_k k' = \mathcal{O}[\gamma]$ in $K'(E) = L(\gamma)$.

The derivation δ vanishes on L and satisfies $\delta(\gamma) = 1$ by Lemma 10.3. Using $\partial(t) = (3x^3 - 2y^2)/x^2 = (x^3 - 2u^5)/x^2$, we compute

$$\delta(\beta) = 5u^{-5}t^4\partial(t) = \frac{5r}{\xi\beta} + \frac{1}{r^2}.$$

The element $\beta\delta(\beta) = 5r/\xi + \beta/r^2$ has nonzero residue $5/\xi$ in $k'(\xi)$, so $v'(\delta(\beta)) = -1$. Thus δ does not preserve $\mathcal{O}'$, and Lemma 2.2 shows that $\mathcal{O} \otimes_k k'$ is not normal. □

Proof of Proposition 10.1. Let $f\colon X \to B$ be the minimal regular model of X_K over B [Liu02, Section 9.3]; it is a regular proper surface over k, hence projective [Liu02, Remark 9.3.5]. Over R it is $\mathcal{Z}$ by Lemma 10.5. By Lemmas 10.6 and 10.7, the rational section η of $\omega_{X/B}$ has divisor $f^*(0) + f^*(\infty)$. Hence $\omega_{X/B} \cong f^*\mathcal{O}_B(2)$, and $\omega_X \cong \omega_{X/B} \otimes f^*\omega_B \cong \mathcal{O}_X$ [Liu02, Theorem 6.4.9]. Since X is normal and X_K is geometrically integral, $f_*\mathcal{O}_X = \mathcal{O}_B$, so $H^0(X, \mathcal{O}_X) = k$. By Lemma 10.8, $X_{k'}$ is not normal at the point over z_0, and X is not geometrically normal [Sta26, Tag 038O].

The separable closure k^{s} also has p-basis $\{c\}$, and $c^9 \notin (k^{\mathrm{s}})^{11}$. These are the only properties of k used above, so the same construction and proofs apply over k^{s}. □

Remark 10.9. (1) The Jacobian of X_K is E. Indeed, both have j-invariant 0 and become isomorphic over K', and the twists of E are classified by K^*/K^{*6}; if $D \in K^*$ is a sixth power z^6 in K', then $z^{11} \in K$ and $D = (D^2)^6/(z^{11})^6$ is a sixth power in K. So E has reduction type II^* at the k-rational point $u = 0$, as in configuration (b) of Remark 7.7.

(2) The valuation v' in the proof of Lemma 10.8 satisfies $v'(u) = 11$; accordingly, the fibre over $u = 0$ of the normalization of $X_{k'}$ is irreducible of multiplicity 11.

(3) For orientation, although we do not use it: X_K is the twist of E by the α_{11}-torsor defined by c/u^{11}, where α_{11} acts on the supersingular curve E through the translations given by ∂. The construction is analogous to the μ_p-twists used in [WZ26, Example 6.2] for $p = 5$ and $p = 7$.

11 Examples and a lift to characteristic zero

The bound $p \geq 23$ in Theorem 8.9 is optimal, both over fields and for regular models in mixed characteristic.

Example 11.1 (Non-smoothness in characteristic $5 \leq p \leq 19$)**.** Let $5 \leq p \leq 19$ be a prime, and let $k = \mathbb{F}_p(c)$, where c is transcendental over $\mathbb{F}_p$. There is a regular projective surface X over k with $H^0(X, \mathcal{O}_X) = k$ and $\omega_X \cong \mathcal{O}_X$ such that $X_{\bar{k}}$ has exactly one singular point, of type A_{p-1}. It admits an elliptic fibration over $\mathbb{P}^1_k$ whose generic fibre has equation

$$y^2 = x^3 + c^2t^m x + t^n, \tag{11.1}$$

where (m, n) is given by

p	5	7	11	13	17	19
m	5	5	5	5	7	7
n	5	4	2	1	2	1

Proof. The table gives $3m - 2n = p$, $m \in \{5, 7\}$ and $n \in \{1, 2, 4, 5\}$. The equations for $p = 13, 17, 19$ are twists of those in [Mat23, Example 9.6 and Table 9]. For every p in the table, the substitution

$$(x_0, y_0, t_0) = (\beta^{2n}x, \beta^{3n}y, \beta^6 t), \qquad \beta^p = c,$$

identifies $y^2 = x^3 + c^2t^m x + t^n$ with $y_0^2 = x_0^3 + t_0^m x_0 + t_0^n$ over $k(\beta)$.

Let $B = \mathbb{P}^1_k$, let $L = \mathcal{O}_B(2)$ and let $g\colon W \to B$ be the projective Weierstrass model of (11.1). The coefficients c^2t^m and t^n are sections of $L^{\otimes 4}$ and $L^{\otimes 6}$, respectively, and adjunction gives $\omega_W \cong g^*(\omega_B \otimes L) \cong \mathcal{O}_W$. The zero section lies in the smooth locus of g.

We check regularity over k by treating c as a variable over $\mathbb{F}_p$. The affine equation is $f = y^2 - x^3 - c^2t^mx - t^n$. On $t \neq 0$, the simultaneous vanishing of $f_c = -2ct^mx$ and $f_x = -3x^2 - c^2t^m$ is impossible. On $t = 0$, the equations $f_x = f_y = 0$ force $x = y = 0$, and this point is smooth if $n = 1$ because $f_t = -1$. Localizing the coordinate ring over $\mathbb{F}_p$ at the nonzero polynomials in c therefore shows that the affine chart of W is regular over k outside the k-rational point $P_0 = (t, x, y) = (0, 0, 0)$.

We resolve the points over $t = 0$ and $t = \infty$ over k. Put $\eta = 1 + c^2t^{m-n}x$, which is a unit at P_0. Since 2 and 3 are invertible, the formal change $x' = x\eta^{-1/3}$, $y' = y\eta^{-1/2}$ changes the equation to

$$y'^2 = x'^3 + t^n.$$

For $n = 1$ this is smooth, and for $n = 2, 4, 5$ it is a split rational double point of type A_2, E_6 or E_8, respectively [Art77]. The standard resolutions are obtained by blowups at k-rational double points and are smooth over k near their exceptional curves; this also holds for $n = p = 5$.

At infinity put $s = t^{-1}$, $u = s^4x$ and $v = s^6y$. The equation is

$$v^2 = u^3 + c^2s^{8-m}u + s^{12-n}.$$

If $m = 7$, its only singular point on $s = 0$ is the ordinary double point $P_\infty = (s, u, v) = (0, 0, 0)$, with quadratic part $v^2 - c^2su$. One blowup resolves it and gives a surface smooth over k near the exceptional curve. If $m = 5$, put $q = 12 - n \geq 7$. The first blowup has one singular point, with equation $V^2 = sU^3 + c^2s^2U + s^{q-2}$. The next blowup has one ordinary double point and, in the chart $s = US$, $V = UJ$, the point with equation

$$J^2 = c^2US^2 + U^2S + U^{q-4}S^{q-2}.$$

Its cubic part factors as $US(c^2S + U)$, and its blowup has three ordinary double points, in the directions $S/U = 0, -c^{-2}$ and $U/S = 0$. All these points are k-rational, and their quadratic parts are nondegenerate. Blowing them up gives a resolution by seven point blowups, smooth over k near the exceptional fibre.

The chart equations and the nondegeneracy assertions in these characteristics are checked by `smooth/small_primes_checks` in Appendix A.4.

Let $\pi\colon X \to W$ be these resolutions over 0 and ∞. Every centre is a double point, so π is crepant; hence X is regular and projective over k with $\omega_X \cong \pi^*\omega_W \cong \mathcal{O}_X$. The discriminant is

$$\Delta = -16t^{2n}(4c^6t^p + 27),$$

which is nonzero, so the generic fibre is a smooth elliptic curve. Stein factorization gives $(g \circ \pi)_*\mathcal{O}_X = \mathcal{O}_B$, and consequently $H^0(X, \mathcal{O}_X) = k$.

Let $a \in \overline{k}$ be the unique root of $4c^6t^p + 27$. It is nonzero, and Δ has order p there. Put $r = -3/(2c^2a^{m-n})$. Then $3r^2 + c^2a^m = 0$ and $2r^3 = a^n$, and at $(t, x, y) = (a, r, 0)$ we have $f = f_x = f_y = 0$ and $f_t = (p/2)a^{n-1} = 0$. Since $c_4 = -48c^2t^m$ is nonzero at a, this is the unique node of the fibre. For $z = t - a$, choose $\rho \in \overline{k}[[z]]$ with $\rho(0) = r$ and $3\rho^2 + c^2t^m = 0$. Writing $x = \rho + \xi$ gives

$$y^2 = \xi^2(3\rho + \xi) + h(z), \qquad h(z) = t^n - 2\rho^3,$$

and

$$\Delta = -432(t^n - 2\rho^3)(t^n + 2\rho^3).$$

The second factor is a unit at $z = 0$, so h has order p. Taking a square root of the unit $3\rho + \xi$ and absorbing the unit factor of h gives the completed local equation $UV = z^p$. Thus the point is of type A_{p-1} in every case. All other fibres away from $0, a, \infty$ are smooth, and X is smooth over k near the fibres over 0 and ∞; hence this is the only singular point of $X_{\bar{k}}$. □

The characteristic-19 surface of Example 11.1 lifts to characteristic zero.

Proposition 11.2. *Let $R = \mathbb{Z}[c]_{(19)}$, with residue field $k = \mathbb{F}_{19}(c)$ and fraction field $\mathbb{Q}(c)$. There is a regular scheme $\mathcal{X}$, flat and projective over R, with $\omega_{\mathcal{X}/R} \cong \mathcal{O}_{\mathcal{X}}$, such that $\mathcal{X} \otimes_R k$ is the surface X of Example 11.1 for $p = 19$ and $\mathcal{X} \otimes_R \mathbb{Q}(c)$ is a K3 surface over $\mathbb{Q}(c)$. In particular $\mathcal{X}$ is not smooth over R, although its special fibre is regular.*

Proof. Let $g\colon \mathcal{W} \to \mathcal{B} = \mathbb{P}^1_R$ be the projective Weierstrass model of $y^2 = x^3 + c^2t^7x + t$, with fundamental line bundle $\mathcal{L} = \mathcal{O}_{\mathcal{B}}(2)$. Its equation remains a nonzerodivisor on the special fibre of the ambient $\mathbb{P}^2$-bundle, so $\mathcal{W}$ is flat over R. At infinity, the coordinates $s = t^{-1}$, $u = s^4x$ and $v = s^6y$ give the equation

$$v^2 - u^3 - c^2su - s^{11} = 0.$$

Blow up the section $s = u = v = 0$ in the smooth ambient scheme, and let $\mathcal{X}$ be the strict transform. The s- and u-charts cover the exceptional conic and have equations

$$v^2 - c^2u - su^3 - s^9 = 0, \qquad v^2 - c^2s - u - u^9s^{11} = 0,$$

respectively. Both are monic in v and hence flat over R, and their derivatives in u and s, respectively, restrict to the unit $-c^2$ along the exceptional fibre. Thus $\mathcal{X}$ is flat and projective over R, and its special fibre is exactly the one-blowup resolution X of Example 11.1. Since X is regular, flatness and properness imply that $\mathcal{X}$ is regular [Sta26, Tags 00NU and 00OE]. Adjunction gives $\omega_{\mathcal{W}/R} \cong g^*(\omega_{\mathcal{B}/R} \otimes \mathcal{L}) \cong \mathcal{O}_{\mathcal{W}}$. The centre has codimension three in the smooth ambient scheme, and $\mathcal{W}$ has multiplicity two along it, so the blowup is crepant and $\omega_{\mathcal{X}/R} \cong \mathcal{O}_{\mathcal{X}}$.

Proposition 9.1 gives a K3 minimal resolution of $X_{\bar{k}}$, so rationality of its singularity gives $H^1(X, \mathcal{O}_X) = 0$. The generic fibre $\mathcal{X}_{\mathbb{Q}(c)}$ is regular, hence smooth over $\mathbb{Q}(c)$, and its canonical sheaf is trivial. By semicontinuity, $h^0(\mathcal{O}_{\mathcal{X}_{\mathbb{Q}(c)}}) = 1$ and $h^1(\mathcal{O}_{\mathcal{X}_{\mathbb{Q}(c)}}) = 0$, so $\mathcal{X}_{\mathbb{Q}(c)}$ is a K3 surface. □

Over $\mathbb{Q}(c)$, the polynomial $4c^6t^{19} + 27$ has nineteen distinct roots, at which the elliptic fibration of $\mathcal{X}_{\mathbb{Q}(c)}$ has fibres of type I_1. Modulo 19 these roots collide, and the nineteen nodes come together at the singular point of type A_{18} of $X_{\bar{k}}$.

Corollary 11.3 (Smoothness over a discrete valuation ring)**.** *Let R be a discrete valuation ring whose residue field k has characteristic $p \geq 23$, and let $\mathcal{X} \to \operatorname{Spec} R$ be a projective flat morphism whose special fibre X_0 is a regular surface with $H^0(X_0, \mathcal{O}_{X_0}) = k$ and $K_{X_0} \equiv 0$. Then $\mathcal{X}$ is smooth over R. The bound $p \geq 23$ is optimal in mixed characteristic, even for regular models with K3 generic fibre and trivial relative canonical sheaf.*

Proof. Theorem 8.9 makes X_0 smooth over k. Since $\mathcal{X} \to \operatorname{Spec} R$ is flat and of finite presentation, it is smooth along X_0 [Liu02, Definition 4.3.35]. Its non-smooth locus is closed, and its image in $\operatorname{Spec} R$ is closed by properness. This image avoids the closed point of $\operatorname{Spec} R$, so it is empty. Proposition 11.2 proves optimality with the stated additional properties. □

Geometric normality does not exclude non-rational singularities in smaller characteristic.

Example 11.4 (Non-rational geometric singularities in characteristics 7 and 11)**.** Let $p \in \{7, 11\}$ and let $k = \mathbb{F}_p(c)$, where c is transcendental over $\mathbb{F}_p$. There is a regular projective surface X over k with $H^0(X, \mathcal{O}_X) = k$ and $\omega_X \cong \mathcal{O}_X$ such that $X_{\bar{k}}$ is normal and has exactly one singular point, which is non-rational. It admits an elliptic fibration over $\mathbb{P}^1_k$ whose generic fibre has equation

$$y^2 = x^3 + t^p + c. \tag{11.2}$$

The completed local ring at the singular point of $X_{\bar{k}}$ is

$$\bar{k}[[x, y, z]]/(y^2 - x^3 - z^p).$$

Proof. Let $B = \mathbb{P}^1_k$, let $L = \mathcal{O}_B(2)$ and let $g\colon W \to B$ be the projective Weierstrass model of (11.2). The coefficient $t^p + c$ is a section of $L^{\otimes 6}$, and adjunction gives $\omega_W \cong g^*(\omega_B \otimes L) \cong \mathcal{O}_W$. The zero section lies in the smooth locus of g. The affine chart over $\mathbb{A}^1_k \subset B$ is regular: eliminating c identifies

$$\mathbb{F}_p[c, t, x, y]/(y^2 - x^3 - t^p - c) \cong \mathbb{F}_p[t, x, y],$$

and its coordinate ring over k is a localization of this polynomial ring.

At infinity put $s = t^{-1}$, $u = s^4x$ and $v = s^6y$. The equation becomes

$$v^2 = u^3 + s^{12-p} + cs^{12}.$$

Its only possible singular point on $s = 0$ is $(s, u, v) = (0, 0, 0)$. For $p = 11$ this point is smooth, because the derivative in s is -1 there. For $p = 7$, put $\eta = 1 + cs^7$. The formal change $u' = u\eta^{-1/3}$, $v' = v\eta^{-1/2}$ gives $v'^2 = u'^3 + s^5$, a split rational double point of type E_8 [Art77]. As in the proof of Example 11.1, its resolution is a sequence of blowups at k-rational double points and is smooth over k near the exceptional fibre. Let $\pi\colon X \to W$ be this resolution for $p = 7$, and the identity for $p = 11$. It is crepant, so X is regular and projective over k with $\omega_X \cong \mathcal{O}_X$. The discriminant $\Delta = -432(t^p + c)^2$ is nonzero, so the generic fibre is a smooth elliptic curve. Stein factorization gives $(g \circ \pi)_* \mathcal{O}_X = \mathcal{O}_B$, hence $H^0(X, \mathcal{O}_X) = k$.

Choose $a \in \overline{k}$ with $a^p = -c$. The partial derivatives of the affine equation over $\overline{k}$ vanish precisely at $(t, x, y) = (a, 0, 0)$. The surface $X_{\overline{k}}$ is integral and is smooth elsewhere, including near the resolved fibre at infinity; hence it is normal. Putting $z = t - a$ gives the asserted completed local ring.

Consider the weighted blowup of this local hypersurface with weights $(2, 3, 1)$ on (x, y, z). Its z-chart has coordinates $x = z^2 u$, $y = z^3 v$ and equation

$$v^2 = u^3 + z^{p-6}.$$

The exceptional curve $E = \{z = 0\}$ is reduced and the chart is smooth at its generic point, where u and v are nonzero. A generator of the original dualizing sheaf pulls back as

$$\frac{dx \wedge dz}{2y} = z^{-1} \frac{du \wedge dz}{2v}.$$

Since the second factor is a local canonical generator at the generic point of E, this divisor has discrepancy -1. Thus the singularity is not canonical, so it is not a rational double point. A rational Gorenstein surface singularity is a rational double point [Sta26, Tag 0BGB]; hence this singularity is non-rational. □

12 Surfaces with nef anticanonical class

We extend Theorems 1.1 and 8.9 to regular surfaces with nef anticanonical class. The new case $K_X \not\equiv 0$ is settled by the minimal model program and the following two lemmas.

Lemma 12.1. *Let k be a separably closed field of characteristic p, let Y be a regular surface over k, and let $P \in Y$ be a closed point with $[k(P) : k] = p$. Write $k(P) = k(\alpha)$, where $a = \alpha^p \in k$, choose $\tilde{\alpha} \in \mathcal{O}_{Y,P}$ lifting α, and put $\varphi = \tilde{\alpha}^p - a \in \mathfrak{m}_P$. Let $\varepsilon\colon Y' \to Y$ be the blowup of P, with exceptional curve E, and let ξ be the generic point of $(E_{\overline{k}})_{\mathrm{red}}$. Then the following are equivalent:*

(1) *Y is smooth over k at P;*

(2) *$\varphi \notin \mathfrak{m}_P^2$;*

(3) *$Y'_{\overline{k}}$ is normal at ξ.*

If these conditions hold, then $Y'_{\overline{k}}$ is regular at every point of $E_{\overline{k}}$ except one, which is a rational double point of type A_{p-1}.

Proof. Since $\overline{k}/k$ is purely inseparable, $Y_{\overline{k}} \to Y$ and $Y'_{\overline{k}} \to Y'$ are homeomorphisms; let $\bar{P}$ be the point of $Y_{\overline{k}}$ over P. Let $(\mathcal{O}', \bar{\mathfrak{m}})$ be the local ring of $Y_{\overline{k}}$ at $\bar{P}$, let $I = \mathfrak{m}_P \mathcal{O}'$, and put $e = \tilde{\alpha} - a^{1/p} \in \mathcal{O}'$, so that $e^p = \varphi$. The ring $\mathcal{O}'/I = k(P) \otimes_k \overline{k}$ is isomorphic to $\overline{k}[t]/(t^p)$, where t is the image of e. Hence $\bar{\mathfrak{m}} = I + e\mathcal{O}'$, and $\bar{\mathfrak{m}}/(I + \bar{\mathfrak{m}}^2)$, the cotangent space of $\overline{k}[t]/(t^p)$, is spanned by the image of e. By [Sta26, Tags 038X and 00OF], Y is smooth at P if and only if $\mathcal{O}'$ is regular.

(2)⇒(1): Since $\varphi \notin \mathfrak{m}_P^2$, there is $g \in \mathfrak{m}_P$ such that (φ, g) is a regular system of parameters of $\mathcal{O}_{Y,P}$. Then $\bar{\mathfrak{m}} = (\varphi, g, e) = (g, e)$, so the two-dimensional local ring $\mathcal{O}'$ is regular.

(1)⇒(2): Let (f, g) be a regular system of parameters of $\mathcal{O}_{Y,P}$. As $\mathcal{O}'$ is regular of dimension two, the image of I in $\bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2$ is a line; after renaming, $g \notin \bar{\mathfrak{m}}^2$, and then $\bar{\mathfrak{m}} = (g, e)$. So $\mathcal{O}'/g\mathcal{O}'$ is a discrete valuation ring with uniformizer e, and since $\mathcal{O}'/(f, g) \cong \overline{k}[t]/(t^p)$ has length p, the image of f in $\mathcal{O}'/g\mathcal{O}'$ is e^p times a unit. Hence $I = (g, e^p)$, and the image of I^2 in $\mathcal{O}'/g\mathcal{O}'$ is generated by e^{2p}. Therefore $\varphi = e^p \notin I^2 \supseteq \mathfrak{m}_P^2$. Moreover, $Y'_{\overline{k}}$ is the blowup of $Y_{\overline{k}}$ in the ideal of $P_{\overline{k}}$ [Sta26, Tag 0805], which is I at $\bar{P}$. Since (g, e^p) is a regular sequence, the blowup of $\operatorname{Spec} \mathcal{O}'$ in I is covered by $\operatorname{Spec} \mathcal{O}'[s]/(g - e^p s)$ and $\operatorname{Spec} \mathcal{O}'[u]/(e^p - gu)$, and it is an isomorphism away from $\bar{\mathfrak{m}}$. At a closed point over $\bar{\mathfrak{m}}$, the element $g - e^p s$, and also $e^p - gu$ when u

takes a nonzero value there, is congruent to a nonzero multiple of g modulo the square of the maximal ideal of $\mathcal{O}'[s]$ or $\mathcal{O}'[u]$; so these points are regular. The remaining point, where $g = e = u = 0$, has completed local ring $\overline{k}[[g, e, u]]/(e^p - gu)$, a rational double point of type A_{p-1}.

(2)⇔(3): Let $\eta \in Y'$ be the generic point of E, let v be the normalized valuation of the discrete valuation ring $\mathcal{O}_{Y',\eta}$, and let π be a uniformizer. Since $\mathrm{gr}_{\mathfrak{m}_P} \mathcal{O}_{Y,P}$ is a polynomial ring over $k(P)$, the value $v(h)$ of a nonzero $h \in \mathcal{O}_{Y,P}$ is the largest n with $h \in \mathfrak{m}_P^n$; so (2) holds if and only if $v(\varphi) = 1$. The point ξ lies over η; let A be the local ring of $Y'_{\overline{k}}$ at ξ, a one-dimensional local ring, which is normal if and only if it is a discrete valuation ring. The ring $A/\pi A = k(\eta) \otimes_k \overline{k}$ is isomorphic to $k(\eta) \otimes_{k(P)} \overline{k}[t]/(t^p)$, where $\alpha \mapsto a^{1/p} + t$. As $k(\eta) = k(P)(s)$ is purely transcendental over $k(P)$, the ring $k(\eta) \otimes_{k(P)} \overline{k}$ is a localization of $\overline{k}[s]$ which is integral over $k(\eta)$, hence a field. So $A/\pi A$ has length p, and its maximal ideal is generated by t, the image of e; thus $\mathfrak{m}_A = (\pi, e)$. If $v(\varphi) = 1$, then $\pi \in \varphi A = e^p A$, so $\mathfrak{m}_A = eA$ and A is a discrete valuation ring. Conversely, if A is a discrete valuation ring with normalized valuation w, then $w(\pi) = \mathrm{length}(A/\pi A) = p \geq 2$, so $\mathfrak{m}_A = eA + \mathfrak{m}_A^2$, and Nakayama's lemma gives $w(e) = 1$. Since $w = pv$ on $\mathcal{O}_{Y',\eta}$, we get $v(\varphi) = w(e^p)/p = 1$. □

Lemma 12.2. *Let k be a separably closed field of characteristic $p \geq 5$, and let X be a regular projective geometrically integral surface over k with $H^0(X, \mathcal{O}_X) = k$, $H^1(X, \mathcal{O}_X) = 0$ and $\rho(X) = 2$, such that $-K_X$ is nef and $K_X^2 = 0$. Let $\Gamma \subset X$ be a curve with $K_X \cdot \Gamma < 0$. Then $X_{\overline{k}}$ is normal at the generic point of $(\Gamma_{\overline{k}})_{\mathrm{red}}$.*

Proof. Let ξ be the generic point of $(\Gamma_{\overline{k}})_{\mathrm{red}}$, and suppose that $X_{\overline{k}}$ is not normal at ξ. Let H be an ample divisor on X. The divisor $-K_X$ is nef and not numerically trivial, and $K_X^2 = 0$, so the Hodge index theorem gives $-K_X \cdot H > 0$.

Choose a finite extension k'/k such that the normalization X' of $X_{k'}$ is geometrically normal. Then $S = X'_{\overline{k}}$ is the normalization of $X_{\overline{k}}$; let $\psi\colon S \to X$ be the induced morphism. If $X_{k'}$ were normal at the point over the generic point of Γ, then $X' \to X_{k'}$, and hence $S \to X_{\overline{k}}$, would be an isomorphism near the points over it, and $X_{\overline{k}}$ would be normal at ξ. So the conductor of $X' \to X_{k'}$ is a proper ideal at the points of codimension one of X' over Γ. Pulling back (2.1) to S gives

$$K_S \sim \psi^* K_X - (p-1)C$$

for an effective Weil divisor C on S with a component Γ_S lying over Γ. Put $N = \psi^*(-K_X)$, a nef Cartier divisor on S. Since $X_{\overline{k}} \to X$ is flat and $S \to X_{\overline{k}}$ is finite and birational, we have $(\psi^* D \cdot \psi^* D')_{\overline{k}} = (D \cdot D')_k$ for Cartier divisors D, D' on X; in particular $(N^2)_{\overline{k}} = 0$. The restriction of $-K_X$ to Γ is ample and ψ is finite, so $N \cdot \Gamma_S > 0$, and hence $N \cdot C > 0$.

Let $\sigma\colon \widetilde{S} \to S$ be the minimal resolution, and put $N' = \sigma^* N$. Then N' is nef with $N'^2 = 0$, and $K_{\widetilde{S}} \cdot N' = K_S \cdot N = -(p-1)\, C \cdot N < 0$. For $n \geq 0$, the pushforward to S of a member of $|K_{\widetilde{S}} - nN'|$ would be an effective Weil divisor numerically equivalent to $-(n+1)N - (p-1)C$, which has negative intersection with the ample divisor $\psi^* H$. Hence $h^2(\widetilde{S}, nN') = h^0(\widetilde{S}, K_{\widetilde{S}} - nN') = 0$, and Riemann–Roch gives

$$h^0(\widetilde{S}, nN') \geq \chi(\mathcal{O}_{\widetilde{S}}) + \frac{n(p-1)}{2}\, C \cdot N.$$

Since N' is nef with $N'^2 = 0$, this inequality gives $\kappa(\widetilde{S}, N') = 1$, so N' is semiample (cf. [Tan20, Lemma 2.7]). Choose $n > 0$ such that $|M|$ is base point free with one-dimensional image, where $M = nN'$. Let $g\colon \widetilde{S} \to B'$ be the Stein factorization of the morphism defined by $|M|$, and write M_b for the fibre over a closed point $b \in B'$. Every σ-exceptional curve Θ satisfies $M \cdot \Theta = nN' \cdot \Theta = 0$, so it lies in a fibre of g. Every fibre of σ is connected, because S is normal, and it is a point or a union of σ-exceptional curves; so it lies in a single fibre of g. Hence $\sigma(M_b) \cap \sigma(M_{b'}) = \emptyset$ for $b \neq b'$, and $\sigma^{-1}(\sigma(M_b))$ is the support of M_b. Since $M_b^2 = 0$ and the σ-exceptional curves span a negative definite lattice, $\sigma(M_b)$ is a connected curve. By Lemma 3.1, ψ is a universal homeomorphism, so the curves $G_b = \psi(\sigma(M_b))$, with their reduced structure, are pairwise disjoint and satisfy $\psi^{-1}(G_b) = \sigma(M_b)$. The effective divisor $\sigma^* \psi^* G_b$ is then supported on M_b, so

$$(-K_X \cdot G_b)_k = (N' \cdot \sigma^* \psi^* G_b)_{\overline{k}} = \frac{1}{n}(M \cdot \sigma^* \psi^* G_b)_{\overline{k}} = 0.$$

As $\rho(X) = 2$ and $-K_X$ is a nonzero class with $(-K_X)^2 = 0$, its orthogonal complement in $N^1(X)_{\mathbb{Q}}$ is spanned by $-K_X$. Hence $G_b \equiv c_b(-K_X)$, with $c_b > 0$ because $G_b \cdot H > 0$.

Every numerically trivial line bundle $\mathcal{L}$ on X is trivial. Indeed, $h^2(X,\mathcal{L}) = h^0(X, K_X - \mathcal{L}) = 0$ because $(K_X - \mathcal{L}) \cdot H < 0$; for $\mathcal{L} = \mathcal{O}_X$ this gives $h^2(X,\mathcal{O}_X) = 0$, so $\chi(\mathcal{O}_X) = 1$ by $H^1(X,\mathcal{O}_X) = 0$. Riemann–Roch then gives $h^0(X,\mathcal{L}) \geq \chi(\mathcal{L}) = \chi(\mathcal{O}_X) = 1$, and a nonzero section of $\mathcal{L}$ has an effective, numerically trivial divisor, which is zero. Fix distinct closed points $b, b' \in B'$ and positive integers m, m' with $mc_b = m'c_{b'}$. Then $mG_b \sim m'G_{b'}$, and these two disjoint members span a base point free pencil. Let $f\colon X \to B$ be the Stein factorization of the resulting morphism $X \to \mathbb{P}^1_k$; then B is a regular projective curve and $f_*\mathcal{O}_X = \mathcal{O}_B$. Every fibre X_t of f over a closed point $t \in B$ is numerically a positive rational multiple of mG_b, hence of $-K_X$. So $K_X \cdot X_t = 0$, and Γ dominates B because $\Gamma \cdot X_t > 0$. The argument for the generic fibre in the proof of Proposition 4.4, which uses only that f is flat with $f_*\mathcal{O}_X = \mathcal{O}_B$, that $K_X \cdot X_t = 0$ for all closed points $t \in B$, and that $p > 3$, shows that the generic fibre X_η is smooth over $K = k(B)$.

Since X is geometrically integral, $K' = K \otimes_k \overline{k}$ is a subring of the domain $k(X) \otimes_k \overline{k}$, and it is integral over K; so K' is a field, and $B_{\overline{k}}$ is integral with generic point $\operatorname{Spec} K'$. The point ξ lies over the generic point of $B_{\overline{k}}$, because Γ dominates B and $B_{\overline{k}} \to B$ is a homeomorphism. Hence the local ring of $X_{\overline{k}}$ at ξ is a local ring of $X_{\overline{k}} \times_{B_{\overline{k}}} \operatorname{Spec} K' = X_\eta \otimes_K K'$, which is smooth over K'. So $X_{\overline{k}}$ is regular at ξ, a contradiction. □

Proposition 12.3. *Let k be a field of characteristic $p \geq 5$, and let X be a regular projective surface over k with $H^0(X,\mathcal{O}_X) = k$ such that $-K_X$ is nef and $K_X \not\equiv 0$.*

(1) *The surface X is geometrically normal over k, and $X_{\overline{k}}$ has at worst rational double points.*

(2) *If $p \geq 11$, then X is smooth over k.*

Proof. We may assume that k is separably closed. Let H be an effective ample divisor on X. Since $-K_X$ is nef, we have $K_X^2 \geq 0$ and $K_X \cdot H \leq 0$. If $K_X \cdot H = 0$, the Hodge index theorem gives $K_X^2 \leq 0$, with equality only if $K_X \equiv 0$; hence $K_X \cdot H < 0$.

By the minimal model program for regular surfaces [Tan18b, Theorem 1.1 and Section 1.2], whose birational steps are contractions to regular points by Castelnuovo's criterion [Kol13, Theorem 10.5], there is a sequence

$$X = Y_N \xrightarrow{\varepsilon_N} Y_{N-1} \to \cdots \xrightarrow{\varepsilon_1} Y_0 = Z$$

of regular projective surfaces, where each ε_i contracts an integral curve $E_i \subset Y_i$ to a closed point $P_i \in Y_{i-1}$, and either K_Z is nef or there is a K_Z-Mori fibre space $f\colon Z \to B$. By [Sta26, Tag 0C5R], ε_i is a composition of blowups in closed points; since its exceptional locus E_i is irreducible, ε_i is the blowup of P_i. Put $\kappa_i = k(P_i)$. The blowup formulas give $K_{Y_i} = \varepsilon_i^* K_{Y_{i-1}} + E_i$ and $K_{Y_i}^2 = K_{Y_{i-1}}^2 - [\kappa_i : k]$. Nefness of the anticanonical divisor descends under each ε_i, since for every curve $C \subset Y_{i-1}$ with strict transform $\widetilde{C}$ we have $-K_{Y_{i-1}} \cdot C = (-K_{Y_i} + E_i) \cdot \widetilde{C} \geq 0$. By descending induction on i, the divisor $-K_Z$ is nef, the composite $\varepsilon\colon X \to Z$ satisfies $K_X = \varepsilon^* K_Z + F$ for an effective ε-exceptional divisor F, and

$$\sum_{i=1}^{N} [\kappa_i : k] = K_Z^2 - K_X^2 \leq K_Z^2. \tag{12.1}$$

If K_Z were nef, then $K_X \cdot H = K_Z \cdot \varepsilon_* H + F \cdot H \geq 0$, contrary to $K_X \cdot H < 0$; so $f\colon Z \to B$ is a Mori fibre space. Since $\varepsilon_* \mathcal{O}_X = \mathcal{O}_Z$ and $f_* \mathcal{O}_Z = \mathcal{O}_B$, we have $H^0(Z,\mathcal{O}_Z) = H^0(B,\mathcal{O}_B) = k$.

We show that $Z_{\overline{k}}$ has at worst rational double points, that Z is smooth over k if $\dim B = 1$ or $p \geq 11$, and that $K_Z^2 \leq 9$. If $\dim B = 0$, then Z is a regular del Pezzo surface with $\rho(Z) = 1$, and $K_Z^2 \leq 9$ by [Tan24, Theorem 1.8]. By [BT22, Theorem 3.7(1)], the surface $Z_{\overline{k}}$ is normal with at worst canonical singularities, that is, rational double points [Kol13, Theorem 2.29 and Example 3.26]; and Z is smooth if $p \geq 11$ by [BT22, Proposition 5.2]. If $\dim B = 1$, then B is a regular projective curve, and f is smooth by [BT22, Proposition 2.18], since k is separably closed and $p \neq 2$. Then Z is smooth over k by [WZ26, Proposition 3.1], applied with $\Delta = 0$, and [WZ26, Lemma 2.18] gives

$$K_Z^2 = -4 \deg_k \omega_B = 8 - 8 \dim_k H^1(B,\mathcal{O}_B) \leq 8.$$

In both cases (12.1) gives $\sum_i [\kappa_i : k] \leq 9$. Each κ_i is a finite purely inseparable extension of the separably closed field k, so $[\kappa_i : k]$ is a power of p. Since $p^2 > 9$ and $2p > 9$, every κ_i equals k except for at most one, which has degree p over k; and if $p \geq 11$, then $\kappa_i = k$ for every i.

Suppose that $(Y_{i-1})_{\bar{k}}$ has at worst rational double points. If $\kappa_i = k$, then P_i is a smooth point of Y_{i-1} [Sta26, Tags 031E and 038X], and $(Y_i)_{\bar{k}}$ is the blowup of $(Y_{i-1})_{\bar{k}}$ at the regular point over P_i [Sta26, Tag 0805]. By [Har77, II, Theorem 8.24], applied to a smooth open neighbourhood of P_i, the surface Y_i is smooth over a neighbourhood of E_i, and $(Y_i)_{\bar{k}}$ has at worst rational double points. If $[\kappa_i : k] = p$ and Y_{i-1} is smooth at P_i, then $(Y_i)_{\bar{k}}$ has at worst rational double points by Lemma 12.1.

(1) Suppose that (1) fails. By the previous paragraph and induction on i, there is an index j with $[\kappa_j : k] = p$ such that Y_{j-1} is not smooth at P_j, and $\kappa_i = k$ for all $i \neq j$. The points $P_1, \dots, P_{j-1}$ are k-rational, so their images in Z are k-rational, hence smooth, points. By the previous paragraph and induction on i, the surface Y_{j-1} is smooth over a neighbourhood in Z of these images, and $Y_{j-1} \to Z$ is an isomorphism over their complement. Hence Z is not smooth at the image P of P_j, and $[k(P) : k] = p$. In particular $\dim B = 0$, and $d = K_Z^2$ satisfies $d \geq p$ by (12.1).

Let $r\colon T \to Z_{\bar{k}}$ be the minimal resolution. By [BT22, Lemma 3.2], $Z_{\bar{k}}$ is rational, hence so is T; and $K_T = r^* K_{Z_{\bar{k}}}$, so $(K_T^2)_{\bar{k}} = (K_Z^2)_k = d$. As T is obtained from $\mathbb{P}^2_{\bar{k}}$ or a Hirzebruch surface by blowups, $\rho(T) = 10 - d$. The r-exceptional curves are linearly independent in $N^1(T)_{\mathbb{Q}}$ and orthogonal to $r^*(-K_{Z_{\bar{k}}})$, which has positive self-intersection; so there are at most $9 - d \leq 9 - p \leq 4$ of them. Let $\bar{P} \in Z_{\bar{k}}$ be the point over P. It is a rational double point with at most four exceptional curves. By the determinants listed in Table 1 and Corollary 3.4(1), applied with $W = Z$ and $S = Z_{\bar{k}}$, the only possibility is $p = 5$ and type A_4. Then $4 \leq 9 - d$, so $d = 5$. Now (12.1) gives $5 \leq \sum_i [\kappa_i : k] = 5 - K_X^2$, so $K_X^2 = 0$ and $N = 1$: the surface X is the blowup of Z at P, with exceptional curve E_1 and $K_X \cdot E_1 = E_1^2 = -5$.

We apply Lemma 12.2 to X and $\Gamma = E_1$. The scheme $X_{\bar{k}}$ is the blowup of the integral scheme $Z_{\bar{k}}$ [Sta26, Tag 0805], hence integral, and $\rho(X) = \rho(Z) + 1 = 2$. Rational double points are rational singularities [Kol13, Corollary 2.30(2)] and T is rational, so $H^1(Z, \mathcal{O}_Z) \otimes_k \bar{k} = H^1(T, \mathcal{O}_T) = 0$. Since $\mathcal{O}_{E_1}(-nE_1) \cong \mathcal{O}_{\mathbb{P}^1_{\kappa_1}}(n)$ has vanishing H^1 for $n \geq 0$, the theorem on formal functions [Har77, III, Theorem 11.1] gives $R^1 \varepsilon_{1*} \mathcal{O}_X = 0$; hence $H^1(X, \mathcal{O}_X) = H^1(Z, \mathcal{O}_Z) = 0$. So Lemma 12.2 shows that $X_{\bar{k}}$ is normal at the generic point of $((E_1)_{\bar{k}})_{\mathrm{red}}$, and Lemma 12.1 shows that Z is smooth at P, a contradiction.

(2) If $p \geq 11$, then Z is smooth over k and $\kappa_i = k$ for every i. By the paragraph before (1) and induction on i, each Y_i is smooth over k; in particular $X = Y_N$ is smooth. □

Remark 12.4. The bound (12.1) is what controls the residue fields κ_i, and without such a bound a blowup can destroy geometric normality. For instance, let $a, b \in k$ satisfy $[k(a^{1/p}, b^{1/p}) : k] = p^2$, and let $P \in \mathbb{P}^2_k$ be the closed point defined in an affine chart by $x^p = a$ and $y^p = b$. The blowup X of $\mathbb{P}^2_k$ at P is regular, because its exceptional divisor $E \cong \mathbb{P}^1_{k(P)}$ is a regular Cartier divisor. By [Sta26, Tag 0805], $X_{\bar{k}}$ is the blowup of $\mathbb{P}^2_{\bar{k}}$ in the ideal (u^p, v^p), where $u = x - a^{1/p}$ and $v = y - b^{1/p}$. It contains the chart $u^p = v^p t$ in $\mathbb{A}^3_{\bar{k}}$, which is singular along the line $u = v = 0$; hence X is not geometrically normal. Here $K_X^2 = 9 - p^2 < 0$, so $-K_X$ is not nef.

A Verification scripts

The complete source, running instructions and recorded outputs are supplied in the companion archive `verification-scripts.zip`, with the directory structure `smooth/` and `p11/` preserved. The Python dependencies are SymPy, NetworkX and NumPy. The local algebra checks use Singular [DGPS25]. Run `python3 path/to/script.py` or `Singular -q path/to/script.sing` from the extracted archive, with the appropriate path. The script `smooth/p13.py` requires the `fork` multiprocessing method.

A.1 Enumeration of exceptional configurations

The script `smooth/configs.py` enumerates the positive definite matrices $A = (-E_i \cdot E_j)$ of size $r = 8 + d$ satisfying Lemmas 8.4 and 8.6. With z_i the multiplicities of the exceptional curves and $a_i = K_T \cdot E_i$, the search imposes

$$Az = a, \qquad z_i \in \mathbb{Z}_{>0}, \qquad z \cdot a = d, \qquad 1 \leq d \leq 4.$$

The prime values of the determinant in its output are

d	r	prime values of $\det A$
1	9	$\varnothing$
2	10	5, 17
3	11	5
4	12	13, 53, 113

In particular the only primes at least 29 occur for $d = 4$, as used in Proposition 8.8. The script `smooth/crosscheck.py` independently enumerates the configurations with transversal pairwise intersections and no triple points, running over all trees with r vertices. Its determinant sets agree with the `simple-tree dets` lines of `configs.py`. The general search also covers tangencies and triple points.

A.2 Characteristics 13 and 17

The scripts `smooth/p13.py` and `smooth/p17.py` use the same search and retain matrices with $\det A \in \{p, p^3\}$. For $1 \le d \le 3$, they give no configuration for $p = 13$ and exactly one for $p = 17$, with $d = 2$ and $r = 10$. These are the outputs used in Proposition 9.1. The script `smooth/p17_local.sing` checks one representative of the surviving graph. The proof of the proposition gives the obstruction for every analytic type with that graph.

A.3 The characteristic-11 counterexample

The script `p11/p11_checks.py` checks the derivative table, the invariant equation, the residues and orders along $u = 0$, the value at infinity and the coordinate identities for geometric nonnormality in Section 10. The script `p11/ekedahl.sing` independently checks the normalized gradient certificate. The identities (10.3) and (10.4) are derived directly in Section 10.

A.4 Examples and the lift to characteristic zero

The scripts `smooth/small_primes_checks.py` and `smooth/small_primes_local.sing` check the equations and local calculations in Example 11.1. The script `smooth/small_primes_lift_checks.py` checks the two blowup charts and their unit derivatives over $\mathbb{Z}[c]_{(19)}$ in Proposition 11.2. The script `smooth/nonrational_checks.py` checks the equations and discrepancy calculation in Example 11.4 and imports `small_primes_checks.py`.

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